UNIVERSITY
OF OSLO

Master's Thesis Spring 2026

Numerical Optimization and Braiding of Poor Man's Majorana Chains

Interactions, Projected Exchange, and Higher-Energy Sectors

Hishem Kløvnes

60 ECTS study points

Computational Science: Physics
Faculty of Mathematics and Natural Sciences

Spring 2026

Hishem Kløvnes

Numerical Optimization and Braiding of Poor Man's Majorana Chains

Interactions, Projected Exchange, and Higher-Energy Sectors

Supervisors:
Viktor Svensson
Morten Hjort-Jensen

Abstract

Poor Man's Majorana (PMM) models provide a compact and tunable platform for quantum computation, but their finite size and sensitivity to interactions make it difficult to identify useful parameter regimes. This thesis investigates whether interacting quantum-dot chains can be numerically optimized to exhibit favorable Majorana-like properties and whether the resulting configurations support the expected Majorana exchange within projected braiding models.

Spinless two-, three-, and four-dot PMM chains are studied at fixed hopping and interaction strengths. The onsite energies and pairing amplitudes are optimized using a customized loss function that combines typical Majorana diagnostics. Selected configurations are then used as building blocks for a three-subsystem braiding geometry. The resulting geometric braid unitaries are computed using Kato transport and compared across ideal Majorana, projected microscopic-junction, energy-level, projected local-operator, and block-matched local-operator constructions.

The optimization reproduces the known non-interacting two-dot sweet spot and finds increasingly inhomogeneous parameter profiles for longer chains. Among the tested interacting configurations, the three-dot system at $\frac{U}{t} = 0.1$ provides a strong overall balance of spectral near-degeneracy, excitation gap, edge-localized Majorana polarization, and suppressed bulk polarization. Within its lowest projected manifold, the projected microscopic junction braid reproduces the expected Majorana exchange with high accuracy. Sector-resolved self-target scans remain accurate across the tested sectors for the non-interacting and weakly interacting configurations, while the selected stronger-interaction local-operator constructions show greater sector dependence.

The block-matching analysis shows that projected local operators can be represented essentially exactly within the energy-level Majorana basis. However, different internal blocks of the same higher-dimensional energy sector can follow opposite exchange orientations. A complete higher-dimensional sector therefore does not necessarily implement one uniform logical gate. These results demonstrate useful Majorana-like exchange within controlled finite projected models, while also showing that extending braiding beyond a known low-energy block requires control of the occupied block and a common-target comparison between local and energy-level exchange channels.

Contents

Chapter 1

Introduction

In 1937, the Italian physicist Ettore Majorana showed that the relativistic wave equation for fermions admits a completely real representation. In his paper *Teoria simmetrica dell'elettrone e del positrone* (*Symmetric theory of the electron and positron*), he described the possibility of a neutral fermion that is identical to its own antiparticle [28, 31]. No elementary Majorana fermion has yet been conclusively identified. However, collective excitations in solid-state systems can behave as Majorana quasiparticles, providing a possible route for studying Majorana physics in controlled devices [9, 29, 45].

The interest in these quasiparticles is closely connected to one of the central challenges of quantum computation: quantum information is highly sensitive to environmental noise. Fluctuations in temperature, electromagnetic fields, and imperfect control can cause decoherence and introduce errors into a calculation [5]. Majorana zero modes offer a possible way to reduce sensitivity to some local disturbances. An ordinary fermionic mode can be decomposed into two Majorana components, and in a one-dimensional topological superconductor these components may become spatially separated at opposite ends of the system [1, 6, 14]. The corresponding fermionic degree of freedom is encoded non-locally. Local perturbations then have limited access to the complete encoded state, provided that fermion parity is preserved, the excitation gap remains open, and overlap between the two Majorana components is sufficiently small [6, 10].

Majorana zero modes are also predicted to exhibit non-Abelian exchange statistics. Adiabatically exchanging two modes transforms the state within a degenerate subspace and can implement a quantum gate [27, 45]. Such an exchange may be realized by physically moving the modes or by tuning couplings in a sequence that produces the same exchange operation. The possibility of combining non-local encoding with exchange-based gates is what makes Majorana systems interesting for topological quantum computation.

The minimal theoretical model illustrating this physics is the Kitaev chain, introduced by Alexei Kitaev in 2000 [20]. It describes a one-dimensional chain of spinless fermions with nearest-neighbor hopping and p-wave superconducting pairing. In its topological phase, the model supports Majorana zero modes at its ends and provides a clear description of how non-local fermionic states and exchange operations can emerge. The Kitaev chain is highly idealized, however, and realizing its ingredients in a controllable physical system remains challenging [36].

One compact engineered platform inspired by the Kitaev chain is the quantum-dot–superconductor setup, often referred to as a Poor Man's Majorana (PMM) system [24, 37]. Quantum dots act as discrete sites, while superconducting segments generate effective hopping and non-local pairing through elastic cotunnelling and crossed Andreev reflection. The dot energies, pairing strengths, tunnel couplings, and interaction strengths can be tuned, making PMM systems useful for exploring Majorana-like physics in finite devices. Their finite size is also a fundamental limitation: the resulting modes are not protected in the same way as the edge modes of an infinite topological system, and low-energy degeneracy alone does not establish that a configuration has useful Majorana character or supports the desired exchange operation [44].

The analytical non-interacting two-dot sweet spot is known, but longer interacting chains contain many tunable parameters and no simple general sweet-spot condition [42]. Numerical optimization offers a way to search this parameter space, but optimizing energy degeneracy alone could also favor ordinary accidental degeneracies. A useful search must therefore combine several complementary criteria, including even–odd parity degeneracy, excitation gaps, local charge indistinguishability, and edge-localized Majorana polarization. Finding a favorable static configuration is still not sufficient: the stronger operational test is whether it supports the expected Majorana exchange under a braiding protocol. This thesis additionally asks whether the effective exchange structure remains meaningful when the projected space is enlarged beyond the lowest manifold.

The investigation is organized around four research questions:

1. Can numerical optimization recover known PMM sweet spots and identify promising Majorana-like configurations in interacting two-, three-, and four-dot chains?

2. Which configurations best balance parity degeneracy, excitation gaps, local charge indistinguishability, and edge-localized Majorana polarization?

3. Can projected microscopic junction couplings reproduce the ideal Majorana exchange within a selected low-energy manifold?

4. How do energy-level and projected local-operator braid constructions behave across higher energy sectors, and what does the internal block structure of these sectors imply for logical gate operations?

1.1 Scope and Interpretation

The systems studied in this thesis are finite, spinless PMM chains represented in the full many-body Fock basis and analyzed by exact diagonalization. Throughout the thesis, the term "Majorana-like" refers to finite-system configurations that satisfy the selected spectral and local diagnostics. It does not imply exact topological protection or prove the existence of perfectly isolated Majorana zero modes, but that is why the PMM system is interesting: it can be tuned to produce low-energy states that are close enough to the Majorana ideal to make the physical interpretation meaningful and the braiding behavior nontrivial. The numerical optimization identifies favorable minima of the chosen loss function, but it cannot guarantee that the global optimum has been found.

The braiding calculations are also performed within controlled projected models. In the sector-resolved scans, the energy-level, direct local-operator, and block-matched local-operator constructions are each compared with a target defined from their own exchange operators. These self-target comparisons test whether each construction executes its intended braid and how that performance depends on the selected energy sector. They do not by themselves establish that the local constructions reproduce the same exchange channel as the energy-level construction. That stronger statement requires a common-target comparison.

1.2 Method Overview

The first part of the numerical study searches for favorable PMM chains. The hopping amplitudes are fixed to set the energy scale, while the onsite energies and pairing amplitudes are optimized for several chain lengths and fixed interaction strengths. The loss function combines parity-pair splitting, excitation gaps, local charge differences, and Majorana-polarization profiles. Local gradient-based optimization is combined with repeated restarts and basin hopping to explore the resulting non-convex parameter landscape. The selected configurations are then reconstructed and evaluated using parity-resolved spectra and local Majorana diagnostics.

The second part tests braiding. The strongest candidate of the interacting configurations is used as the building block for a three-subsystem junction. Within its lowest projected manifold, an ideal four-Majorana braid is compared with braids generated by projected microscopic junction operators and projected local Majorana components. The analysis is then extended sector by sector using energy-level Majoranas, direct projected local operators, and block-matched local operators. For each construction, the geometric unitary generated by the braid path is computed using Kato transport and compared with the corresponding target exchange.

1.3 Thesis Outline

Chapter 2 presents the theoretical background, including Majorana zero modes, the Kitaev chain and PMM systems, diagnostics of Majorana-like states, Majorana exchange, projected subspaces, and Kato transport. Chapter 3 describes the numerical PMM representation, optimization strategy, post-optimization diagnostics, effective Majorana construction, braiding models, and braid-validation procedure. Chapter 4 presents the optimized configurations, selects the strongest braiding candidate, benchmarks the braid in the lowest projected manifold, and extends the analysis across the identified energy sectors. Chapter 5 discusses the physical interpretation, quantum-computing implications, and limitations of the results. Chapter 6 summarizes the principal conclusions and outlines directions for future work.

Chapter 2

Theory

Somewhere in this section write more clearly about the difference between non-interacting and interacting theories.

2.1 Majorana Zero Modes and Topological Motivation

Topological quantum computing is an approach to fault-tolerant quantum computation in which quantum gates are implemented by braiding non-Abelian anyons. The central idea is that information can be encoded non-locally in a degenerate subspace, so that local perturbations have limited access to the encoded quantum state. In this setting, the result of a gate depends on the topology of the braid rather than on microscopic details of the path.

Anyons are quasiparticles that, so far, have only been observed in effectively two-dimensional systems. In three spatial dimensions, identical particles are classified as either bosons or fermions, corresponding to symmetric and antisymmetric exchange statistics. In two dimensions, however, the topology of particle exchange allows for a richer possibility: exchanging two particles can produce statistics that are neither bosonic nor fermionic [16]. In particular, exchanging two identical particles twice is not necessarily topologically equivalent to doing nothing, because the exchange path cannot always be continuously shrunk to a point without crossing another particle. Particles with such exchange statistics are called anyons, and they are classified as Abelian or non-Abelian depending on how the quantum state transforms under exchange. Abelian anyons acquire only a phase factor, while non-Abelian anyons transform the state within a degenerate subspace. This latter property is what makes non-Abelian anyons especially interesting for quantum computation [41].

A collection of non-Abelian anyons with fixed positions and fixed local quantum numbers can possess a non-trivial topological degeneracy. Quantum information can then be encoded in this degenerate subspace, while braiding the anyons implements unitary transformations within it. Since the resulting transformation depends only on the braid topology, and not on small local deformations of the path, this provides a possible route toward fault-tolerant quantum gates. In suitable schemes, braiding operations can be combined with additional ingredients to implement the gates needed for quantum computation.[38]

One of the most promising candidates for realizing non-Abelian exchange statistics in condensed-matter systems is the Majorana zero mode (MZM), predicted to occur

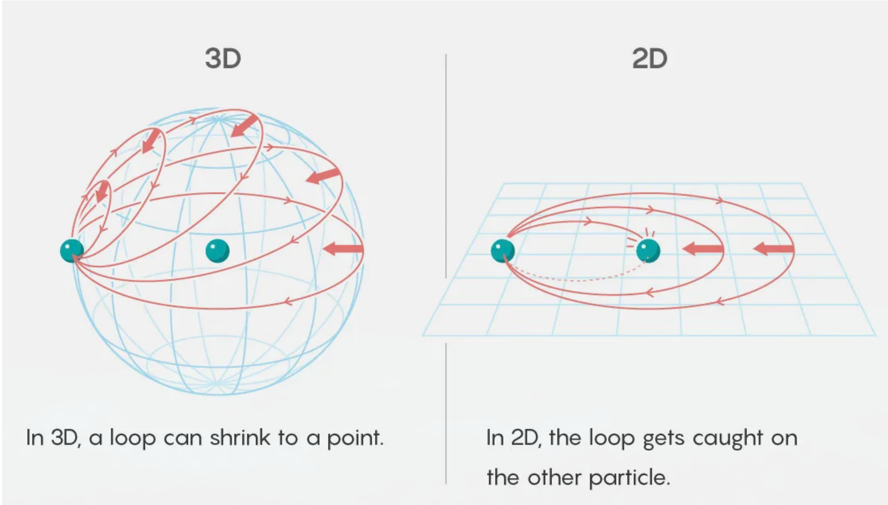

Figure 2.1: Visualization of particle exchange in two and three dimensions. Illustration by 5W Infographics for Quanta Magazine, reproduced from [30].

in certain topological superconductors. A Majorana zero mode is a zero-energy quasiparticle described by an operator that is equal to its own Hermitian conjugate. In one-dimensional topological superconductors, such modes can appear at the ends of the system. Unlike an ordinary fermionic mode, a single MZM can be viewed as "half" of a Dirac fermion: two Majorana operators can be combined to form one conventional fermionic operator,

$$c = \frac{1}{2}(\gamma_1 + i\gamma_2) \tag{2.1}$$

with the inverse relations

$$\gamma_1 = c + c^\dagger, \qquad \gamma_2 = -i(c - c^\dagger). \tag{2.2}$$

Here γ_1 and γ_2 are Majorana operators satisfying

$$\gamma_i = \gamma_i^\dagger, \quad \{\gamma_i, \gamma_j\} = 2\delta_{ij} \tag{2.3}$$

The non-Abelian character of MZMs is reflected in the fact that exchanging two of them produces a non-trivial unitary transformation on the degenerate ground-state manifold [39].

The primary motivation for studying MZMs and their braiding properties is their combination of non-local encoding and spectral protection. When a fermionic state is formed from two spatially separated MZMs, local environmental noise cannot easily determine or flip the encoded state, because such an error would require coupling to both separated Majorana components. Therefore, local perturbations are strongly suppressed as a source of decoherence [32].

MZMs also give rise to ground-state degeneracies. A system with $2n$ Majorana zero modes spans a 2^n-dimensional fermionic Hilbert space associated with the occupations of n non-local fermions. If the total fermion parity is fixed, the physical ground-state degeneracy is reduced to 2^{n-1}. In either case, the useful low-energy manifold must be separated from excited states by a finite spectral gap, which protects the encoded information against thermal excitations[21]

The remaining question is how such Majorana operators can emerge from an ordinary fermionic Hamiltonian. The Kitaev chain provides the minimal model where this

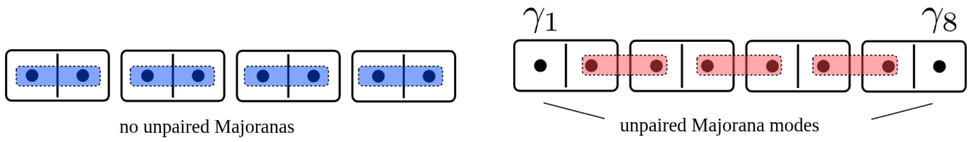

Figure 2.2: Schematic representation of the Kitaev chain in its trivial and topological phases. Left: Majorana modes pair on the same sites. Right: Majorana modes pair between adjacent sites, leaving unpaired modes at the chain ends. Adapted from the *Topology in Condensed Matter* course [2], licensed under CC BY-SA 4.0.

mechanism can be seen explicitly, and it also gives the conceptual starting point for the Poor Man's Majorana systems studied in this thesis.

2.2 Kitaev Chain and Poor Man's Majorana Systems

Before discussing the Majorana models used in this thesis, we first introduce their theoretical origin: the Kitaev chain. The Kitaev chain, or Kitaev-Majorana chain, is a simplified model for a topological superconductor. First proposed by Alexei Kitaev in 2000 [21], the Kitaev chain is a one-dimensional superconducting system that can host Majorana zero modes at its ends under certain conditions, illustrated in Figure 2.2. The model is defined on a lattice of spinless fermions with nearest-neighbor hopping and superconducting pairing. The Hamiltonian of the Kitaev chain is

$$H = -\mu \sum_{j=1}^{N} \left(c_j^\dagger c_j - \frac{1}{2} \right) - t \sum_{j=1}^{N-1} \left(c_j^\dagger c_{j+1} + c_{j+1}^\dagger c_j \right) + \Delta \sum_{j=1}^{N-1} \left(c_j c_{j+1} + c_{j+1}^\dagger c_j^\dagger \right), \tag{2.4}$$

Each term in Eq. 2.4 has a clear physical interpretation. The first term represents the chemical potential μ, which controls the filling of the fermionic states. The second term describes the hopping of fermions between neighboring sites with amplitude t. The third term represents p-wave superconducting pairing, where pairs of fermions are created or annihilated on adjacent sites with amplitude Δ.
As shown schematically in Fig. 2.2, the Kitaev chain has two distinct phases depending on the parameters μ, t, and Δ. In the trivial phase, the Majorana modes on the same site pair up to form conventional fermions, leaving no unpaired modes. In the topological phase, Majoranas on adjacent sites couple, leaving unpaired modes at the chain ends. These unpaired Majoranas combine into a non-local fermionic mode that can be occupied or empty without costing energy, leading to a twofold degeneracy in the ground state. This degeneracy is protected as long as local perturbations do not close the bulk gap or directly hybridize the edge modes.

The topological phase of the Kitaev chain is the basic model for the Majorana zero modes studied in this thesis. The goal is then to understand how similar Majorana-like physics can be engineered in more realistic systems.

2.2.1 Bulk Spectrum and Topological Phase

In order to determine when the Kitaev chain is topological, we need to consider the properties of the bulk spectrum.

For a translationally invariant chain, we can perform a Fourier transform to momentum space, which allows us to analyze the energy spectrum and identify the conditions for the topological phase. The Fourier transform is given by

$$c_j = \frac{1}{\sqrt{N}} \sum_p e^{ipaj} c_p \tag{2.5}$$

This transformation allows the Hamiltonian to be expressed in terms of momentum-space operators c_p and $c_p^\dagger$. The resulting Hamiltonian in momentum space is

$$H = \sum_p \xi_p \left(c_p^\dagger c_p - \frac{1}{2} \right) - \sum_p t \cos pa + \Delta \sum_p \left(c_p^\dagger c_{-p}^\dagger e^{ipa} + c_{-p} c_p e^{-ipa} \right), \tag{2.6}$$

with $\xi_p = -\mu - 2t \cos pa$ and a being the lattice spacing. In the Bogoliubov–de Gennes formalism, the quasiparticle spectrum becomes

$$E_{p,\pm} = \pm\sqrt{(\mu + 2t \cos pa)^2 + 4\Delta^2 \sin^2 pa}. \tag{2.7}$$

Even though the Bogoliubov–de Gennes formalism describes a non-interacting quasiparticle problem, it still provides the basic topological picture that the interacting models in this thesis build on. In Fig. 2.3, we can see how the spectrum changes as the chemical potential μ is tuned. The presence of a gap is crucial for the stability of the Majorana modes, and the closing and reopening of this gap signals a topological phase transition.

The quasiparticle spectrum is gapped for most parameter values, but the gap closes when

$$|\mu| = 2|t|.$$

This is also visible in Fig. 2.3. The gap closing marks the transition between the trivial and topological phases. When μ crosses this boundary, the ground state changes its topological character without requiring ordinary symmetry breaking [1].

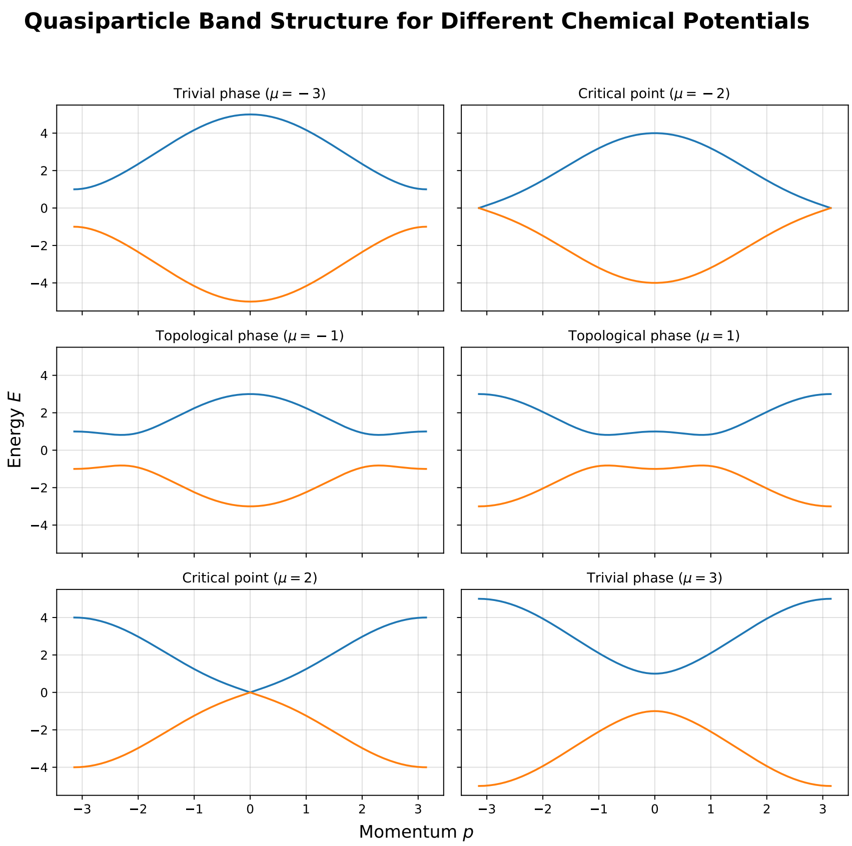

Figure 2.3: Quasiparticle energy spectrum of the Kitaev chain for different chemical potentials μ. Left: In the trivial phase ($|\mu| > 2|t|$), the spectrum is fully gapped with no zero-energy modes. Right: In the topological phase ($|\mu| < 2|t|$), the gap closes and reopens, signaling the emergence of zero-energy edge modes.

2.2.2 Majorana Representation and Zero Modes

Looking back at Fig. 2.2, we want a representation that makes the unpaired edge modes more explicit. To do so, we express the fermionic operators in terms of Majorana operators. For each site j, we define two Majorana operators γ_j^A and γ_j^B as follows:

$$c_j = \frac{1}{2}(\gamma_j^A + i\gamma_j^B), \quad c_j^\dagger = \frac{1}{2}(\gamma_j^A - i\gamma_j^B), \tag{2.8}$$

where $\gamma_j^{A,B}$ satisfy $\{\gamma_j^\alpha, \gamma_k^\beta\} = 2\delta_{jk}\delta_{\alpha\beta}$ and $(\gamma_j^\alpha)^\dagger = \gamma_j^\alpha$. In the trivial phase, the Majorana operators pair on the same site. In the topological phase, and especially at the ideal point $t = \Delta$ and $\mu = 0$, γ_j^B couples to γ_{j+1}^A. This leaves two edge modes unpaired, γ_1^A and γ_N^B, at the chain ends. These two unpaired Majoranas define a non-local fermionic mode,

$$f = \frac{1}{2}(\gamma_1^A + i\gamma_N^B), \tag{2.9}$$

which satisfies $\{f, f^\dagger\} = 1$ and costs zero energy to occupy. This non-local fermionic mode creates the basis for the exploration of Majorana physics in this thesis, and it is the key to understanding how Majorana modes can be used for quantum computation. The occupation of this non-local mode corresponds to a twofold degeneracy in the ground state, which is protected as long as local perturbations do not close the bulk gap or directly hybridize the two edge modes. This non-local encoding of information is what gives Majorana modes their potential for fault-tolerant quantum computation, as local errors cannot easily access or change the encoded state[1].

2.2.3 Poor Man's Majorana Systems

The ideal Kitaev chain is a useful theoretical model, but it is not directly realizable in most physical systems, as it ideally prefers to be infinite, homogeneous, and non-interacting. The reason for this, is that the Majorana modes are localized at the edges of the chain, and their energy splitting decreases exponentially with the chain length $\exp\left(-\frac{L}{l_0}\right)$[21]. This would make the Majorana modes robust against any local perturbation. However, in practice we have to work with finite systems, and the Majorana modes will generally have a small energy splitting due to their overlap. However, a Poor mans Majorana system is a finite system, which under the right parameter conditions, can mimic the low-energy physics of a Kitaev chain, including the presence of Majorana-like modes, in a much more compact and experimentally accessible setup.
While the Kitaev chain is the minimal theoretical model, it assumes spinless fermions and intrinsic p-wave pairing. Real superconductors are usually spinful and dominated by conventional s-wave pairing. The Poor Man's Majorana (PMM) platform is a compact engineered system whose low-energy behavior can mimic a short Kitaev chain. The simplest version is a double quantum dot coupled through a superconducting segment. Each dot acts as a site, elastic cotunneling produces effective hopping, and crossed Andreev reflection produces an effective non-local pairing term.

The effective two-dot Hamiltonian can be written as

$$H = \sum_{i=L,R} \epsilon_i n_i + t(d_L^\dagger d_R + d_R^\dagger d_L) + \Delta(d_L^\dagger d_R^\dagger + d_R d_L) + U n_L n_R, \tag{2.10}$$

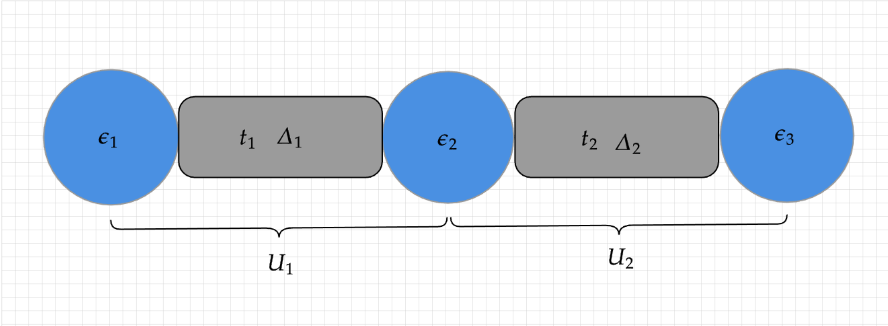

Figure 2.4: Schematic representation of the PMM setup. The three quantum dots are represented by the blue circles, and the superconducting segments are represented by the grey rectangles. The tunable parameters of the system, such as the dot energy levels, ECT, CAR and the Coulomb interaction, are indicated by their respective labels. By tuning these parameters, we can access different regimes of the system and identify the sweet spots where Majorana-like modes emerge.

where $d^\dagger_{L/R}$ and $d_{L/R}$ are creation and annihilation operators on the left and right dots, n_i is the number operator, $\epsilon_{L,R}$ are the dot energy levels, t is the elastic cotunneling amplitude, Δ is the crossed Andreev reflection amplitude, and U is the interdot Coulomb interaction energy. The parameters map directly onto the Kitaev-chain language:

- **Elastic cotunneling (ECT):** An electron tunnels coherently from one dot to the other through a virtual process in the superconductor. This produces the effective hopping amplitude t.
- **Crossed Andreev reflection (CAR):** A Cooper pair splits, with one electron entering each dot. This produces the non-local pairing amplitude Δ.
- **On-site energy:** The dot energies ϵ_L and ϵ_R play the role of local chemical potentials.

At the ideal non-interacting sweet spot, $|\epsilon_L| = |\epsilon_R| = 0$ and $|t| = |\Delta|$, the two-dot system develops Majorana-like states localized primarily on opposite dots. Since the system is finite and lacks a true bulk, these states are not topologically protected in the same sense as an infinite Kitaev chain. This is why they are referred to as Poor Man's Majoranas [1].

For the purposes of this thesis, the PMM idea is generalized from two dots to interacting n-dot chains. A schematic representation is shown in Figure 2.4. Adding more dots introduces more tunable parameters and more possible low-energy structure, but it also makes the analytic sweet-spot problem much harder. This motivates the numerical optimization strategy discussed later in the thesis.

2.3 Diagnostics of Majorana-like States

In a finite PMM system, the presence of low-energy or near-zero-energy states is necessary, but not sufficient, evidence for Majorana physics. As discussed in the previous section, the ideal Kitaev chain has zero-energy Majorana modes at its ends, producing a degenerate low-energy manifold. In a short and finite PMM chain, however, the Majorana modes can overlap, hybridize, and split away from zero energy. Other ordinary low-energy states can also appear because the dot energies, tunnel couplings, pairing amplitudes, and interactions are all tunable.

For this reason, the identification of Majorana-like states must rely on several complementary diagnostics rather than a single spectral feature. The diagnostics used here can be divided into two broad groups. Spectral diagnostics test whether the low-energy manifold has the expected parity degeneracy and excitation gap. Local diagnostics test whether the corresponding states are non-local, locally indistinguishable, and Majorana-like in their particle-hole structure. Together, these quantities define what is meant in this thesis by a promising Majorana-like configuration.

2.3.1 Parity-Resolved Spectrum

An essential first step in identifying Majorana-like states is resolving the energy spectrum by fermion parity. The mean-field Hamiltonian of a superconductor does not conserve particle number because superconducting pairing terms connect states whose particle numbers differ by two. It does, however, conserve the parity of the total particle number. The parity operator can be written as

$$P = (-1)^N = \prod_n (1 - 2c_n^\dagger c_n),$$

where N is the total particle number and n labels the single-particle orbitals. Its eigenvalues are $+1$ for even parity and -1 for odd parity.

Two Majorana components combine to form a non-local fermionic mode. Occupying or emptying this mode changes the total fermion parity, connecting states in the even- and odd-parity sectors. At the ideal two-dot sweet spot, where the dot energies vanish and $|t| = |\Delta|$, the ground states in these two sectors become degenerate [2, 12, 25].

Parity resolution does not by itself identify Majorana-like states. Instead, it makes it possible to search for degeneracies between opposite-parity states, as expected when a zero-energy fermionic mode connects the two sectors. In this thesis, parity resolution is additionally used to examine whether analogous even-odd pairing persists at higher energy levels and how including these levels affects the projected braiding dynamics.

2.3.2 Energy Degeneracy and Spectral Gap

Once the states have been separated into parity sectors, the next diagnostic is the energy splitting between parity partners. Majorana-like physics requires a near-degenerate pair of states, usually one even and one odd, associated with the occupation of the non-local fermionic mode. In an ideal infinite Kitaev chain, this degeneracy is exact. In a finite PMM chain, the edge modes can overlap, producing a small but nonzero splitting. The goal is therefore not only to find low-energy states, but to find parameter regimes where the even-odd splitting is small compared with all other relevant energy scales.

The excitation gap is equally important. The gap separates the low-energy parity manifold from quasiparticle excitations and suppresses thermal or dynamical leakage out of the computational subspace. In PMM and quantum-dot-chain proposals, the useful gap is controlled by the same physical couplings that create the effective Kitaev chain, especially the elastic cotunneling and crossed-Andreev pairing amplitudes. If the gap closes, the Majorana-like signature is no longer protected, and the low-energy

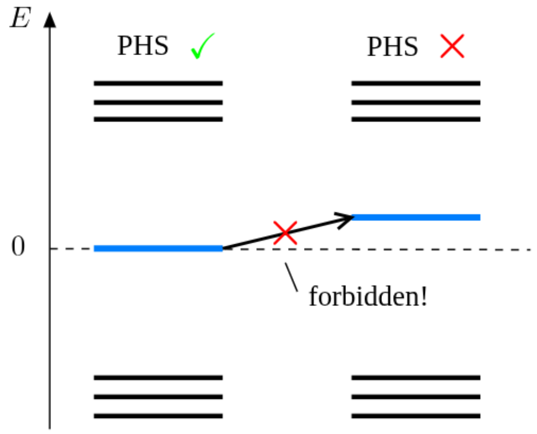

Figure 2.5: Energy spectrum showing how the energy eigenvalues of the system must be symmetric around zero energy due to particle-hole symmetry. FIgure from.[2]

interpretation becomes unreliable [40]. A sizable gap is also essential for the braiding calculations in this thesis, because it justifies projecting the dynamics onto a low-energy subspace.

This thesis also explores whether finite interacting chains can display approximate even-odd degeneracy across several energy levels, not only in the ground state. If such a structure survives, then some Majorana-like features may remain meaningful even when excited states are included in the projected Hilbert space. Figure 2.5 illustrates the particle-hole symmetric structure expected in the idealized superconducting spectrum. In practice, the relevant diagnostic is whether the candidate parity partners are both nearly degenerate and well separated from the rest of the spectrum [2].

2.3.3 Local Distinguishability and Charge Expectation

A defining feature of spatially separated Majorana modes is that the encoded information is non-local. Local measurements should therefore not reveal whether the system is in the even or odd ground state. This principle is central to the topological motivation for Majorana qubits: the qubit state is stored globally, while local perturbations should have only limited access to it [32]. In the PMM setting, the same idea appears in a finite-system form: the parity qubit can only be fully read out by probing both Majorana components, rather than a single dot alone [26].

One way to quantify this property is through the local distinguishability parameter

$$\mathrm{LD} = \frac{1}{N}\sum_{j=1}^{N} ||\rho_j^o - \rho_j^e|| = \frac{1}{N}\sum_{j=1}^{N} ||\delta\rho_j||.$$

[43]. Here, $\rho_j^{o/e}$ are the reduced density matrices of site j for the odd and even parity ground states, respectively, and $||\cdot||$ is the trace norm. A value close to zero indicates that the two states are locally indistinguishable, while a larger value indicates that local measurements can tell the two parity states apart.

In the spinless model considered in this thesis, the local charge occupation is the only non-trivial parity-preserving observable on a single site. Local distinguishability can therefore be evaluated directly from the charge expectation values in the even and odd states,

$$\langle e| n_{L(R)} |e\rangle , \quad \langle o| n_{L(R)} |o\rangle .$$

An ideal Majorana mode has no ordinary local charge identity: it is an equal particle-hole superposition rather than a conventional electron or hole. In a PMM system, this means that the local charge distribution should not reveal the parity state of the non-local fermion. At the sweet spot, the relevant charge signatures become symmetric between the two parity states, corresponding to the "half-electron" or charge-neutral character of the Majorana components [13, 26].

For the diagnostics used here, the important quantity is not the absolute charge on a dot, but the difference between the even- and odd-parity partners. For a spinless site, this charge difference contains the same information as the difference between the corresponding local reduced density matrices. A small charge difference therefore means that the two states are locally indistinguishable and that the parity information is stored non-locally rather than being visible on a single dot. In a spinful system, additional local observables could distinguish the states even when their local charge expectation values agree.

2.3.4 Majorana Polarization

Local indistinguishability tests whether the information is hidden from local measurements, but it does not by itself say whether the operator connecting the two parity states has the particle-hole structure expected of a Majorana mode. For this, we use the Majorana polarization (MP). The MP is a local measure of how Majorana-like a state is: an ideal Majorana mode should be self-conjugate, with balanced particle and hole character, and should be localized near the ends of the chain. In an ideal PMM configuration, the polarization should therefore be concentrated at the edge dots, with opposite signs on the two ends and little or no weight in the bulk[43].

One useful definition is

$$M_\alpha = \frac{W_\alpha^2 - Z_\alpha^2}{W_\alpha^2 + Z_\alpha^2},$$

where

$$W_\alpha = \sum_\sigma \langle o| c_{\alpha\sigma} + c_{\alpha\sigma}^\dagger |e\rangle , \quad Z_\alpha = i \sum_\sigma \langle o| c_{\alpha\sigma} - c_{\alpha\sigma}^\dagger |e\rangle .$$

Here, α denotes the site index, σ is the spin index, and $|o\rangle$ and $|e\rangle$ are the odd- and even-parity ground states. In an ideal sweet spot with two perfectly localized Majoranas, the edge values would satisfy $M_L = -M_R = \pm 1$ [1]. In the spinless model used later in this thesis, the same idea is applied without the spin sum: the diagnostic checks whether the parity-changing Majorana matrix elements are localized at the chain ends with the expected relative sign.

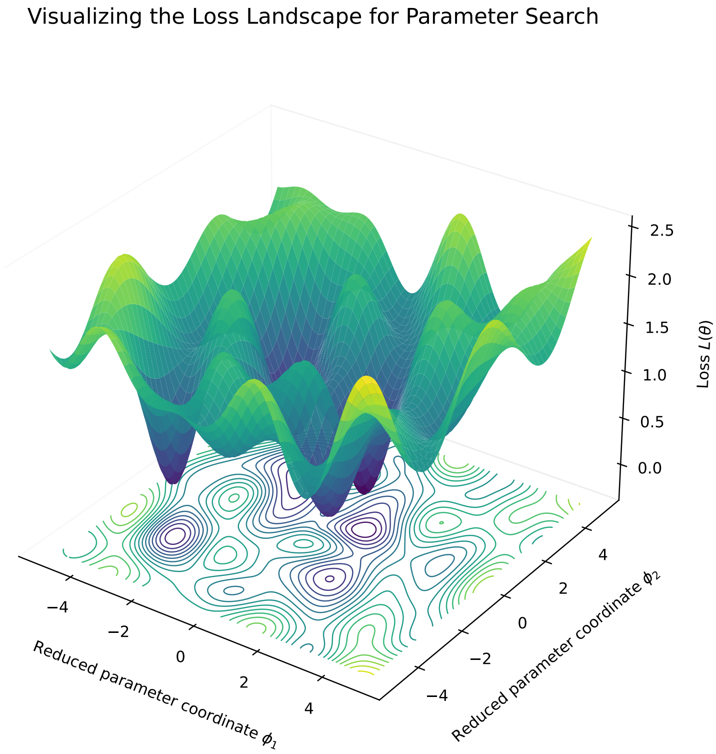

Figure 2.6: Visualization of the parameter space for an n-dot system, showing how multiple parameters and their combinations create a complex landscape with multiple local minima. This figure does not represent the actual parameter space, but serves to illustrate the concept of a high-dimensional optimization problem with many local minima and maxima.

2.3.5 Optimization Motivation

Together, these diagnostics provide the physical ingredients used later to build the optimization loss function. For a simple two-dot system, the relevant tunable parameters are the dot energies, elastic cotunneling, crossed Andreev reflection, and interaction strength. For longer chains, the number of tunable parameters grows quickly, visualized in Figure 2.6. Since interactions and inhomogeneous parameters make it difficult to identify sweet spots analytically, the diagnostics above can be combined into a loss function and used to search numerically for promising Majorana-like regimes. The resulting loss landscape may contain many local minima, which motivates the repeated-restart and basin-hopping strategy described in the methods chapter.

2.4 Majorana Exchange and Braiding

Non-Abelian braiding is one of the strongest operational signatures of Majorana bound states [38]. When two Majorana modes are exchanged, the system is transformed by a non-trivial unitary operation within its degenerate low-energy subspace. Since different exchanges generally correspond to non-commuting unitary operations, sequences of braids can be used to manipulate quantum information[34]. For perfectly isolated Majorana zero modes, the resulting operation depends on the ordering of the exchanges rather than on details of the path, providing the basis for topological quantum computation [32].

In quantum-dot-based minimal Kitaev chains, braiding does not require the Majorana-like modes to be physically moved. Instead, this thesis considers coupling-based braiding, in which three Majorana systems are coupled together in the shape of a Y-junction. Each arm of the Y junction represent a subsystem of PMMs. By cyclically turning the relevant couplings on and off, the dominant Majorana pairing is transferred through the junction and produces an effective exchange of two outer Majorana modes, with an additional sign change for one of the nodes [34]. For finite Poor Man's Majorana systems, the modes are imperfect and lack complete topological protection. The resulting operation must therefore be verified by monitoring the low-energy manifold and explicitly calculating the generated braid unitary.

2.4.1 From Qubit Gates to Fermionic Operators

In quantum computation, quantum gates are unitary operations that manipulate the state of qubits. Quantum gates, or quantum logic gates, are the building blocks of quantum circuits, the same way classical logic gates are for digital circuits. Unlike many classical logic gates, quantum logic gates are reversible, meaning that they can be undone by applying the inverse operation. This reversibility is a fundamental property of quantum mechanics and is essential for quantum computation.

Quantum gates are unitary operations and can be described as unitary matrices, relative to some orthonormal basis. For example, the Pauli gates X, Y, and Z are represented by the following matrices:

$$X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad Y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}. \tag{2.11}$$

The Pauli gates (X, Y, Z) are fundamental single-qubit gates that correspond to rotations around the x, y, and z axes of the Bloch sphere, respectively. They are also examples of involutory gates, meaning that applying them twice yields the identity operation (e.g., $X^2 = Y^2 = Z^2 = I$).

In order to replicate these quantum logic gates using Majorana modes, we need to understand how to express these gates in terms of fermionic operators. This is because the Majorana modes are described by fermionic operators, and we want to find a way to translate the abstract quantum gates into operations that can be implemented in a Majorana system. By expressing the Pauli gates in terms of fermionic creation and annihilation operators, we can identify how to manipulate the Majorana modes to achieve the desired quantum gate operations. This connection is crucial for implementing

quantum computation using Majorana-based qubits.
Let us start by introducing two quantum states $|0\rangle$ and $|1\rangle$, which we can simply represent as column vectors:

$$|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad |1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}. \tag{2.12}$$

These two vectors represent the two possible states of a qubit, where $|0\rangle$ is the state where the qubit is in the "0" state and $|1\rangle$ is the state where the qubit is in the "1" state. These states form the computational basis for a single qubit. The Pauli gates act on these basis states as follows:

$$\begin{aligned} X|0\rangle &= |1\rangle, \quad X|1\rangle = |0\rangle, \\ Y|0\rangle &= i|1\rangle, \quad Y|1\rangle = -i|0\rangle, \\ Z|0\rangle &= |0\rangle, \quad Z|1\rangle = -|1\rangle. \end{aligned}$$

To put these operations in the context of fermionic systems, the X gate maps the two computational basis states into each other. The Y gate performs the same state exchange while introducing relative phase factors of i and $-i$. The Z gate, on the other hand, does not exchange the states, but applies a phase of -1 to $|1\rangle$ while leaving $|0\rangle$ unchanged.

A single fermionic mode has the occupation states $|0_f\rangle$ and $|1_f\rangle = c^\dagger |0_f\rangle$. However, these states have opposite fermion parity, and the fermion-parity superselection rule forbids coherent superpositions between them[7]. They therefore cannot directly serve as the computational basis of a qubit. Instead, a qubit can be encoded using two fermionic modes while fixing their total parity. In the odd-parity sector, the logical basis states are

$$|0_L\rangle = |1,0\rangle = c_1^\dagger |\mathrm{vac}\rangle, \qquad |1_L\rangle = |0,1\rangle = c_2^\dagger |\mathrm{vac}\rangle.$$

where $|n_1, n_2\rangle = (c_1^\dagger)^{n_1}(c_2^\dagger)^{n_2}|\mathrm{vac}\rangle$. Both states have odd total fermion parity, so coherent superpositions $\alpha|0_L\rangle + \beta|1_L\rangle$ are allowed. Within this encoded subspace, the Pauli operators can be written as

$$\begin{aligned} X &= c_1^\dagger c_2 + c_2^\dagger c_1, \\ Y &= -i(c_1^\dagger c_2 - c_2^\dagger c_1), \\ Z &= c_1^\dagger c_1 - c_2^\dagger c_2. \end{aligned} \tag{2.13}$$

We can now see how easily the quantum logic gates can be transformed into fermionic language, which is the first step towards understanding how to implement these gates in a Majorana system. By expressing the gates in terms of fermionic operators, we can identify the necessary interactions and couplings in a Majorana-based setup that would allow us to perform the desired quantum gate operations. This is crucial for the development of Majorana-based quantum computation, as it provides a clear pathway for translating abstract quantum operations into physical manipulations of Majorana modes.

2.4.2 Jordan-Wigner Transformation

Jordan-Wigner transformation is a mathematical technique used to map spin operators to fermionic creation and annihilation operators. Proposed by Pascual Jordan and Eugene Wigner in 1928 [17], where they originally introduced it to solve the one-dimensional spin-1/2 chain problem, but later was extended to two-dimensional systems and beyond. The transformation is particularly useful in the study of quantum many-body systems, as it allows for the analysis of spin systems using fermionic techniques.
The Jordan-Wigner transformation is useful here because it lets us represent qubit or spin operators in terms of fermionic operators while preserving the correct anticommutation relations between different sites. This is not only a formal point: in the numerical braiding calculations, Majorana operators from different sites and subsystems must be embedded into a common Hilbert space with the correct fermionic algebra. The Jordan-Wigner strings provide this bookkeeping. The transformation allows us to express the spin-1/2 Pauli operators in terms of fermionic creation and annihilation operators[33]. For a chain of spin-1/2 particles, the transformation is given by:

$$\begin{aligned} \sigma_j^+ &= \frac{1}{2}(\sigma_j^x + i\sigma_j^y) \equiv f_j^\dagger, \\ \sigma_j^- &= \frac{1}{2}(\sigma_j^x - i\sigma_j^y) \equiv f_j, \\ \sigma_j^z &= 2f_j^\dagger f_j - 1, \end{aligned}$$

Leaving us with the correct same-site fermionic relations, but operators on different sites commute instead of anticommute. To fix this, we take a chain of fermions, and define a new set of fermionic operators:

$$\begin{aligned} a_j^\dagger &= e^{i\pi \sum_{k=1}^{j-1} f_k^\dagger f_k} \cdot f_j^\dagger \\ a_j &= e^{-i\pi \sum_{k=1}^{j-1} f_k^\dagger f_k} \cdot f_j, \\ a_j^\dagger a_j &= f_j^\dagger f_j \end{aligned}$$

The exponential term, can be simplified to a string of σ^z operators, giving us

$$e^{\pm i\pi \sum_{k=1}^{j-1} f_k^\dagger f_k} = \Pi_{k=1}^{j-1}(-\sigma_k^z). \tag{2.14}$$

The transformed spin operators now satisfy the correct fermionic anticommutation relations:

$$\begin{aligned} \{a_j^\dagger, a_k\} &= \delta_{jk}, \\ \{a_j^\dagger, a_k^\dagger\} &= 0, \\ \{a_j, a_k\} &= 0. \end{aligned}$$

The Jordan-Wigner transformation is a powerful tool for analyzing spin systems using fermionic techniques, and it provides a clear example of how quantum gates can be expressed in terms of fermionic operators. By applying this transformation, we can gain insights into the behavior of spin systems and their connections to fermionic systems, which is essential for understanding the physics of Majorana modes and their applications in quantum computation.

2.4.3 Pauli Matrices in the Majorana Representation

The reason for rewriting the Pauli operators in terms of Majorana bilinears is that braiding acts naturally on Majorana operators, while the gate implemented by the braid is usually compared with a Pauli operation on an encoded qubit. By expressing the Pauli gates in terms of Majorana bilinears, we can directly relate the physical operations performed on the Majorana modes to the abstract quantum gates that we want to implement. This connection is important to both understand and implement quantum gates in a Majorana-based quantum computing platform. We start by considering four Majorana operators, grouped into two ordinary fermions:

$$\begin{aligned} c_1 &= \frac{1}{2}(\gamma_0 + i\gamma_1), \quad c_1^\dagger = \frac{1}{2}(\gamma_0 - i\gamma_1), \\ c_2 &= \frac{1}{2}(\gamma_2 + i\gamma_3), \quad c_2^\dagger = \frac{1}{2}(\gamma_2 - i\gamma_3). \end{aligned} \tag{2.15}$$

These four Majorana operators γ_0, γ_1, γ_2, and γ_3 span a four-dimensional Hilbert space, which can be thought of as the state space of two fermionic modes. By fixing the total fermion parity, we can select a two-dimensional subspace that serves as the computational basis for a qubit. Within the odd-parity encoding defined above, the Pauli operators can be represented as bilinears of the Majorana operators:

$$\begin{aligned} X &= i\gamma_0\gamma_3 = -i\gamma_1\gamma_2, \\ Y &= -i\gamma_0\gamma_2 = -i\gamma_1\gamma_3, \\ Z &= i\gamma_0\gamma_1 = -i\gamma_2\gamma_3. \end{aligned}$$

These expressions can be derived by substituting the fermionic operators in Eq. 2.13 with the Majorana representation of the fermionic operators. The resulting bilinears of Majorana operators correspond to the Pauli gates acting on the encoded qubit.
Thus, controlled Majorana couplings of the form $i\gamma_i\gamma_j$ act as Pauli operations on the encoded qubit. This is the operator-level connection between Majorana braiding and gate implementation[4, 22, 32]

2.4.4 Majorana Exchange Gates

An anticlockwise exchange of two Majoranas γ_i and γ_j is represented by the unitary

$$B_{ij} = \exp\left(-\frac{\pi}{4}\gamma_i\gamma_j\right) = \frac{1}{\sqrt{2}}(1 - \gamma_i\gamma_j),$$

up to orientation conventions and an overall phase, as was shown by M. Leijnse and K. Flensberg [23] and C.W.J Beenakker [8]. With this convention, the unitary acts on the Majorana operators as

$$\begin{aligned} B_{ij}\gamma_i B_{ij}^\dagger &= \gamma_j, \\ B_{ij}\gamma_j B_{ij}^\dagger &= -\gamma_i, \\ B_{ij}\gamma_k B_{ij}^\dagger &= \gamma_k, \quad k \neq i, j. \end{aligned}$$

The important point is that exchanges are generated by Majorana bilinears. Therefore, the same $i\gamma_i\gamma_j$ operators that represent Pauli operations in the encoded qubit space also determine the target braid gates used later in this thesis. For example, squaring an exchange produces a parity-dependent bilinear operation,

$$B_{ij}^2 = -\gamma_i \gamma_j,$$

up to phase and orientation conventions. This is why the numerical braid unitary can be checked by comparing its action on the Majorana operators and by comparing its projected parity blocks with the expected target gate [3, 32].

The remaining question is how to compute the unitary generated by an adiabatic coupling protocol, which is the role of Kato transport in the next section.

2.5 Projected Subspaces, Kato Transport and Braid Unitaries

This section connects the high-level topological theory of braiding with the microscopic details of the PMM system. The goal is to understand how to compute the braid unitary generated by a time-dependent parameter protocol, and how to compare it with the ideal target gate. In these systems, braiding is not performed by physically moving particles through space, but rather by tuning Hamiltonian parameters in a way that mimics the effect of an exchange.
An adiabatic parameter protocol defines a path in the space of Hamiltonians. If the relevant low-energy subspace remains isolated along this path, the geometric transport of that subspace gives the braid unitary. The Kato formulation provides a convenient way to compute this unitary without having to track individual eigenstates and their arbitrary phases. By projecting the dynamics onto the low-energy subspace defined by the Majorana-like states, we can extract the effective braid unitary and compare it with the ideal target gate expressed in terms of Majorana bilinears.

2.5.1 Kato Transport in a Degenerate Subspace

The Berry phase is a geometric phase acquired by a quantum state during adiabatic evolution around a closed loop in parameter space. For a non-degenerate eigenstate, the state returns to itself up to a phase. This phase has a dynamical part, which depends on the time scale and the energy of the state, and a geometric part, which depends only on the path in parameter space. The geometric contribution is the Berry phase.

For a degenerate subspace, the situation is richer. The adiabatic evolution can mix states within the degenerate manifold, so the Berry phase becomes a matrix rather than a scalar phase. In this thesis, this non-Abelian Berry matrix is what determines the braid unitary acting on the Majorana qubit space.

The Kato formulation provides a convenient way to compute this adiabatic transport. Instead of tracking individual eigenvectors and their phases, one works directly with the projector $P(t)$ onto the relevant low-energy subspace. In the convention used in the numerical implementation, the Kato evolution satisfies

$$\partial_t U_t = [P, \partial_t P]\, U_t, \qquad U_0 = I.$$

[18]. The final unitary U_T is the geometric transport associated with the path in the chosen subspace. A full physical time evolution would also contain dynamical phases depending on the instantaneous energies and total protocol time. The Kato construction isolates the adiabatic geometric part, which is the component relevant for identifying the braid gate. In numerical calculations, $P(t)$ is obtained by diagonalizing the Hamiltonian at each time step and constructing the projector onto the chosen low-energy manifold.

This unitary matrix is the braid unitary generated by the parameter protocol. By comparing U_T with the ideal target gate expressed in terms of Majorana bilinears, we can assess how well the parameter-tuning braid approximates the desired quantum gate operation. This comparison can be done both at the operator level, by checking how U_T transforms the Majorana operators, and at the level of the projected parity blocks, by

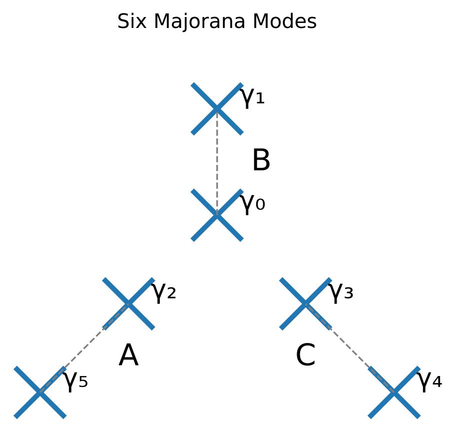

Figure 2.7: Schematic representation of a system hosting six majoranas. The illustrations shows pairs of two and two Majoranas coupling together, in a Y-junction, each pair form a subsystem that can be described by a Poor Man's Majorana setup, A, B and C.

checking how U_T acts on the encoded qubit states.

2.5.2 Effective Braiding Hamiltonian

As a reference model, assume that the system contains four Majorana modes γ_0, γ_1, γ_2, and γ_3 involved in the braid. A useful physical picture is a Y-junction made from three PMM subsystems, where one central Majorana is coupled sequentially to different neighboring Majoranas.

In Fig. 2.7, the four Majoranas γ_0, γ_1, γ_2, and γ_3 are the modes actively used in the braid, while γ_4 and γ_5 form a spectator pair. The ideal effective Hamiltonian can be written as

$$\begin{aligned} H =&\Delta_{01} i\gamma_0\gamma_1 + \\ &\Delta_{02} i\gamma_0\gamma_2 + \\ &\Delta_{03} i\gamma_0\gamma_3, \end{aligned} \tag{2.16}$$

where Δ_{ij} is a tunable coupling between Majoranas γ_i and γ_j [8, 15]. By turning these couplings on and off in a controlled sequence, the system can implement an effective exchange operation. This model is not meant to describe all microscopic details of the dot device directly. Instead, it serves as a reference model where the intended Majorana exchange can be tested without additional complications from the full interacting Hamiltonian.

2.5.3 Projected Microscopic Junctions

In a microscopic PMM device, the tunable quantities are not abstract Majorana bilinears, but physical hopping and pairing couplings between dots or subsystems. A junction between two PMM subsystems can be modeled by

$$H_{JuncAB} = t_{AB}(d_A^\dagger d_B + d_B^\dagger d_A) + \Delta_{AB}(d_A^\dagger d_B^\dagger + d_B d_A), \tag{2.17}$$

which is the same type of elastic cotunneling and crossed-Andreev pairing structure appearing in the PMM Hamiltonian. After such microscopic junction operators are projected into the retained low-energy subspace, they can be compared with the effective Majorana bilinears of Eq. 2.16. This motivates comparing two related braid constructions: one built from ideal or energy-level Majorana operators, and one built from projected microscopic local operators.

2.5.4 Energy-Level Versus Local-Operator Braids

The central comparison in the braiding analysis is between two ways of replacing the ideal energy-level operators γ_2 and γ_3 in Eq. 2.16. The effective Hamiltonian assumes that the Majorana operators are ideal and perfectly localized, while the optimized PMM system only provides approximate Majorana-like degrees of freedom. To test how close the microscopic system comes to the ideal exchange picture, we compare an energy-level construction with a local-operator construction.

In Majorana algebra, the target exchange operation used in this thesis is

$$B_{\gamma_2\gamma_3} = \exp\left(-\frac{\pi}{4}\gamma_2\gamma_3\right). \tag{2.18}$$

The resulting braid unitary is later compared with this target gate, both through its action on Majorana operators and through its projected parity-block representation.

The local-operator braid is built from microscopic endpoint operators rather than from idealized Majorana operators. In principle, one can project either the real or imaginary local Majorana component, corresponding to $c^\dagger + c$ and $i(c^\dagger - c)$, respectively. In the calculations presented here, the physical braid uses the projected minus component, $P\,i(c^\dagger - c)\,P$, for the relevant B and C subsystem sites. After projection into an N-dimensional low-energy subspace, these operators are normalized and used as the effective braid channels that replace γ_2 and γ_3. The driven Hamiltonian therefore takes the schematic form

$$\begin{aligned} H_{\text{local}} =& \Delta_{01} i\gamma_0\gamma_1 + \\ & \Delta_{02} i\gamma_0\Gamma_B^P + \\ & \Delta_{03} i\gamma_0\Gamma_C^P, \end{aligned} \tag{2.19}$$

with

$$\Gamma_B^P = \pm P\,i(c_B^\dagger - c_B)\,P, \qquad \Gamma_C^P = \pm P\,i(c_C^\dagger - c_C)\,P. \tag{2.20}$$

Here P is constructed by diagonalizing the full Hamiltonian and selecting the N states, corresponding to the desired low-energy subspace [19, 35]. The local-operator braid therefore tests whether the physically projected microscopic endpoint operators access the same exchange channel as the ideal target gate.

The energy-level construction starts from the optimized subsystem Hamiltonian. For clarity, this is the n-dot extension of the effective two-dot model 2.10,

$$H = \sum_i^n \epsilon_i n_i + \sum_j^{n-1} t_j(c_j^\dagger c_{j+1} + c_{j+1}^\dagger c_j) + \sum_j^{n-1} \Delta_j(c_j^\dagger c_{j+1}^\dagger + c_{j+1}c_j) + \sum_j^{n-1} U n_j n_{j+1}, \tag{2.21}$$

Diagonalizing this Hamiltonian 2.21, we obtain the energy eigenstates, which can be classified by parity. By pairing even and odd states that are close in energy, we can construct effective Majorana operators as combinations of these parity partners. In its simplest form, this construction looks like:

$$\begin{aligned} \gamma_+ &= \sum_i^n \left(|e_i\rangle \langle o_i| + |o_i\rangle \langle e_i| \right), \\ \gamma_- &= i \sum_i^n \left(|e_i\rangle \langle o_i| - |o_i\rangle \langle e_i| \right). \end{aligned} \tag{2.22}$$

Here n is the number of even-odd energy pairs included in the construction [19, 35]. For $n = 1$, only the ground-state parity pair contributes. Larger values of n retain more of the low-energy spectrum and allow the braid construction to be tested in enlarged projected spaces. The paired-state construction is designed to realize the Majorana algebra within the selected parity-pair subspace; after embedding and projection into the full retained low-energy space, the resulting operators are checked numerically for normalization and anticommutation errors.

In the higher-energy analysis, the relevant projected Majorana channel is not assumed to be purely γ_+ or purely γ_-. Instead, the implementation constructs the accumulated projected γ_+ and γ_- components from the chosen parity pairs, and then fits a global linear combination to the projected local reference operator:

$$\gamma_{\text{fit}} = a\,\gamma_+^P + b\,\gamma_-^P. \tag{2.23}$$

The coefficients a and b are chosen to minimize the mismatch with the projected microscopic local operator in the retained subspace. In a well defined system, either a or b will be significant, but not both. This fitting procedure can be interpreted as selecting the effective Majorana direction that best aligns the energy-level construction with the physically projected local channel.

The comparison between these two constructions is one of the central checks in this thesis. If the energy-level and local-operator braids agree, then the optimized spectrum and the physical endpoint couplings select the same Majorana exchange channel. If they differ, the spectrum may still contain Majorana-like parity pairs, but the local microscopic operators do not access the ideal braid cleanly. This distinction is what allows the braiding analysis to go beyond asking whether Majorana-like states exist, and instead test whether they can be used to implement the intended exchange operation.

Chapter 3

Method

This chapter describes the methods used to optimize the Majorana modes in the junction setup and to analyze their properties. We first discuss the parameter optimization process, including the model parameterization, loss function design based on Majorana criteria, and the optimization strategy. Next, we detail the post-optimization analysis of the setup, covering Hamiltonian reconstruction, parity classification, and various diagnostics. Finally, we describe the braiding protocol implemented in an effective Majorana model and how we validate the results using gate-validation checks.

3.1 Numerical PMM Hamiltonian and Basis

The numerical calculations are built from the spinless n-dot PMM Hamiltonian introduced in the theory chapter. This section specifies how that model is represented in the code: which Hamiltonian is diagonalized, which basis is used, how the fermionic operators are constructed, and how the model parameters are organized before optimization and braiding calculations. The goal here is not to rederive the PMM model, but to define the concrete numerical object used throughout the rest of the method chapter.

3.1.1 Single-Chain Hamiltonian

For a single n-dot PMM chain, the Hamiltonian used in the optimization and diagnostic calculations is

$$H_{\mathrm{PMM}} = \sum_{i=1}^{n} \epsilon_i n_i + \sum_{i=1}^{n-1} \left[-t_i \left(c_i^\dagger c_{i+1} + c_{i+1}^\dagger c_i \right) + \Delta_i \left(c_i c_{i+1} + c_{i+1}^\dagger c_i^\dagger \right) + U_i n_i n_{i+1} \right]. \quad (3.1)$$

Here $c_i^\dagger$ and c_i create and annihilate a spinless fermion on dot i, and $n_i = c_i^\dagger c_i$ is the corresponding number operator. The onsite energies ϵ_i play the role of local chemical potentials, t_i are elastic cotunneling amplitudes, Δ_i are crossed-Andreev pairing amplitudes, and U_i are nearest-neighbor density-density interaction strengths. This is the same operator structure used both when searching for optimized PMM chains and when reconstructing selected chains for later braiding calculations.

3.1.2 Many-Body Fock Basis

All operators are represented in the full many-body Fock basis. For a chain with n dots, a basis state is specified by the occupation pattern

$$|n_1 n_2 \cdots n_n\rangle, \qquad n_i \in \{0, 1\}, \tag{3.2}$$

so the Hilbert-space dimension of a single chain is 2^n. Diagonalizing Eq. 3.1 in this basis gives the full many-body spectrum and eigenstates. This is important because the diagnostics used later, such as parity classification, charge expectation values, and Majorana-polarization matrix elements, are all many-body quantities.

For braiding calculations, several optimized chains are combined into a larger Hilbert space. If D identical PMM subsystems are used, each containing n dots, the total number of fermionic sites is $N_{\text{tot}} = Dn$ and the full Hilbert-space dimension is $2^{N_{\text{tot}}}$. In the three-arm braiding geometry used later, $D = 3$, corresponding to subsystems A, B, and C.

3.1.3 Jordan-Wigner Construction of Fermionic Operators

The fermionic creation and annihilation operators must satisfy the canonical anticommutation relations

$$\{c_i, c_j^\dagger\} = \delta_{ij}, \qquad \{c_i, c_j\} = \{c_i^\dagger, c_j^\dagger\} = 0. \tag{3.3}$$

In the code, these operators are constructed with Jordan-Wigner strings. Starting from local spin raising and lowering operators on site j, the fermionic operators are represented schematically as

$$c_j^\dagger = \left[\prod_{k<j}(-\sigma_k^z)\right]\sigma_j^+, \qquad c_j = \left[\prod_{k<j}(-\sigma_k^z)\right]\sigma_j^-. \tag{3.4}$$

The string of σ^z operators keeps track of the fermionic sign picked up when operators on different sites are exchanged. This same construction is used for single-chain calculations and for the enlarged three-subsystem Hilbert space. In the latter case, the site ordering fixes the Jordan-Wigner strings across subsystems, ensuring that operators belonging to different PMM chains still anticommute correctly.

Once the creation and annihilation operators have been constructed, the code precomputes the basic operator products needed repeatedly:

$$n_i = c_i^\dagger c_i, \qquad h_{ij} = c_i^\dagger c_j + c_j^\dagger c_i, \qquad p_{ij} = c_i^\dagger c_j^\dagger + c_j c_i, \qquad d_{ij} = n_i n_j. \tag{3.5}$$

These building blocks are then combined with the parameter values in Eq. 3.1 to assemble the Hamiltonian matrix. The same Jordan-Wigner ordering is also used when several PMM chains are embedded into the combined braiding Hilbert space, so that the fermionic operators of different subsystems anticommute correctly. For the three-subsystem geometry, sites are ordered as

$$A_1, A_2, \ldots, A_n, B_1, B_2, \ldots, B_n, C_1, C_2, \ldots, C_n.$$

Operators acting on subsystem B therefore carry a Jordan-Wigner string over all sites in subsystem A, while all operators acting on subsystem C carry a string over all sites in both A and B. This ensures that the fermionic algebra is preserved across the entire device, which is crucial for correctly defining the Majorana operators and their braiding properties later on.

3.1.4 Parameter Conventions

The numerical implementation allows each parameter family in Eq. 3.1 to be fixed, homogeneous, or inhomogeneous. A fixed parameter is held at a prescribed value and is not included in the optimization vector. A homogeneous parameter has the same value on every site or bond, while an inhomogeneous parameter is allowed to vary independently from site to site or bond to bond. This convention makes it possible to compare restricted ansätze with more flexible parameter searches using the same Hamiltonian-building routine.

In the main optimization runs used in this thesis, the hopping amplitudes are fixed to $t_i = 1$, and the interaction strength is fixed to selected values of U. The onsite energies ϵ_i and pairing amplitudes Δ_i are optimized. Fixing the hopping scale prevents the optimizer from lowering the loss through a trivial global rescaling of the kinetic energy, and instead forces the search to find nontrivial onsite and pairing profiles for each interaction regime. This also rules out the possibility of blowing up all other parameters in order to suppress the relative effect of interactions, which would be an unphysical way to achieve a low loss in the strongly interacting regime. The optimized parameters are stored in a reduced vector form, where only the free parameters are included. After optimization, the full set of physical parameters is reconstructed by reinserting fixed values and expanding homogeneous parameters as needed.

After optimization, the selected physical parameters are stored and later reloaded for post-optimization diagnostics and braiding calculations. The same parameter convention is also used when constructing the three-subsystem braiding geometry: the optimized single-chain parameters define each subsystem, while additional inter-subsystem couplings are introduced only in the braiding-model sections below.

3.2 Parameter Optimization and Loss Function

The purpose of the optimization is to find parameter regimes where the finite PMM chain has the spectral and local signatures of Majorana-like states. The theory chapter defined the physical criteria; here they are turned into a numerical objective. For each trial parameter vector, the Hamiltonian in Eq. 3.1 is constructed, diagonalized, separated into parity sectors, and assigned a scalar loss. The optimizer then updates only the free parameters, while fixed parameters are reinserted when the physical Hamiltonian is rebuilt.

3.2.1 Optimization Vector and Physical Parameters

The optimizer does not act directly on a full dictionary of physical parameters. Instead, it acts on a reduced real vector θ containing only the entries that are not fixed by the chosen parameter configuration. For example, if t_i is fixed and U_i is fixed to a selected interaction strength, then those entries are omitted from θ. If a parameter is homogeneous, only one number is included; if it is inhomogeneous, one number is included for each site or bond.

The reduced vector is converted back into a parameter dictionary before the Hamiltonian is evaluated. Schematically,

$$\theta \longrightarrow \{t_i, U_i, \epsilon_i, \Delta_i\} \longrightarrow H_{\mathrm{PMM}}(\theta). \tag{3.6}$$

During this conversion, fixed values are reinserted, homogeneous values are expanded across all sites or bonds when needed, and positive couplings such as t_i, U_i, and Δ_i are constrained to be non-negative by taking their absolute values. This separation between the optimization vector and the physical parameter dictionary makes it possible to use the same code for different ansätze.

In the main optimization runs, the hopping amplitudes are fixed to $t_i = 1$, while the interaction strength U is fixed to the value being studied. The remaining parameters, primarily the onsite energies ϵ_i and pairing amplitudes Δ_i, are optimized. For each system size n and interaction strength U, the lowest-loss physical parameter set is stored and later reused in the diagnostic and braiding calculations.

3.2.2 Parity Classification Inside the Loss

For each trial Hamiltonian, the full many-body spectrum is obtained by exact diagonalization. The eigenstates are then classified by the total fermion parity operator,

$$P_{\text{tot}} = \prod_{i=1}^{n} (1 - 2n_i). \tag{3.7}$$

This gives two ordered energy lists, one in the even sector and one in the odd sector,

$$\{E_k^{(e)}\}, \qquad \{E_k^{(o)}\}. \tag{3.8}$$

The loss function compares these parity partners level by level. If the diagonalization produces an invalid parity structure, for example if one sector is missing or if the two sectors cannot be paired consistently, the trial point is discarded and the optimizer proceeds to another restart or candidate point.

3.2.3 Even-Odd Degeneracy Term

The first loss contribution penalizes energy splitting between even- and odd-parity partners,

$$\delta_k = \left| E_k^{(e)} - E_k^{(o)} \right|. \tag{3.9}$$

For non-interacting theories, this step shouldnt be necessary, but explain why These splittings are weighted more strongly at low energy, because the ground-state degeneracy is the most important Majorana signature. In the implementation, the weights decrease from a maximum value for the lowest parity pair to a smaller value for higher levels. The degeneracy contribution therefore has the schematic form

$$\mathcal{L}_{\text{deg}} = \sum_k w_k \left| E_k^{(e)} - E_k^{(o)} \right|, \qquad w_0 > w_1 > \cdots. \tag{3.10}$$

This term encourages the optimizer to produce an approximately parity-paired spectrum, with particular emphasis on the lowest-energy states.

3.2.4 Excitation-Gap Term

Degeneracy alone is not sufficient: the parity-paired states must also be separated from nearby levels. The second loss contribution penalizes small gaps within the parity-resolved spectra. For adjacent levels, the relevant gap is taken as the smaller of the even-sector and odd-sector spacings,

$$\Delta_k = \min\left(E_{k+1}^{(e)} - E_k^{(e)}, E_{k+1}^{(o)} - E_k^{(o)} \right). \tag{3.11}$$

Small gaps are penalized using a smooth threshold function,

$$\mathcal{L}_{\text{gap}} = \sum_{k} \tilde{w}_k \, \text{softplus}\left(\Delta_{\text{target}} - \Delta_k\right), \tag{3.12}$$

where Δ_{target} is the desired minimum gap scale, and the softplus function is defined as $\text{softplus}(x) = \log(1 + e^x)$[11]. This term favors spectra where the relevant parity-paired states are not only close to each other, but also isolated from the rest of the spectrum.

3.2.5 Charge and Majorana-Polarization Terms

The loss also includes local diagnostics that test whether the parity partners resemble Majorana states rather than accidental degeneracies. The charge penalty measures how differently the even and odd partners occupy each site,

$$\mathcal{L}_Q = \sum_{k} \sum_{j} \left| \langle e_k | \, n_j \, | e_k \rangle - \langle o_k | \, n_j \, | o_k \rangle \right|. \tag{3.13}$$

A small value means that the parity partners have similar local charge distributions, consistent with non-local encoding.

The Majorana-polarization penalty compares the site-resolved Majorana-polarization profile with the expected edge-localized pattern. In the target configuration, the polarization is concentrated on the two ends of the chain with opposite sign, while the bulk sites carry little weight. Numerically, the penalty is evaluated from matrix elements of local Majorana operators between the even and odd parity partners. The sign of the target pattern is allowed to flip, since exchanging the labels of the two Majorana components changes the overall sign but not the physical localization pattern.

3.2.6 Total Normalized Loss

The full loss is a weighted sum of the spectral and local contributions. In the implementation, the degeneracy and gap terms are first collected into a single weighted spectral penalty array, after which the charge and Majorana-polarization penalties are added. Written schematically, the implemented loss is

$$\mathcal{L} = \frac{\omega_{\text{deg}} \mathcal{L}_{\text{deg}} + \omega_{\text{gap}} \mathcal{L}_{\text{gap}} + \mathcal{L}_Q + \mathcal{L}_{\text{MP}}}{\overline{|E|} + \eta}, \tag{3.14}$$

where $\overline{|E|}$ is the mean absolute energy scale of the spectrum and η is a small numerical offset. ω_{deg} and ω_{gap} are weight arrays, created to favor the low-energy part of the spectrum, which are the most essential states. The degeneracy and gap contributions contain level-dependent weights that emphasize the low-energy part of the spectrum, where the Majorana signatures are most important.

The normalization by $\overline{|E|}$ makes the spectral part of the loss less sensitive to the overall energy scale of a trial Hamiltonian. However, this normalization is not invariant under a uniform energy shift $H \mapsto H + \alpha \mathbb{1}$, since such a shift changes the absolute energies but leaves gaps and eigenvectors unchanged. In the parametrization used here there is no independent identity-shift parameter, so the optimizer does not explore such shifts directly.

A further limitation is that the same denominator also rescales the charge and Majorana-polarization penalties. These terms are dimensionless eigenvector diagnostics rather than energy penalties, so dividing them by an energy scale can make their contribution smaller when the spectrum becomes larger, even if the local Majorana quality does not improve. This possible bias is partly mitigated by fixing the hopping scale to $t = 1$ and fixing U during each optimization run, but the scalar loss should still be interpreted together with the separate spectral and local diagnostics.

3.2.7 Search Strategy

The resulting loss landscape is non-convex, and in practise, we find that the interacting loss landscape is more complicated. The numerical search therefore combines local gradient-based optimization with repeated restarts and basin hopping. Local optimization is performed with Adam. At each iteration, the reduced vector θ is converted into physical parameters, the Hamiltonian is built and diagonalized, the loss is evaluated within the PyTorch optimization loop and used to update the free parameters with Adam.

Because a single local run can converge to a poor local minimum, the local optimization is repeated from randomly perturbed initial vectors. The lowest-loss result among these restarts is retained as the representative of that region of parameter space. Basin hopping is then used to explore different regions: a random displacement is applied to the current parameter vector, local optimization with restarts is performed again, and the new candidate is accepted if it improves the loss. Candidates with slightly worse loss can also be accepted with a Metropolis probability, allowing the search to escape shallow local minima.

For each pair (n, U), the best configuration found by this procedure is stored together with its loss, parameter configuration, and reconstructed physical parameters. These stored configurations form the input to the post-optimization diagnostics described in the next section.

3.3 Post-Optimization Diagnostics

The optimization procedure returns candidate parameter sets ranked by a scalar loss. Since this loss is a soft weighted objective rather than a set of hard constraints, a low value does not by itself prove that the Hamiltonian realizes an ideal Majorana regime. Different loss contributions can trade off against each other, finite weights allow residual even–odd splittings, and the local charge and Majorana-polarization penalties may be small without giving a perfectly isolated ground-state doublet.

For this reason, after optimizing many configurations, the lowest-loss result for each pair (n, U) is selected and analyzed separately. The stored optimized parameters $\Phi_{\text{opt}}(n, U)$ are used to rebuild $H_{\text{PMM}}(\theta_{\text{opt}})$, which is then diagonalized and evaluated using the diagnostics below. This post-optimization step resolves the single scalar loss into physical quantities such as the ground-state splitting, excitation gap, charge difference, and Majorana-polarization profile.

3.3.1 Parity-Resolved Spectrum

Using the parity classification introduced in Sec. 2.3.1, the reconstructed Hamiltonian is diagonalized and separated into ordered even- and odd-parity spectra using the total fermion parity operator

$$P_{\text{tot}} = \prod_{i=1}^{n} (1 - 2n_i). \tag{3.15}$$

For plotting and comparison, all energies are shifted so that the lowest state in either parity sector has energy zero.

The most important scalar spectral diagnostic is the splitting of the lowest even-odd pair,

$$\delta_0 = \left| E_0^{(e)} - E_0^{(o)} \right|. \tag{3.16}$$

The corresponding low-energy gap is defined as

$$\Delta_{\text{gap}} = \min\left(E_1^{(e)}, E_1^{(o)}\right) - \max\left(E_0^{(e)}, E_0^{(o)}\right), \tag{3.17}$$

after the common ground-state energy shift. This quantity measures the distance from the lowest parity pair to the nearest excited state in either parity sector.

3.3.2 Local Charge Difference

Following the local-distinguishability discussion in Sec. 2.3.3, the diagnostic routine evaluates the site-resolved charge difference for the lowest parity pair,

$$q_i = \left| \langle e_0 | \, n_i \, | e_0 \rangle - \langle o_0 | \, n_i \, | o_0 \rangle \right|. \tag{3.18}$$

The total charge difference reported in the results is

$$Q_0 = \sum_{i=1}^{n} q_i. \tag{3.19}$$

In the diagnostic figures, the individual q_i values are plotted on top of the Majorana-polarization profile so that charge visibility and spatial localization can be compared site by site.

3.3.3 Majorana-Polarization Profile

Using the Majorana-polarization diagnostic defined in Sec. 2.3.4, the post-optimization routine computes the lowest-pair profile M_i from matrix elements of local Majorana operators between parity-partner states.

To summarize this profile in the results, two scalar quantities are extracted. The edge polarization is defined as the mean absolute polarization on the two end sites,

$$\overline{|M_{\text{edge}}|} = \frac{1}{2}\left(|M_1| + |M_n|\right), \tag{3.20}$$

while the bulk polarization is defined as the largest absolute value on the interior sites,

$$\max |M_{\text{bulk}}| = \max_{2 \le i \le n-1} |M_i|. \tag{3.21}$$

For two-dot systems there are no interior sites, so the bulk contribution is set to zero.

3.3.4 Selection for Braiding Calculations

The post-optimization diagnostics determine which optimized chains are used in the braiding calculations. Since the later sector-resolved braiding analysis tests projected manifolds beyond the lowest parity pair, the selected configurations are not judged only by the ground-state splitting, but also by the excitation gap, charge difference, and Majorana-polarization profile across the relevant low-energy structure.

3.4 Effective Majorana Operator Construction

The braiding calculations require explicit matrix representations of the Majorana operators associated with the optimized PMM chains. The theory chapter described why these operators should connect even and odd parity states. Here the focus is narrower: how the operators are constructed from the optimized eigenstates, embedded into the three-subsystem Hilbert space, projected to a chosen low-energy space, and checked before they are used in the braid.

3.4.1 Energy-Level Majoranas from Parity Pairs

For each optimized subsystem, the Hamiltonian is diagonalized and the eigenstates are separated into even and odd parity sectors. The states are then paired by their order in energy. From the jth even-odd pair, two parity-changing operators are formed,

$$\gamma_{+,j} = |e_j\rangle \langle o_j| + |o_j\rangle \langle e_j| , \qquad \gamma_{-,j} = i \left(|e_j\rangle \langle o_j| - |o_j\rangle \langle e_j|\right). \tag{3.22}$$

The effective Majorana operators used in the projected calculations are built by summing these contributions over a selected number of parity pairs,

$$\gamma_+^{(m)} = \sum_{j=0}^{m-1} \gamma_{+,j}, \qquad \gamma_-^{(m)} = \sum_{j=0}^{m-1} \gamma_{-,j}. \tag{3.23}$$

Here m is the number of even-odd pairs included in the operator construction. In the code this is controlled separately from the total projection dimension: the projection dimension determines how much of the full three-subsystem Hilbert space is retained, while m determines how many subsystem parity pairs are used to build the Majorana operator itself.

3.4.2 Embedding Into the Three-Subsystem Hilbert Space

Using the Jordan–Wigner ordering specified in Sec. 3.1.3, subsystem operators are embedded into the full three-subsystem Hilbert space as

$$\gamma_A \mapsto \gamma_A \otimes I_B \otimes I_C, \tag{3.24}$$

$$\gamma_B \mapsto JW_A \otimes \gamma_B \otimes I_C, \tag{3.25}$$

and

$$\gamma_C \mapsto JW_{AB} \otimes \gamma_C, \tag{3.26}$$

where JW_A is the Jordan-Wigner string over subsystem A, and JW_{AB} is the corresponding string over subsystems A and B. The strings preserve the fermionic anticommutation relations between subsystems.

3.4.3 Projection to the Retained Space

The full three-subsystem Hamiltonian is diagonalized, and a projection basis V is formed from a selected set of eigenvectors. In the low-energy benchmark, V contains the eigenvectors of the retained lowest-energy manifold. In the sector-resolved scans, a separate basis V_i is instead formed from all eigenvectors belonging to energy sector i. Any full-space operator O is then represented in the selected space by

$$O_P = V^\dagger O V. \tag{3.27}$$

This projection is applied to the embedded Majorana operators before they are used in the projected braid models. The same projection basis is also used for all other operators in a given calculation, so that the energy-level Majoranas, local operators, static Hamiltonian, and driven braid terms act in the same retained space.

3.4.4 Projected Local Components

In addition to the energy-level construction in Eq. 3.23, the code also constructs projected local Majorana components from the microscopic fermion operators on the relevant subsystem sites,

$$\Gamma_{+,P} = V^\dagger (c^\dagger + c) V, \qquad \Gamma_{-,P} = V^\dagger i (c^\dagger - c) V. \tag{3.28}$$

These projected local components provide a more microscopic reference for the Majorana operator at a chosen end of a subsystem. The direct projected local-operator model uses the normalized minus component $\Gamma_{-,P}$. When this operator is compared with the corresponding energy-level Majorana, only an overall sign is allowed,

$$s_{\text{opt}} = \underset{s \in \{-1,+1\}}{\operatorname{argmin}} \left\| \Gamma_{-,P} - s \gamma_-^P \right\|_F . \tag{3.29}$$

This sign alignment is used only to compare the two operator constructions; it does not allow the local operator to rotate into a different energy-level Majorana direction.

For higher-dimensional energy sectors, a single overall sign is generally insufficient. Degenerate sectors contain several independent blocks associated with different combinations of subsystem energy-level labels. A label operator is therefore constructed from the paired eigenstates of each subsystem and projected into the selected sector. Diagonalizing this projected label operator decomposes the sector into labelled blocks. Within each block r, the projected local operator is fitted independently to the two available energy-level Majorana components,

$$\Gamma_{-,P}^{(r)} \approx a_r \gamma_{+,P}^{(r)} + b_r \gamma_{-,P}^{(r)}. \tag{3.30}$$

The coefficients a_r and b_r are continuous least-squares coefficients and are allowed to differ between blocks. The fitted blocks are then reassembled into a block-matched local operator on the full selected sector. This construction tests whether the local operator can be represented by the energy-level Majorana pair after the independent degenerate blocks have been identified. The detailed construction of the label operator and the resulting block structure is presented together with the sector-resolved analysis, and explained in the appendix A.

3.4.5 Normalization and Algebra Checks

After projection, the Majorana operators are no longer assumed to satisfy the ideal algebra exactly. The projected operators are therefore checked numerically before being used in the braid. The diagnostics test Hermiticity, normalization,

$$\gamma_i^2 \approx I, \tag{3.31}$$

and mutual anticommutation,

$$\{\gamma_i, \gamma_j\} \approx 0 \qquad (i \neq j). \tag{3.32}$$

The matrix mismatch between the direct projected local component and the block-matched operator is also stored. These checks are important because the later braid errors should not be interpreted only as failures of the pulse protocol: they can also reflect imperfect Majorana operators after projection.

3.5 Braiding Models and Pulse Protocol

The braiding calculations use the optimized PMM chains as building blocks for a three-subsystem junction. The purpose of this section is to define the Hamiltonians that are driven during the braid and the common pulse sequence used for all model variants. The following section then describes how the resulting time-dependent evolution is transported and compared with the target exchange gate.

3.5.1 Three-Subsystem Geometry

The junction consists of three optimized PMM subsystems, labelled A, B, and C. Subsystem A acts as the reference arm and supplies two Majorana operators, denoted γ_0 and γ_1. The Majorana operators to be exchanged are associated with subsystems B and C, and are denoted γ_2 and γ_3. In this notation, the intended exchange is generated by moving the coupling of γ_0 from γ_1 to γ_2, then to γ_3, and finally back to γ_1.

All braiding models are built in the projected Hilbert space introduced in Sec. 3.4. The projection basis determines which states of the full three-subsystem Hamiltonian are retained. Within this common projected space, different choices of coupling operators can be compared directly.

In the projection scans, the driven braid terms can also be supplemented by the projected static Hamiltonian of the uncoupled subsystem background,

$$H_{\text{static}} = V^\dagger \left(H_B + H_C\right) V. \tag{3.33}$$

This term keeps the internal optimized-chain structure of subsystems B and C while the time-dependent junction couplings are varied. Since it is included in the same way for the model comparisons, the differences between the braid results come from the choice of driven coupling operators rather than from a different static background.

3.5.2 Ideal Four-Majorana Model

The simplest braid model uses only four Majorana operators and hand-chosen bilinear couplings. In the projected space, the time-dependent Hamiltonian is

$$H_{\text{ideal}}(t) = \lambda_1(t)\, i\gamma_0\gamma_1 + \lambda_2(t)\, i\gamma_0\gamma_2 + \lambda_3(t)\, i\gamma_0\gamma_3. \tag{3.34}$$

This model is used as a reference calculation. Since the couplings are written directly as Majorana bilinears, it isolates the braid protocol itself from the additional complications of microscopic junction operators and imperfect projected Majoranas.

3.5.3 Projected Microscopic Junction Model

To connect the braid more directly to the optimized dot-chain Hamiltonian, the ideal bilinears can be replaced by microscopic junction operators. The junction between subsystems A and B is constructed from ordinary hopping and pairing terms between the endpoint dots,

$$H_{AB}^{\text{junc}} = -t_{\text{j}} \left(c_A^\dagger c_B + c_B^\dagger c_A\right) + \Delta_{\text{j}} \left(c_A c_B + c_B^\dagger c_A^\dagger\right), \tag{3.35}$$

with an analogous operator H_{AC}^{junc} for the A–C junction. These full-space junction operators are projected into the retained low-energy space,

$$T_{AB} = V^\dagger H_{AB}^{\text{junc}} V, \qquad T_{AC} = V^\dagger H_{AC}^{\text{junc}} V. \tag{3.36}$$

The corresponding projected microscopic braid Hamiltonian is then

$$H_{\text{junc}}(t) = \lambda_1(t)\, T_A + \lambda_2(t)\, T_{AB} + \lambda_3(t)\, T_{AC}, \tag{3.37}$$

where $T_A = i\gamma_0\gamma_1$ fixes the initial pairing inside subsystem A. This model tests whether the physical junction couplings reproduce the desired Majorana exchange after projection.

3.5.4 Projected Local-Operator Model

A second microscopic comparison keeps the bilinear braid structure but replaces the ideal exchanged Majoranas by projected local Majorana components. For example, on the relevant endpoint of subsystem B one constructs

$$\Gamma_{B,-} = V^\dagger i(c_B^\dagger - c_B)V, \tag{3.38}$$

and similarly for subsystem C. After normalization, these local projected operators define $\gamma_{2,\text{loc}}$ and $\gamma_{3,\text{loc}}$. The local-operator braid Hamiltonian is

$$H_{\text{loc}}(t) = \lambda_1(t)\, i\gamma_0\gamma_1 + \lambda_2(t)\, i\gamma_0\gamma_{2,\text{loc}} + \lambda_3(t)\, i\gamma_0\gamma_{3,\text{loc}}. \tag{3.39}$$

This model is useful because it sits between the ideal and fully microscopic descriptions. The projected microscopic model first projects the full junction operator, while the projected local-operator model first projects local Majorana components and then forms bilinears from them. It therefore keeps a clear Majorana-bilinear form, but the operators being coupled are obtained from projected local fermionic operators rather than only from energy-level parity pairs.

3.5.5 Block-Matched Local-Operator Model

The sector-resolved scans also use a block-matched local-operator model. It has the same bilinear form as Eq. 3.39, but replaces the direct projected local operators by the block-matched operators defined in Eq. 3.30. The labelled blocks are fitted independently and then reassembled before the braid Hamiltonian is constructed. This model tests whether resolving the internal degeneracy structure of a sector improves the braid generated from the local channel.

3.5.6 Pulse Sequence

The same pulse sequence is used for each braid model. To avoid confusing the braid amplitudes with the superconducting pairing parameters Δ_i, the time-dependent braid couplings are denoted by $\lambda_i(t)$. The coupling envelope is a smooth pulse constructed from two sigmoid functions,

$$\lambda(t;T_p) = \lambda_{\min} + (\lambda_{\max} - \lambda_{\min}) \frac{1}{1+\exp\left[-s(t-T_p+w/2)\right]} \frac{1}{1+\exp\left[s(t-T_p-w/2)\right]}, \tag{3.40}$$

where T_p is the pulse center, w is the pulse width, and s controls the steepness of the rise and fall. The three braid couplings are then chosen as

$$\lambda_1(t) = \lambda(t;0) + \lambda(t;T) - \lambda_{\min}, \qquad \lambda_2(t) = \lambda(t;T/3), \qquad \lambda_3(t) = \lambda(t;2T/3). \tag{3.41}$$

Thus the A-arm coupling is active at the beginning and end of the cycle, while the A–B and A–C couplings are activated in sequence during the middle of the protocol. In the numerical calculations used here, the total braid time is set to $T = 1$, with $\lambda_{\max} = 1$, $\lambda_{\min} = 0$, pulse width $w = T/3$, and steepness $s = 20/w$.

At each time step, the Hamiltonian for the chosen model is assembled from the same three time-dependent coefficients. The instantaneous spectrum is monitored along the path, and the isolated low-energy band is then transported using the Kato procedure described in the next section.

3.6 Kato Transport and Braid Validation

After a braid Hamiltonian has been defined, the remaining task is to compute the unitary generated by the adiabatic path and to check whether it implements the desired Majorana exchange. The calculation is performed in the projected Hilbert space, but the transported subspace can be smaller than the full projection space. This section describes the transported bands used in the calculations, how the Kato evolution is computed, and which diagnostics are used to compare the resulting unitary with the target braid.

3.6.1 Transported Bands

In the low-energy benchmark, the transported subspace is the isolated four-dimensional lowest-energy manifold. Its gap to the remaining projected states is monitored throughout the braid path.

In the sector-resolved scans, each energy sector is projected and tested independently. Activating the braid Hamiltonian splits the initially degenerate sector into a lower and an upper band of equal dimension. The Kato calculation transports the lower band, so that for a sector projection of dimension d_{P_i} the transported dimension is fixed to

$$d_{T_i} = \frac{d_{P_i}}{2}. \tag{3.42}$$

The spectral gap between these two bands is monitored along the path. The existence of this isolated lower band makes the transport well defined, while the final target comparison determines whether the transported unitary actually performs the intended exchange.

3.6.2 Numerical Kato Transport

The braid path is discretized into time points $t_0, \dots, t_N$. At each time step, the Hamiltonian is diagonalized and the projector onto the transported band is constructed,

$$P(t_\ell) = \sum_{a=1}^{d_T} |\psi_a(t_\ell)\rangle \langle\psi_a(t_\ell)| \,. \tag{3.43}$$

The Kato generator is approximated from consecutive projectors,

$$K(t_\ell) = P(t_\ell)\dot{P}(t_\ell) - \dot{P}(t_\ell)P(t_\ell), \qquad \dot{P}(t_\ell) \approx \frac{P(t_{\ell+1}) - P(t_\ell)}{\Delta t}. \tag{3.44}$$

The transport unitary is updated as

$$U_K(t_{\ell+1}) = \exp[-\Delta t\, K(t_\ell)]U_K(t_\ell), \qquad U_K(t_0) = I. \tag{3.45}$$

This convention matches the numerical implementation and gives the geometric transport of the selected subspace along the braid path. The instantaneous energies and pulse amplitudes are stored at the same time, so that the quality of the transported band can be checked together with the final braid unitary.

3.6.3 Target Exchange Gate

The target unitary for exchanging γ_2 and γ_3 is taken to be

$$U_{\text{target}} = \exp\left(-\frac{\pi}{4}\gamma_2\gamma_3\right). \tag{3.46}$$

This convention corresponds to the exchange action

$$\gamma_2 \mapsto -\gamma_3, \qquad \gamma_3 \mapsto \gamma_2, \tag{3.47}$$

up to the overall sign convention chosen for the Majorana operators. Since a global phase of the transported unitary is physically irrelevant, comparisons with the target gate are made after aligning the overall phase. In practice, the reported gate error is a Frobenius-norm distance between the transported unitary and the phase-aligned target unitary, restricted to the initial transported basis.

3.6.4 Operator-Action Checks

The first validation step checks the induced action on the Majorana operators directly. For each model, the transported unitary is applied as

$$\gamma_i' = U_K^\dagger \gamma_i U_K. \tag{3.48}$$

The resulting operators are compared with the expected single-exchange map,

$$\gamma_2' \approx -\gamma_3, \qquad \gamma_3' \approx \gamma_2, \qquad \gamma_0' \approx \gamma_0, \qquad \gamma_1' \approx \gamma_1. \tag{3.49}$$

The mismatch is measured using a normalized Frobenius norm,

$$\|A\|_{\text{norm}} = \frac{\|A\|_F}{\sqrt{d}}, \tag{3.50}$$

where d is the dimension of the matrix being compared. These checks are useful because they test the braid as an operator transformation, rather than only as a matrix acting on a chosen basis.

3.6.5 Parity-Resolved Gate Checks

The transported unitary is also analyzed in a parity basis. The total parity operator is projected into the initial transported space and diagonalized, separating the unitary into odd- and even-parity blocks. The off-block leakage is defined from the norm of the odd-to-even and even-to-odd blocks,

$$\mathcal{L}_{\text{parity}} = \|U_{oe}\| + \|U_{eo}\|. \tag{3.51}$$

A small value indicates that the braid approximately preserves fermion parity inside the transported manifold.

After the parity blocks have been separated, each block is compared with the corresponding block of the target exchange gate in Eq. 3.46. This produces separate odd- and even-sector target-gate errors. Reporting the parity blocks separately is useful because the same Majorana exchange can appear with different matrix representations in the two parity sectors, while still describing the same physical braid.

3.6.6 Energy-Level and Projected-Local Targets

The final comparison distinguishes between two possible choices of target Majoranas. The energy-level target uses the Majorana operators constructed from even-odd parity pairs, as described in Sec. 3.4. The projected-local target instead uses the normalized projected local operators obtained from microscopic combinations such as $i(c^\dagger - c)$. These two targets need not be identical after projection, especially when more excited states are retained or when the optimized Majorana operators are not perfectly local.

For this reason, the physical braid is compared in several ways: against the energy-level target, against the projected-local target, and against the ideal reference braid. The mismatch between the two target constructions is also recorded. This separation is important for interpreting the results. If the physical braid disagrees with the energy-level target but agrees with the projected-local target, the issue may be the identification of the effective Majorana operators rather than the adiabatic pulse itself. Conversely, if both target comparisons fail, the projected braid Hamiltonian is not implementing the desired exchange within the chosen transported subspace.

Chapter 4

Results

This chapter presents the main numerical results of the thesis. The first part analyzes the optimized PMM chains obtained from the basin-hopping optimization procedure. The optimized parameters are compared across system sizes and interaction strengths, and the resulting Hamiltonians are evaluated using the spectral and local Majorana diagnostics introduced in the previous chapters.

The second part focuses on braiding. Based on the post-optimization diagnostics, the most promising optimized configuration is selected as the building block for a three-subsystem braiding geometry. The braid analysis begins with a controlled low-energy comparison between the ideal four-Majorana model, the projected microscopic junction model, and the projected local-operator model. These calculations serve as benchmarks for whether the optimized microscopic system reproduces the exchange behavior expected from the effective Majorana description.

The final part extends the braid analysis to perform braiding in all identified energy sectors of the full three-subsystem system. This tests whether the optimized PMM chains can support Majorana-like exchange beyond the lowest projected manifold and how braids constructed from local microscopic operators behave across different sectors. The sector scans reported here compare each construction with its own target; a common-target comparison is required to determine whether the local constructions reproduce the energy-level exchange channel.

4.1 Optimized PMM Chains

This section summarizes the parameter configurations obtained from the basin-hopping optimization procedure and compares how they depend on chain length and interaction strength. The optimization was performed for chain lengths $n = 2, 3$, and 4, and for fixed interaction strengths $U = 0, 0.1, 0.5, 1.0, 2.0$, and 5.0. In all cases, the hopping amplitudes were fixed to $t_i = 1$, while the onsite energies ϵ_i and pairing amplitudes Δ_i were optimized.

Fixing the hopping scale makes the optimized parameters easier to compare across system sizes and interaction strengths. It also prevents the optimizer from reducing the relative importance of interactions through a trivial global rescaling of the hopping and pairing amplitudes. The purpose of this section is therefore to identify the lowest-loss parameter sets and to extract the main trends in the optimized onsite and pairing profiles before applying the post-optimization diagnostics.

4.1.1 Lowest-loss configurations

Table 4.1 summarizes the lowest-loss configuration found for each combination of chain length n and fixed interaction strength U. Since the hopping amplitudes were fixed to $t_i = 1$ in these runs, the table lists only the optimized pairing amplitudes and onsite energies. Empty entries correspond to parameters that are absent for shorter chains.

n	U	Loss	Δ_1	Δ_2	Δ_3	ϵ_1	ϵ_2	ϵ_3	ϵ_4
2	0.0	5.657e-03	1.000	–	–	-0.001	0.002	–	–
2	0.1	2.617e-01	1.047	–	–	-0.050	-0.047	–	–
2	0.5	1.083e+00	1.249	–	–	-0.255	-0.249	–	–
2	1.0	1.940e+00	1.498	–	–	-0.499	-0.496	–	–
2	2.0	3.203e+00	2.000	–	–	-1.006	-1.004	–	–
2	5.0	4.004e+00	3.499	–	–	-2.508	-2.506	–	–
3	0.0	3.442e-01	1.010	1.348	–	0.019	1.774	-0.466	–
3	0.1	5.603e-01	1.001	1.002	–	-0.101	-0.607	-0.003	–
3	0.5	9.359e-01	1.004	1.012	–	-0.046	-0.889	-0.454	–
3	1.0	1.452e+00	1.025	1.130	–	-0.195	-1.530	-0.802	–
3	2.0	2.326e+00	1.584	0.978	–	-1.511	-3.077	-0.484	–
3	5.0	3.142e+00	0.846	1.746	–	-4.814	1.009	-0.184	–
4	0.0	3.161e-01	2.698	5.332	1.004	-0.059	-4.252	0.236	-0.003
4	0.1	4.840e-01	1.790	5.002	1.013	0.156	2.376	-1.966	-0.050
4	0.5	1.006e+00	1.212	3.490	1.206	-0.246	-0.499	-0.503	-0.250
4	1.0	5.670e-01	2.238	18.807	1.072	-0.500	-1.028	-1.025	-0.496
4	2.0	1.824e+00	1.492	5.537	1.588	-0.869	3.526	-6.508	-1.220
4	5.0	2.600e+00	3.389	9.513	3.336	-2.645	5.032	-15.431	-2.293

Table 4.1: Lowest-loss optimized configurations extracted from the optimization procedure for each combination of system size n and fixed interaction strength U.

The two-dot results provide a useful reference point. In the non-interacting case, the optimizer reproduces the expected sweet spot, with $\Delta_1 \approx t_1 = 1$ and onsite energies close to zero. As the interaction strength is increased, the optimized pairing amplitude grows above the fixed hopping scale, while the onsite energies move approximately toward $\epsilon_i \approx -U/2$. This indicates that the optimizer deforms the known two-dot solution in a smooth and physically interpretable way.

At the same time, the loss increases substantially with interaction strength. This suggests that the interacting two-dot system becomes progressively less compatible with the full set of optimization criteria, even though the optimized parameters continue to follow the expected interacting trend. The table therefore motivates the more detailed spectral and local diagnostics presented below, rather than by itself establishing whether the optimized states have useful Majorana character.

For the three-dot system, the weak- and intermediate-interaction configurations remain close to the homogeneous pairing limit, with $\Delta_1 \approx \Delta_2 \approx 1$ for $U = 0.1$ and $U = 0.5$. The onsite energies, however, become inhomogeneous, with the middle dot often shifted more strongly than the edge dots. This suggests that the optimizer uses the additional site to reshape the effective potential profile while keeping the pairing amplitudes near the hopping scale.

The four-dot results show a more complex optimization landscape. Several of the lowest-loss configurations have strongly enhanced central pairing amplitudes, while the outer pairing amplitudes remain closer to unity. The onsite energies also become increasingly inhomogeneous, especially at larger interaction strengths. These trends indicate that longer PMM chains do not simply reproduce the two-dot sweet spot, but instead favor structured parameter profiles that depend sensitively on both n and U.

One recurring feature in the longer chains is that interior onsite energies are often shifted more strongly than the edge onsite energies. A possible interpretation is that this interior detuning helps separate the Majorana-like modes toward the edges. The parity-resolved spectra, charge diagnostics, and Majorana-polarization profiles below test whether the resulting configurations have the desired properties, although a controlled variation of the interior potentials would be needed to establish the mechanism.

4.1.2 Representative optimized spectra

The representative spectrums in Figs. 4.1 and 4.2 illustrate how interactions affect the optimized two-dot system. In the non-interacting case, both parity pairs are nearly degenerate, consistent with the expected two-dot sweet spot. When a weak interaction is introduced at $U = 0.1$, the lowest parity pair remains close to degenerate, while the excited pair begins to show a visible splitting. At $U = 2.0$, this excited-state splitting becomes much larger. Meaning that although the ground-state pair can retain a low-energy Majorana-like signature, it still does not maintain clean parity-pair degeneracy across the full finite spectrum.

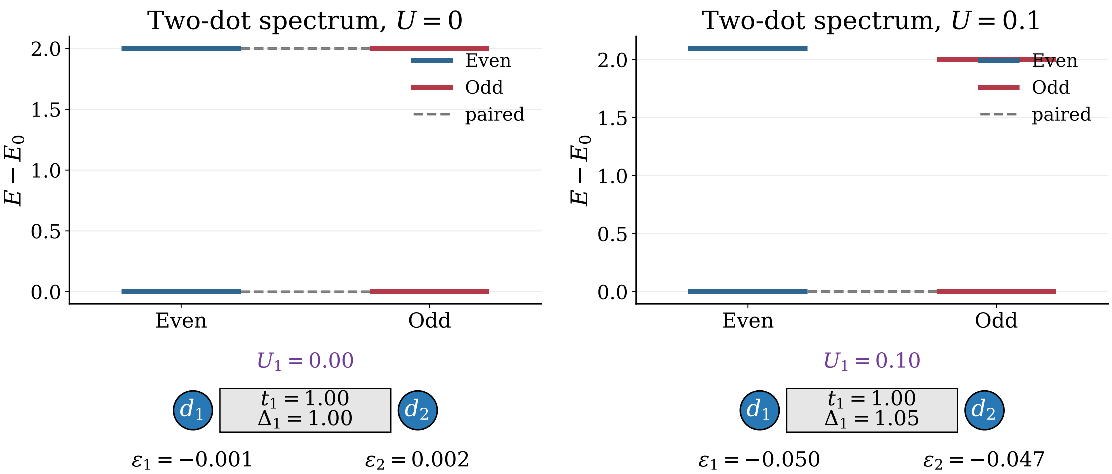

Figure 4.1: Parity-resolved spectra after optimization for the two-dot system at $U = 0$ and $U = 0.1$. Energies are shifted by the ground-state energy in each panel, and the optimized parameters are shown schematically below each spectrum.

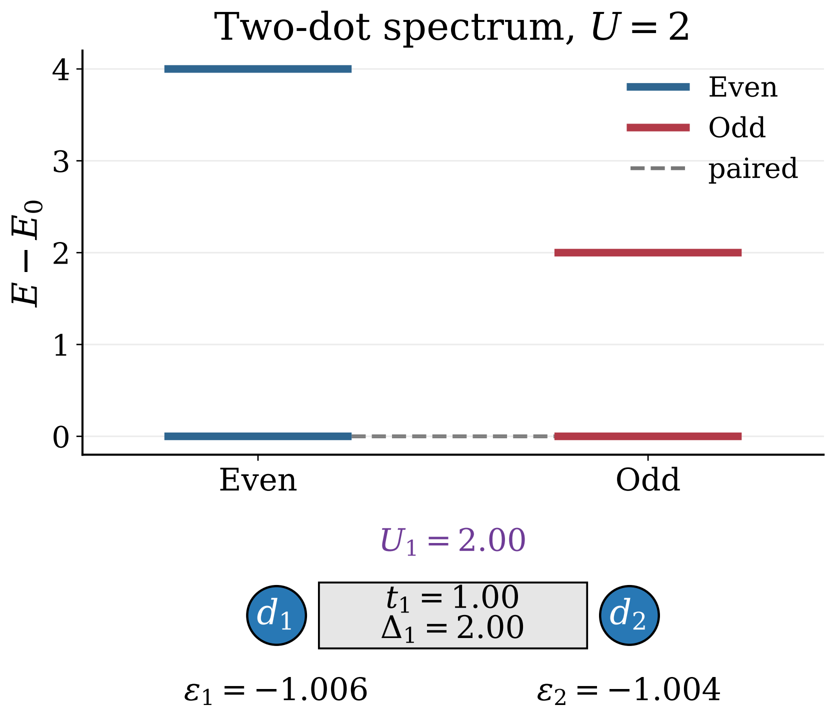

Figure 4.2: Parity-resolved spectrum after optimization for the two-dot system at $U = 2.0$. Energies are shifted by the ground-state energy, and the optimized parameters are shown schematically below the spectrum.

The optimized three-dot system at $U = 0.1$ provides a more promising representative configuration. As shown in Fig. 4.3, the parity-resolved spectrum remains nearly paired, while the Majorana-polarization profile has the expected edge-localized structure. The polarization is large and opposite in sign on the two outer dots, and it is strongly suppressed on the middle dot. This combination of spectral pairing and spatial Majorana character motivates using this configuration as the main candidate for the braiding calculations below. Comparing the three-dot configuration for $U = 0.1$ with the two-dot configuration for $U = 0.1$ also illustrates how the additional site can help stabilize the Majorana-like features in the presence of interactions, by noticing the excited-states retaining a better energy degeneracy in the three-dot case than in the two-dot case, even though both configurations have similar ground-state splitting and similar optimized parameters. The additional site therefore provides more freedom to optimize the excited states, which is important for maintaining a clean low-energy manifold that can support braiding.

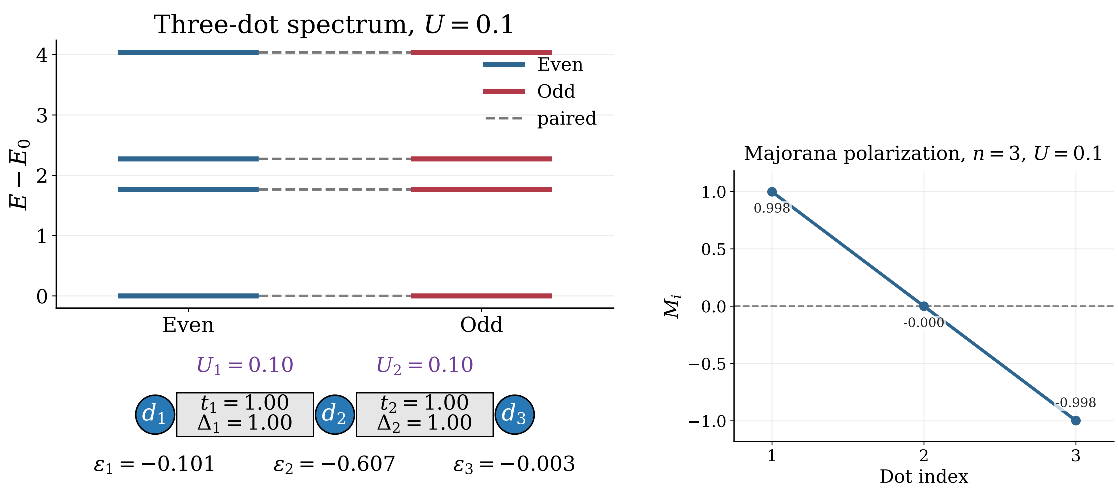

Figure 4.3: Parity-resolved spectrum and Majorana-polarization profile for the optimized three-dot system at $U = 0.1$. The left panel shows the optimized spectrum and parameters, while the right panel shows the Majorana polarization of the lowest even-odd pair.

4.2 Post-Optimization Diagnostics

The optimized parameter values alone do not determine whether a configuration is useful for Majorana braiding. After optimization, the candidate Hamiltonians must therefore be tested through spectral and local diagnostics. The parity-resolved spectra and low-energy gaps show whether the optimizer has produced an isolated nearly degenerate low-energy manifold, while the charge difference and Majorana-polarization profile test whether this manifold has the expected non-local and edge-localized Majorana character.

4.2.1 Parity-resolved spectra and low-energy gaps

Figure 4.4 compares the parity-resolved spectra for $n = 2, 3$, and 4 at selected interaction strengths. A useful Majorana-like configuration should have a nearly degenerate lowest even-odd pair, together with a finite gap separating this pair from higher excited states. The dashed connectors in the figure indicate parity pairs that remain close in energy after optimization.

The non-interacting cases give the cleanest spectra, as expected. For the two-dot system, the parity pairing is nearly ideal at $U = 0.0$, but the excited-state splitting becomes visible already at weak interaction and increases further at stronger interaction. This shows that the interacting two-dot system can retain a low-energy degeneracy while losing clean parity pairing across the full finite spectrum.

The three-dot system is more stable across all interaction strengths shown, compared to both the two-dot and four-dot systems. For all three interaction strengths, $U = 0.0$, $U = 0.1$, and $U = 2.0$, we can see that all energy levels are degenerate in pairs, up to a numerical threshold, with a decently large gap separating the energy levels into distinct energy sectors. The four-dot system, shows a promising structure for weak and no interactions, while we can clearly see that for stronger interactions $U = 2.0$, the energy degeneracy falls apart. Most importantly, the ground state energies are no longer degenerate, and thus there is no longer a clear low-energy manifold that can support braiding. The three-dot system therefore appears to be the most robust candidate for maintaining a clean low-energy manifold across different interaction strengths, but

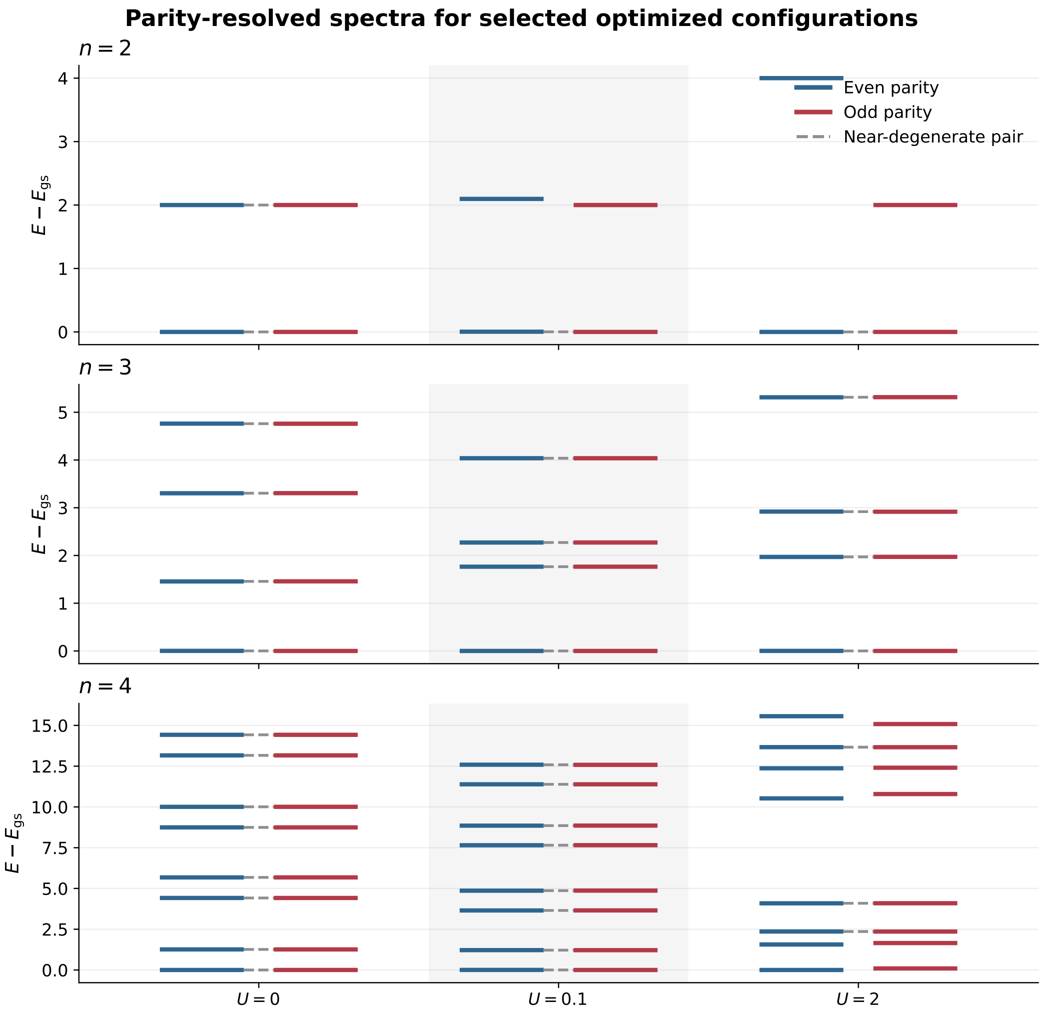

Figure 4.4: Parity-resolved spectra for the best optimized configurations at selected interaction strengths. Each panel corresponds to a different system size n, and energies are shifted so that the ground state is at zero.

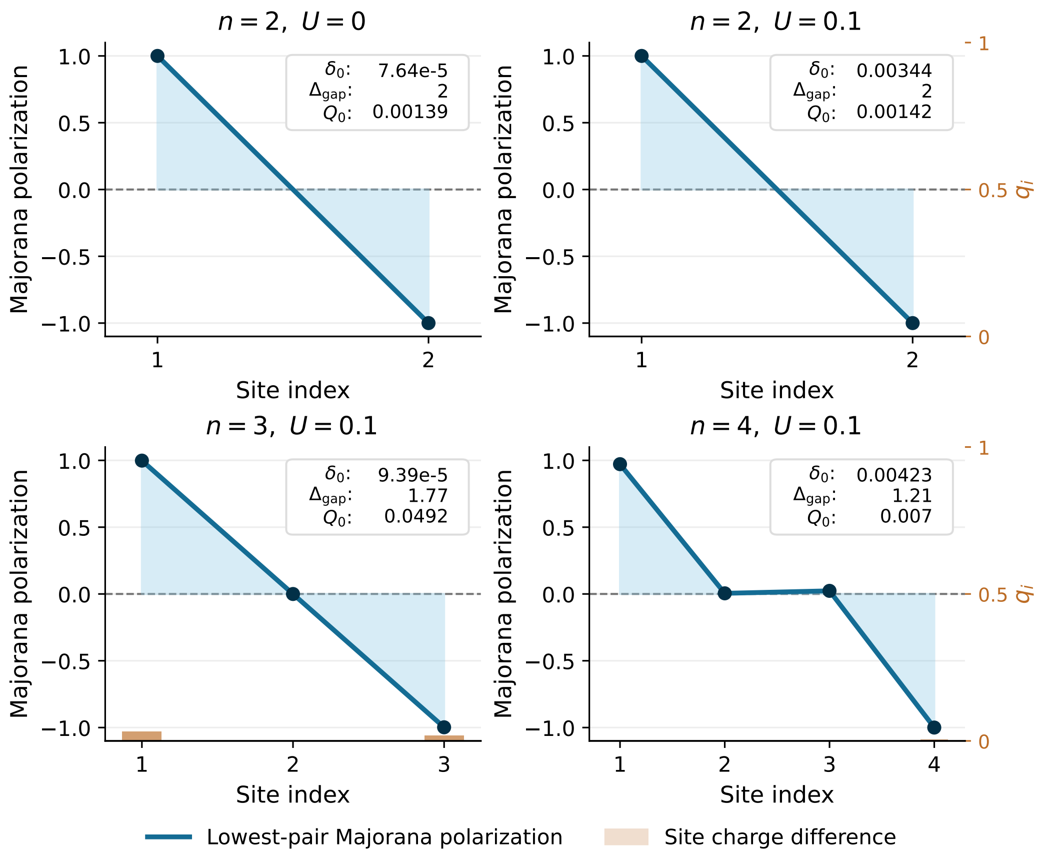

Figure 4.5: Local diagnostics for representative optimized configurations. The blue curve shows the Majorana-polarization profile of the lowest even-odd pair, while the orange bars, read against the 0–1 right axis, show the local charge difference $q_i = |\langle e|n_i|e\rangle - \langle o|n_i|o\rangle|$ on each site. The inset in each panel reports the ground-state splitting δ_0, the low-energy gap Δ_{gap}, and the total charge difference Q_0.

in order to determine if it is possible to braid using the three-dot system in a weak interaction regime, we need to also check the local diagnostics and Majorana character.

4.2.2 Local diagnostics and Majorana character

As discussed in the methodology chapter, Majorana-like states cannot be identified from spectral degeneracy alone. The parity-resolved spectra and low-energy gaps test whether the optimized Hamiltonian contains an isolated, nearly degenerate low-energy manifold. The present subsection examines the complementary local diagnostics: the Majorana-polarization profile and the local charge difference between the lowest even-odd parity partners.

A convincing Majorana-like configuration should have opposite Majorana polarization on the two ends of the chain, with strongly suppressed polarization in the interior, as quantified by Eqs. 3.20 and 3.21. In addition, the even and odd parity partners should be locally difficult to distinguish. This is tested by the charge-difference diagnostic in Eq. 3.19; smaller values indicate that the parity information is less visible to local charge measurements.

n	U	δ_0	Δ_{gap}	Q_0	$\overline{\lvert M_{\text{edge}}\rvert}$	$\max\lvert M_{\text{bulk}}\rvert$
2	0	7.64e-05	2	0.00139	1	0
2	0.1	0.00344	2	0.00142	1	0
3	0.1	9.39e-05	1.77	0.0492	0.998	0.000202
4	0.1	0.00423	1.21	0.007	0.986	0.0233

Table 4.2: Summary of local diagnostics for the representative configurations shown in Fig. 4.5. Here δ_0 is the splitting of the lowest even-odd pair, Δ_{gap} is the gap separating this pair from the next excited states, Q_0 is the total charge difference between the parity partners, $\overline{\lvert M_{\text{edge}}\rvert}$ is the mean absolute Majorana polarization on the first and last sites, and $\max\lvert M_{\text{bulk}}\rvert$ is the largest absolute polarization on the interior sites.

Figure 4.5 compares the spatial profiles directly, while Table 4.2 summarizes the corresponding scalar diagnostics. For the two-dot system, the edge polarization is essentially ideal, with $\overline{\lvert M_{\text{edge}}\rvert} \approx 1$ and no bulk contribution by construction. The total charge difference is also very small, $Q_0 \approx 1.4 \times 10^{-3}$, both at $U = 0$ and at $U = 0.1$. These two-dot cases are useful references, but because they have no interior sites they do not test suppression of Majorana weight in the bulk.

The three-dot configuration at $U = 0.1$ gives the clearest interacting Majorana-polarization pattern. The ground-state splitting is only $\delta_0 = 9.39 \times 10^{-5}$, the excitation gap remains large, $\Delta_{\text{gap}} = 1.77$, and the bulk polarization is almost completely suppressed, with $\max\lvert M_{\text{bulk}}\rvert = 2.02 \times 10^{-4}$. Its charge difference, however, is larger than in the two-dot and four-dot reference cases, with $Q_0 = 0.0492$. This shows that the three-dot state is not ideal in every diagnostic, even though its spectral and polarization properties are the strongest.

The four-dot configuration at $U = 0.1$ shows the opposite tradeoff. It has a smaller charge difference, $Q_0 = 0.0070$, and comparably strong edge polarization, $\overline{\lvert M_{\text{edge}}\rvert} \approx 0.986$. However, the ground-state splitting increases to $\delta_0 = 4.23 \times 10^{-3}$ and the bulk polarization rises to $\max\lvert M_{\text{bulk}}\rvert = 2.33 \times 10^{-2}$.

Taken together, these diagnostics show that different system sizes optimize different aspects of the Majorana criteria. The four-dot case performs better in terms of local charge indistinguishability, while the three-dot case performs better in terms of spectral near-degeneracy and suppression of bulk Majorana weight. Since the braiding analysis requires a low-energy subspace that is both nearly degenerate and spatially well localized, the interacting three-dot system remains the most convincing candidate for the rest of the thesis.

4.3 Selection of Braiding Candidate

The braiding calculations are performed using the optimized three-dot configuration at $U = 0.1$. This choice is based on the combined post-optimization diagnostics above. Although the four-dot configuration has a smaller local charge difference, the three-dot configuration has the most favorable combination of small ground-state splitting, large excitation gap, clear edge-localized Majorana-polarization profile, and strongly suppressed bulk polarization. Since the braiding analysis relies on a nearly degenerate and spatially localized low-energy subspace, the three-dot system is chosen as the main microscopic building block.

The optimized parameters used for each three-dot subsystem are

$$\Delta_1 = 1.001, \qquad \Delta_2 = 1.002, \qquad \epsilon_1 = -0.101, \qquad \epsilon_2 = -0.607, \qquad \epsilon_3 = -0.003, \tag{4.1}$$

with fixed hopping amplitudes $t_i = 1$ and fixed interaction strength $U = 0.1$.

In the braiding geometry, three identical copies of this optimized three-dot chain are combined. Each subsystem has Hilbert-space dimension $2^3 = 8$, giving a total microscopic Hilbert space of dimension $2^9 = 512$. This system is large enough to test braiding beyond the lowest projected manifold, while still being small enough for exact diagonalization and systematic sector-resolved scans. The selected configuration therefore provides a useful setting for studying how interactions affect braiding across the many-body spectrum.

4.4 Braiding in Projected Majorana Models

The optimized three-dot chain is now used as the microscopic building block for the braiding calculations. Before studying the higher energy sectors, I first establish a controlled low-energy benchmark by comparing three related braid models: the ideal four-Majorana reference model, the projected microscopic junction model, and the projected local-operator model. This comparison tests whether the optimized microscopic system reproduces the exchange behavior expected from the effective Majorana description.

In the ideal reference model, the braid is generated directly by tunable Majorana bilinears built from the four Majorana operators defined in Eq. 2.15. This isolates the braid protocol itself from microscopic details. In the projected microscopic model, these ideal bilinears are replaced by projected junction operators constructed from ordinary hopping and crossed-Andreev pairing terms between endpoint dots. For example, the A–B junction is

$$H_{AB}^{\mathrm{junc}} = -t_{\mathrm{j}} \left(c_A^\dagger c_B + c_B^\dagger c_A \right) + \Delta_{\mathrm{j}} \left(c_A c_B + c_B^\dagger c_A^\dagger \right), \tag{4.2}$$

as introduced in Eq. 3.35. After projection into the retained low-energy subspace, this operator gives the effective coupling T_{AB} used in the driven braid Hamiltonian of Eq. 3.37. The projected microscopic braid therefore tests whether physically motivated junction couplings reproduce the same exchange channel as the ideal Majorana model.

4.4.1 Ideal four-Majorana reference

Figure 4.6 shows the reference braid generated by four active Majorana operators in the projected low-energy space. The three couplings are turned on sequentially, so that the dominant coupling is first between γ_0 and γ_1, then between γ_0 and γ_2, and finally between γ_0 and γ_3 before returning to the initial configuration. Throughout this cycle, the transported low-energy manifold remains essentially four-fold degenerate. The largest instantaneous splitting within this manifold is only $\max(E_3 - E_0) = 2.28 \times 10^{-12}$, while the minimum gap to the excited states remains open at $\min(E_4 - E_3) = 1.414$.

Using Kato transport, the resulting unitary reproduces the expected single-exchange action on the Majorana operators. The largest error in the transformations $\gamma_2 \to -\gamma_3$ and $\gamma_3 \to \gamma_2$ is 1.33×10^{-5}. In the parity-resolved gate blocks, the odd and even 2×2 blocks agree with the target exchange gate to within 6.53×10^{-8}, while the off-block leakage is only 1.02×10^{-11}. The effective model therefore provides a clean benchmark for the more microscopic projected calculations.

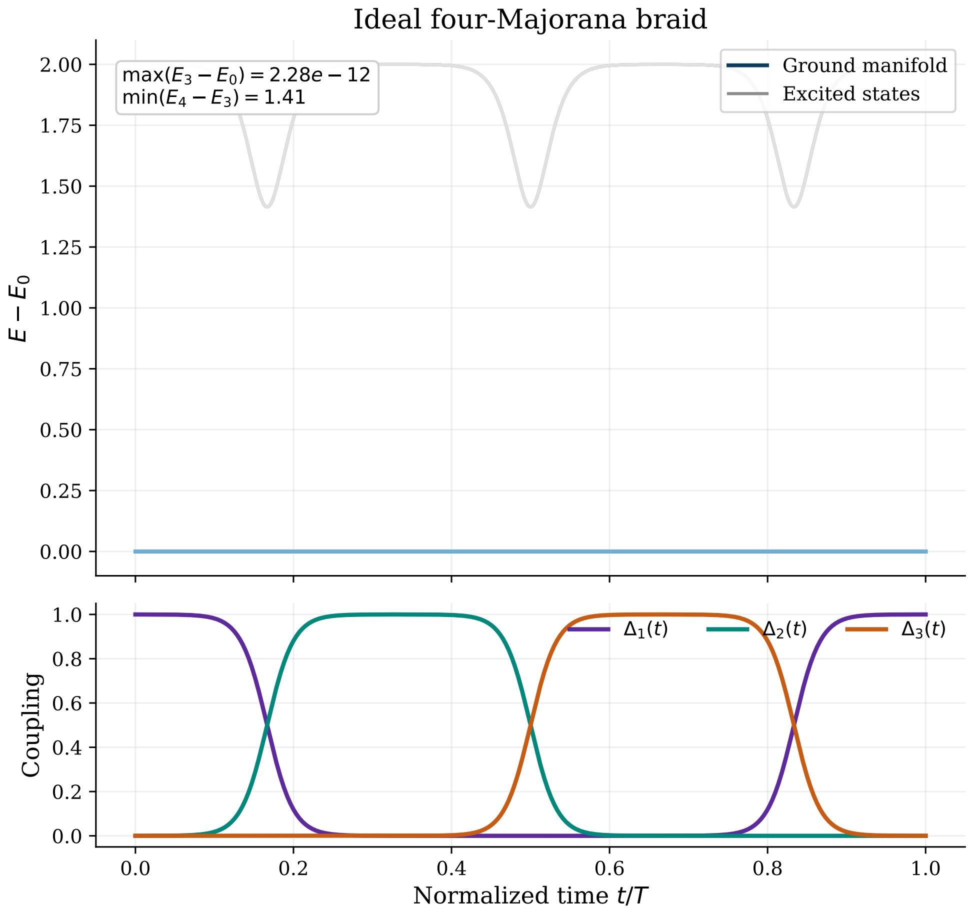

Figure 4.6: Idealized four-Majorana braiding protocol. The top panel shows the instantaneous spectrum of the effective Hamiltonian during the braid, with all energies shifted so that the lowest state remains at zero. The bottom panel shows the three smooth pulse couplings that move the dominant Majorana coupling through the network.

4.4.2 Projected microscopic and local-operator braids

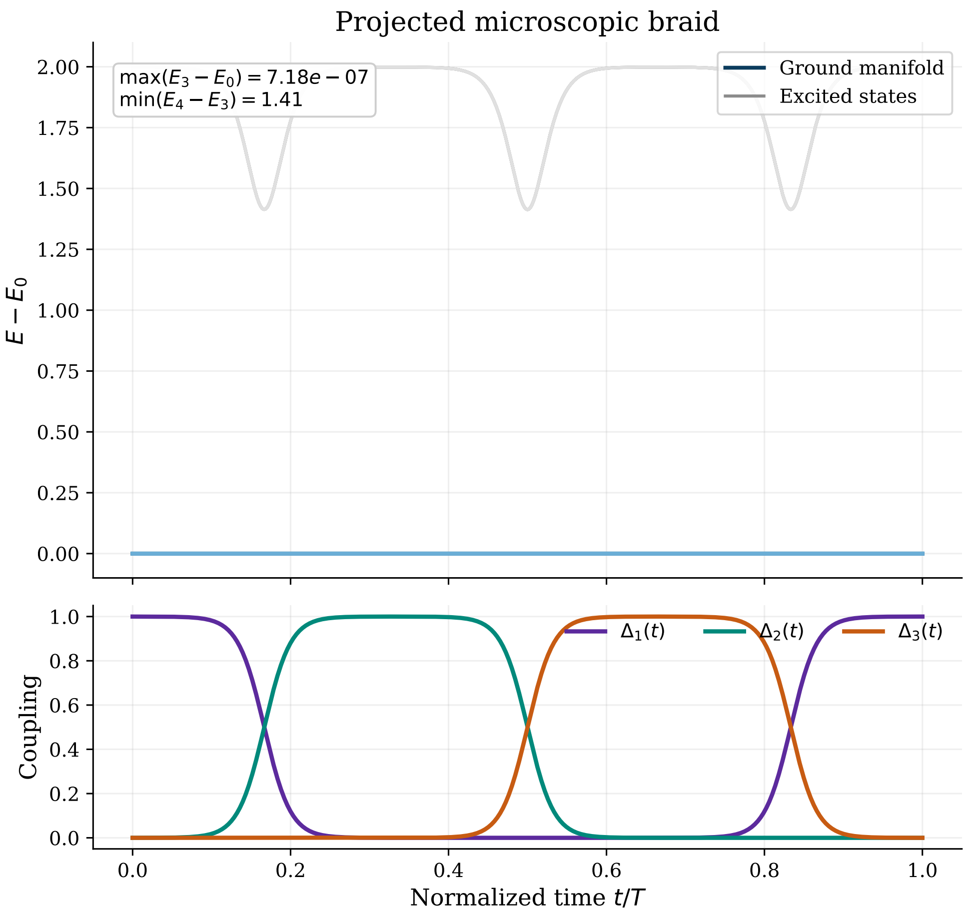

Figure 4.7: Braiding protocol after replacing the idealized couplings by projected microscopic junction operators. The spectral evolution remains almost identical to the effective-model benchmark, with a well-isolated low-energy manifold throughout the cycle.

After replacing the ideal Majorana-bilinear couplings by projected microscopic junction operators, the braid remains close to the reference result. Figure 4.7 shows that the four-dimensional low-energy manifold is still well separated from higher states throughout the cycle. The maximum splitting inside this manifold increases only to $\max(E_3-E_0) = 7.18\times10^{-7}$, while the minimum gap remains large, $\min(E_4-E_3) = 1.413$.

The junction decomposition in Fig. 4.8 explains why the projected microscopic braid agrees so well with the ideal reference. The projected AB coupling is dominated by the bilinear $i\gamma_{A1}\gamma_{B2}$ with coefficient -0.999, and the projected AC coupling is dominated by $i\gamma_{A1}\gamma_{C2}$ with approximately the same coefficient. This identifies γ_{B2} and γ_{C2} as the Majoranas that participate in the microscopic braid, while γ_{A1} acts as the shared Majorana through which the exchange is mediated.

As an additional consistency check, I also compare with a projected local-operator braid. In this construction, the exchanged Majorana channels are obtained by projecting local endpoint Majorana operators rather than by projecting the full microscopic junction Hamiltonian. This tests whether the same low-energy exchange channel is accessed by local microscopic operators.

Table 4.3 shows that the microscopic braid reproduces the ideal benchmark extremely

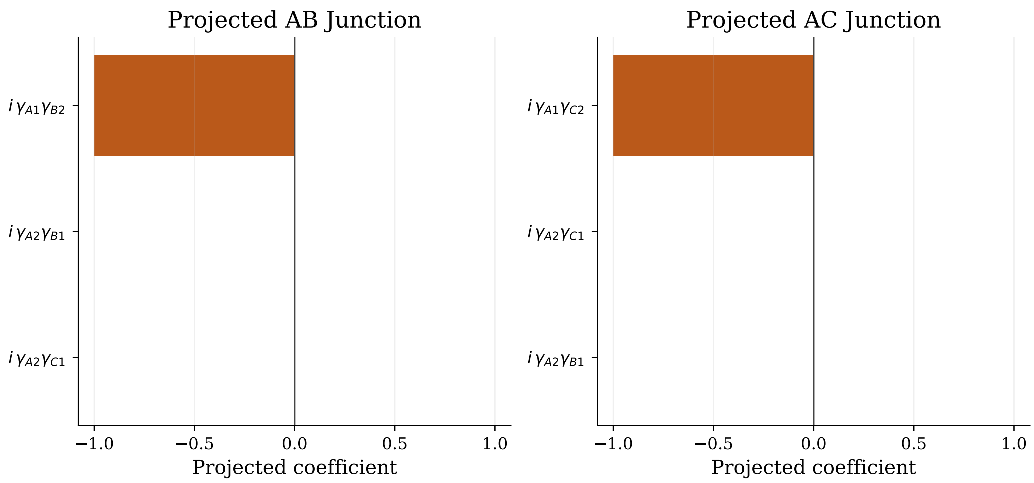

Figure 4.8: Dominant Majorana-bilinear components of the projected microscopic junction operators. The AB junction is almost entirely $i\gamma_{A1}\gamma_{B2}$, while the AC junction is almost entirely $i\gamma_{A1}\gamma_{C2}$. All subleading bilinears are strongly suppressed.

Model	$\max(E_3 - E_0)$	$\min(E_4 - E_3)$	Max 1x err.	Leakage	Odd err.	Even err.
Ideal four-Majorana reference model	2.275e-12	1.414	1.334e-05	1.016e-11	6.535e-08	6.535e-08
Projected microscopic junction model	7.175e-07	1.413	1.334e-05	8.844e-16	6.538e-08	6.538e-08
Projected local-operator model	4.663e-15	1.414	1.334e-05	7.287e-16	6.537e-08	6.537e-08

Table 4.3: Comparison of the main braid diagnostics for the ideal four-Majorana reference model, the projected microscopic junction model, and the alternative projected local-operator model. Here $\max(E_3 - E_0)$ measures the largest instantaneous splitting inside the four-dimensional low-energy manifold, $\min(E_4 - E_3)$ is the minimum gap to the excited sector, "Max 1x err." is the largest deviation in the expected single-exchange Majorana map, "Leakage" is the off-block norm in the parity basis, and the last two columns compare the odd and even parity blocks with the target exchange gate.

Projected operator	Dominant Majorana	Coefficient
$\chi_{B1,y}$	γ_{B2}	-1.000
$\chi_{B2,y}$	γ_{B2}	1.000
$\chi_{C1,y}$	γ_{C2}	-1.000
$\chi_{C2,y}$	γ_{C2}	1.000
Pairwise relation errors		
$\chi_{B1,y} \approx -\chi_{B2,y}$	opposite sign	8.969e-14
$\chi_{C1,y} \approx -\chi_{C2,y}$	opposite sign	5.787e-14

Table 4.4: Projected local-site operators used in the alternative braid model. The table shows that the y-type local operators on the first and last sites of subsystems B and C project onto the same low-energy Majorana channels, γ_{B2} and γ_{C2}, up to an overall sign.

well. The largest single-exchange error stays at 1.33×10^{-5} in all three models, and the odd and even parity blocks continue to match the target gate to approximately 6.54×10^{-8}. The projected local-operator version produces the same conclusion, which makes it a useful consistency check rather than an independent competing protocol.

Table 4.4 explains why the local-operator version appears insensitive to whether the first or last site of subsystem B or C is used. After projection, both $\chi_{B1,y}$ and $\chi_{B2,y}$ reduce to γ_{B2} up to opposite sign, and similarly both $\chi_{C1,y}$ and $\chi_{C2,y}$ reduce to γ_{C2}. The different microscopic site choices therefore feed into the same low-energy braid channel rather than producing genuinely different braiding operations.

4.5 Braiding in Multiple Energy Sectors

In order to explore if braiding is a posible operation beyond the lowest projected manifold, we had to come up with a way of identifying distinct energy sectors in the three-dot, three-subsystem system.
This is done by first, combining all single system Hamiltonians into one single full system Hamiltonian:

$$H = H_A \otimes I_B \otimes I_C + I_A \otimes H_B \otimes I_C + I_A \otimes I_B \otimes H_C, \tag{4.3}$$

This Hamiltonian captures the entire $2^9 \times 2^9$ Hilbert space of the Y-junction system. These three subsystems, A, B and C, are at the initial time decoupled, and identical.

This property of well defined single subsystem Hamiltonians transfers over to the full system Hamiltonian, where if we look at the complete even − odd energy spectre of the full system, we can identify distinct energy sectors, where each sector is made up of states that are degenerate in energy, and seperated by a gap from other sectors Fig. 4.9.

In order to test whether the optimized PMM system can support braiding in multiple energy sectors, we can now perform the same braiding protocol as before, but instead of projecting onto the lowest energy subspace, we can pick any of the identified energy sectors. In order to pick out which sector to explore, we can first identify which eigenvalues and eigenvectors belong to each sector. The eigenvectors will then be used to construct the projection operator P that projects onto the selected sector. This makes it possible to test, sector by sector, whether each braid construction produces its intended exchange behavior. The present self-target scans test braiding beyond the lowest projected manifold, while agreement between the local and energy-level exchange channels requires a separate common-target comparison.

When exploring the different energy sectors, we can create projections of different dimensions. For example $P_0 = P_{\text{GS}}$, the ground state projection, as described by the group size in Fig. 4.9 will have a dimension of 8×512, we get 8 from number of states in the ground state sector, and 512 from the total number of states in the full system. If we look at the first excited sector, we can create a projection P_1 that has a dimension of 24×512, and so on. An interesting observation occured when we projected the system down to any of the sectors in the system. After the braid couplings were activated, the projected spectrum split into a lower and an upper band of equal dimension, separated by a gap Fig. 4.10. This split provides a well-defined lower band for Kato transport, but does not by itself establish that the resulting unitary performs a successful Majorana exchange. The target-overlap results below provide that separate test.

For the sector-resolved scans, the transported dimension was therefore fixed to half of the sector projection dimension, $\dim T_i = \dim P_i/2$, as defined in Eq. (3.42). In each

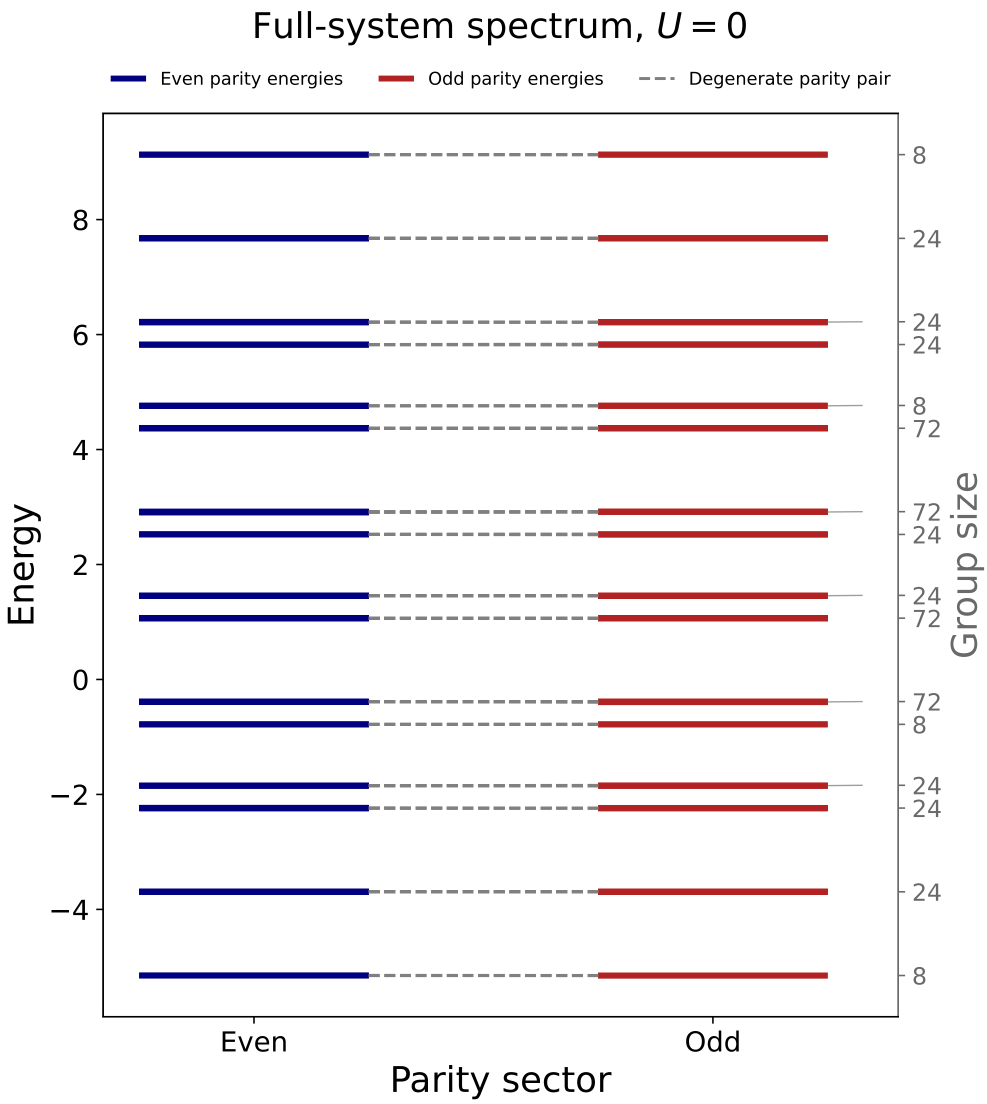

Figure 4.9: Full system Hamiltonian energy spectrum. The spectre is split into even and odd eigenenergies. The dashed lines represent the degeneracy link between the even and odd sectors. The left hand y-axis shows the energy levels, while the right hand y-axis shows how many states reside in each energy sector. This figure was created by a single non-interacting system, similar plots for the interacting model can be found in appendix B

sector, the Kato calculation transports the lower band and measures whether its final unitary reproduces the intended exchange.

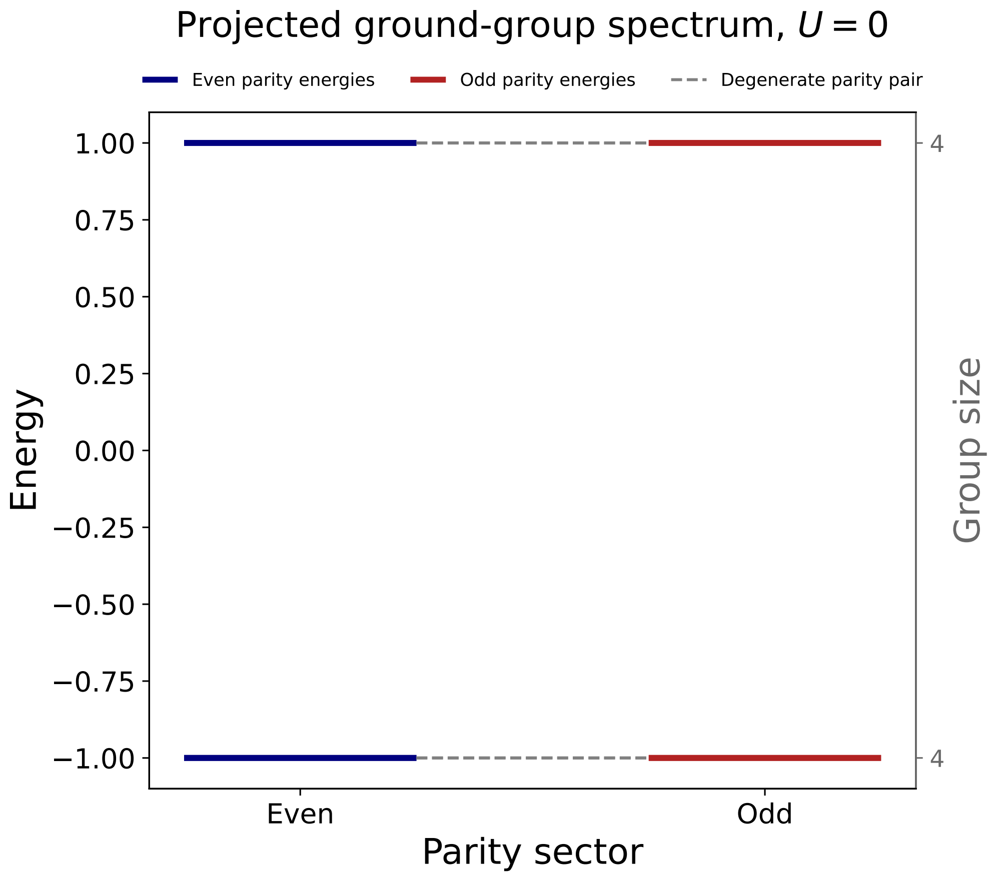

Figure 4.10: This plot shows how the initially degenerate energy sector splits into two distinct bands when the braid Hamiltonian is initialized. This representative example is the ground-state sector of the projected local-operator braid in the non-interacting system.

The first scan provides an energy-level Majorana baseline, starting with the ground-state sector and then moving through the higher energy sectors while simultaneously checking different interaction strengths. In this construction, the Majorana operators are built directly from paired even and odd eigenstates of each optimized subsystem. The non-interacting case provides the cleanest reference, while the interacting cases test whether the energy-level exchange channel extracted from the optimized spectra remains stable across sectors. Figure 4.11 displays the energy-level Majorana baseline. Here the γ operators are created directly from the paired eigenstates by the construction $\gamma = |e\rangle \langle o| + \text{h.c.}$ before being projected into each sector and placed in the braid Hamiltonian. The high-deviation panel shows that the $U = 1.0$ configuration is strongly sector dependent: some sectors remain accurate, while others have errors of order 10^{-1}. The $U = 2.0$ results are much more stable across sectors and generally lie around 10^{-6}. For $U = 0.0$, $U = 0.1$, and $U = 0.5$, the errors remain below the 10^{-6} threshold across all sectors. The unusually poor and irregular $U = 1.0$ result, despite the stronger $U = 2.0$ configuration performing well in this baseline, shows that interaction strength alone does not necessarily explain the ordering of the selected configurations.

Since the low-energy junction decomposition identifies γ_{B2} and γ_{C2} as the active exchange channels, the direct projected local-operator model uses the endpoint operators

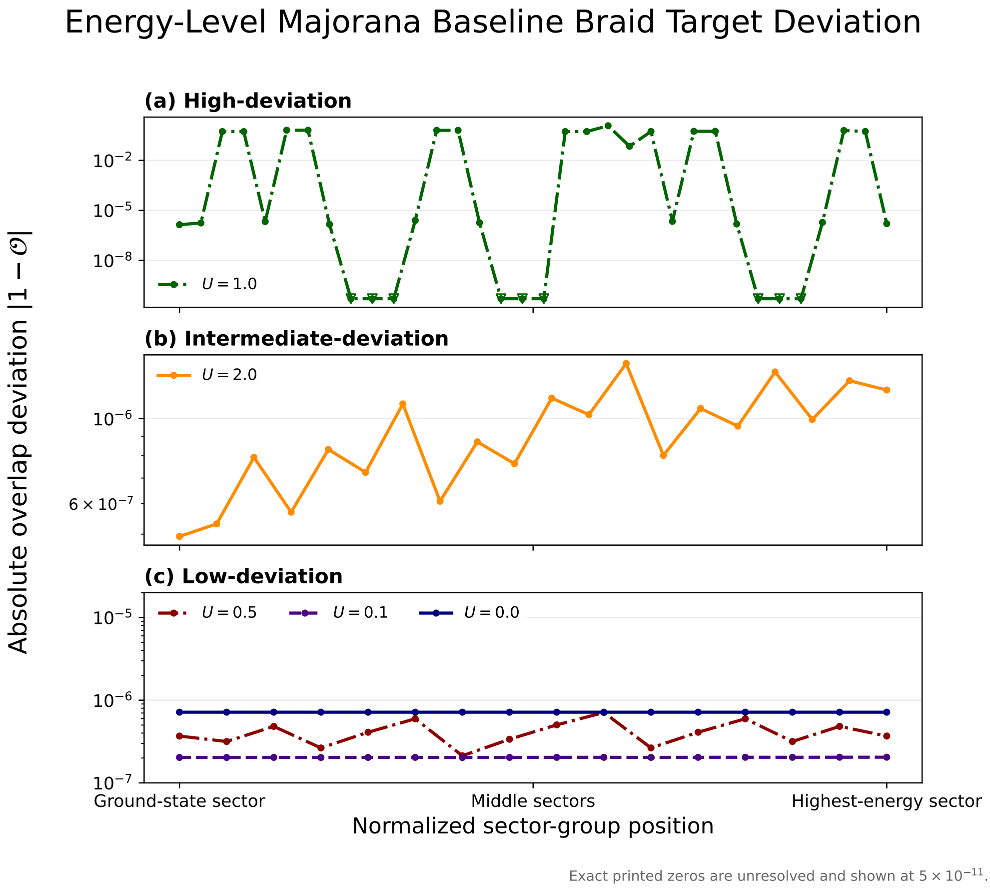

Figure 4.11: Self-target braid errors for the energy-level Majorana baseline across different energy sectors and interaction strengths $U = 0.0$, $U = 0.1$, $U = 0.5$, $U = 1.0$, and $U = 2.0$. The figure is split into three deviation regimes: a) $U = 1.0$, b) $U = 2.0$, and c) $U = 0.0$, $U = 0.1$, and $U = 0.5$. The energy-level construction supports accurate braiding across all sectors for the non-interacting and weakly interacting configurations, while the $U = 1.0$ configuration shows strong sector-dependent failures.

$\Gamma_{B,i} = P_i\, i(c_B^\dagger - c_B)P_i$ and $\Gamma_{C,i} = P_i\, i(c_C^\dagger - c_C)P_i$. When these local operators are compared with γ_{B2} and γ_{C2}, only an overall plus or minus sign is allowed, meaning the direct local operators are not rotated into a different linear combination of energy-level Majoranas.

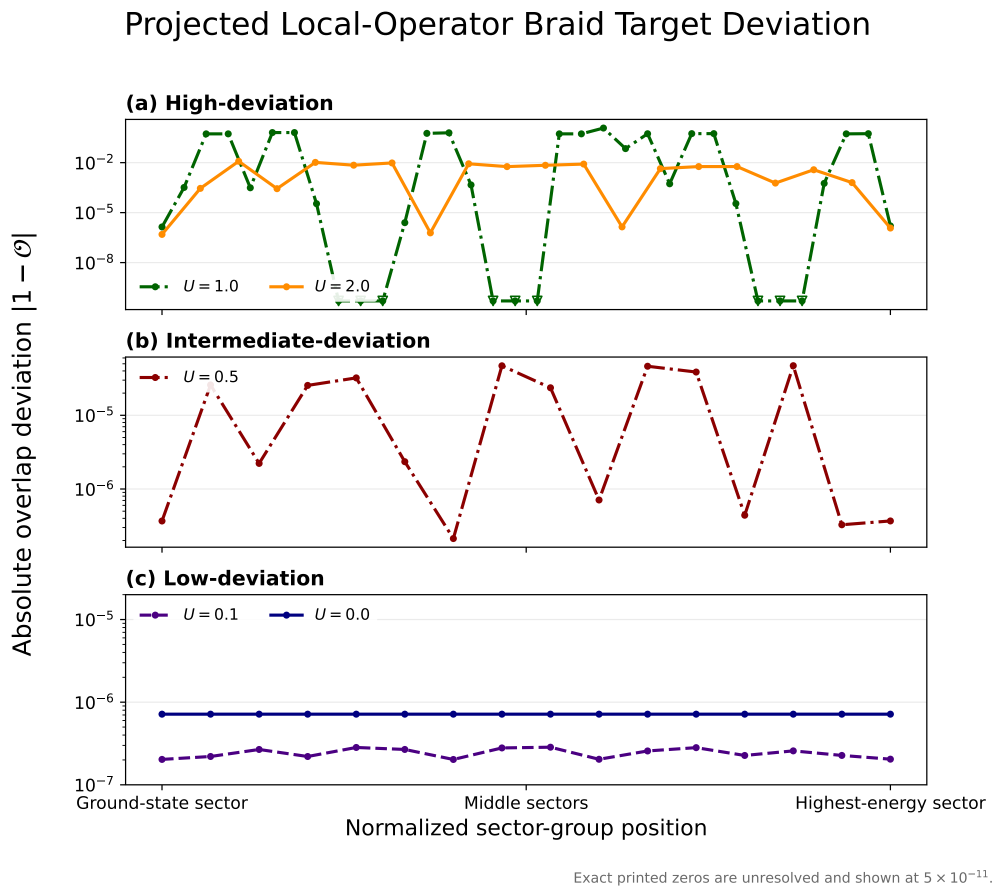

Figure 4.12: Self-target braid errors for the projected local-operator model across different energy sectors and interaction strengths $U = 0.0$, $U = 0.1$, $U = 0.5$, $U = 1.0$, and $U = 2.0$. The figure is split into three deviation regimes: a) $U = 1.0$ and $U = 2.0$, b) $U = 0.5$, and c) $U = 0.0$ and $U = 0.1$. The selected weak-interaction configurations have low deviations across all sectors, while the selected stronger-interaction configurations contain sectors with larger self-target errors.

Figure 4.12 shows how the self-target braid errors vary among the projected local-operator configurations. The high-deviation panel shows that the selected $U = 1.0$ and $U = 2.0$ configurations contain sectors with errors as large as order 10^{-1} and 10^{-2}, respectively. Compared with the energy-level Majorana baseline in Fig. 4.11, the $U = 0.5$ configuration is also more sensitive to sector choice in the projected local-operator model, with errors reaching approximately 10^{-5}. For $U = 0.0$ and $U = 0.1$, the errors remain below 10^{-6} across all tested sectors. Because each interaction strength uses a separately optimized parameter set, this trend does not isolate interaction strength as the sole cause.

During the scans, many higher-dimensional energy sectors, particularly for the non-interacting and weakly interacting configurations, were found to decompose into multiples of the eight-dimensional three-subsystem ground sector. For example, a 24-dimensional sector can contain three independent 8×8 blocks, as illustrated in Figs. 4.13 and 4.14. At stronger interaction strengths, an energy sector can instead contain only part of a labelled block. A single overall sign therefore cannot generally align the projected local operator with the energy-level Majorana across an entire higher-dimensional sector.

To account for this structure, we use the block-matching procedure described in Chapter A. The procedure first identifies the independent blocks using subsystem energy-level labels. Within each block, the projected local operator is fitted independently to a continuous linear combination of the two available energy-level Majorana components. The independently fitted blocks are then reassembled into one operator on the full sector. The diagrams below provide a short visual summary of this decomposition.

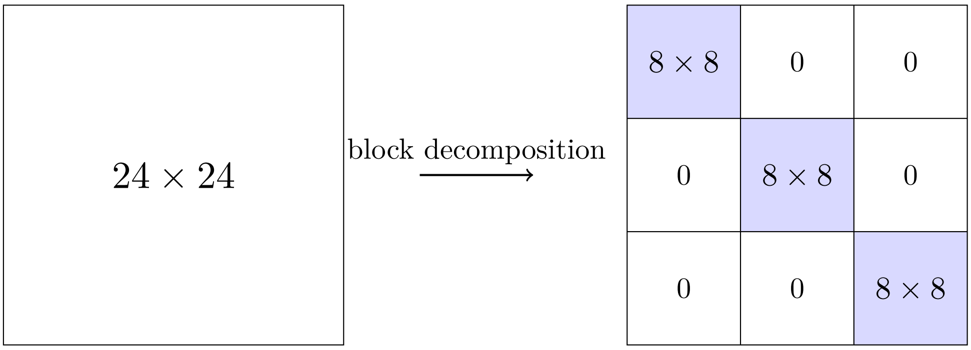

Figure 4.13: Decomposition of a 24×24 sector into three independent 8×8 blocks.

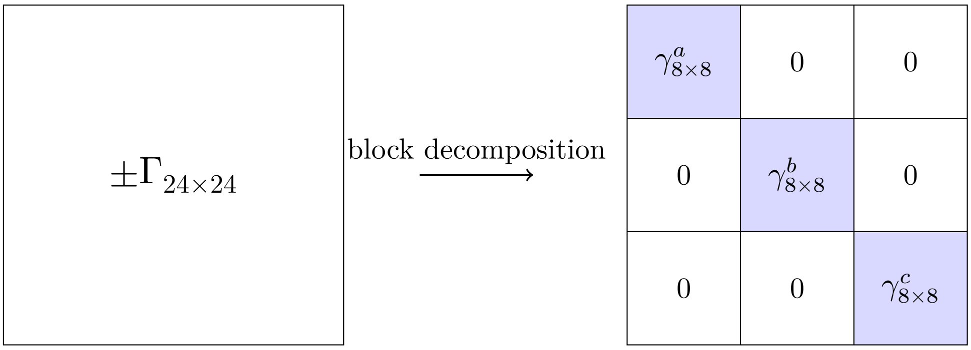

Figure 4.14: Decomposition of a $\pm\Gamma_{24\times24}$ into three independent $\gamma_{8\times8}$ blocks.

With this method in mind, we ran a final scan using the block-matched local operators. Each sector scan begins by performing the independent least-squares fits within its labelled blocks, after which the reassembled block-matched operators are used in the braid Hamiltonian.

Figure 4.15 shows that the block-matched construction retains the same overall self-target error regimes as the direct projected local-operator model. The $U = 0.0$ and $U = 0.1$ configurations remain accurate across the tested sectors, while the deviations increase at $U = 0.5$ and become larger and more sector dependent for $U = 1.0$ and $U = 2.0$. Resolving and independently fitting the internal blocks therefore does not by itself remove the sensitivity observed in the local-operator construction.

The resulting blockwise coefficients are reported in Table C.1. The numerical fitting errors remain between approximately 10^{-15} and 10^{-13} across all tested interaction strengths, showing that the block-matched operators reproduce the projected local

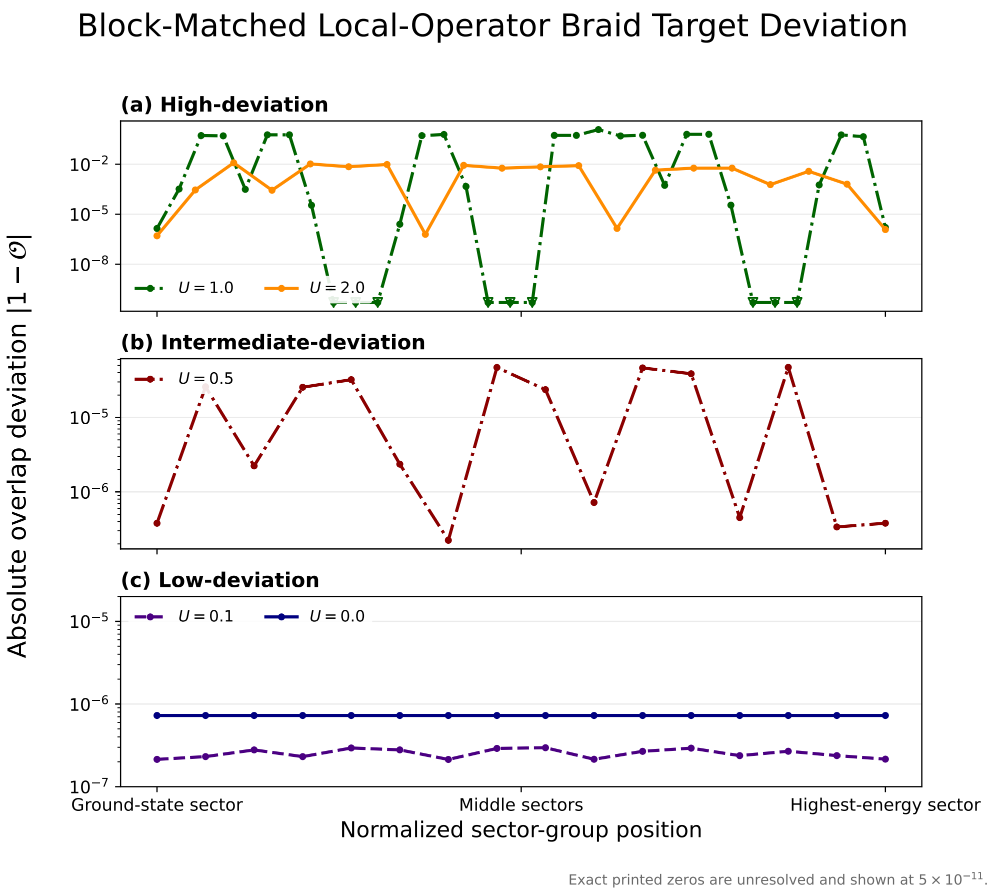

Figure 4.15: Self-target braid errors for the block-matched local-operator model. The same three deviation regimes as in Fig. 4.12 remain visible: a) $U = 1.0$ and $U = 2.0$, b) $U = 0.5$, and c) $U = 0.0$ and $U = 0.1$.

operators essentially exactly. The fitted operator within each block:

$$\Gamma_{r,local} = a_r \gamma_{r1} + b_r \gamma_{r2}$$

finds the coefficients a_r and b_r that best match the projected local operator within block r. The first-component coefficients a_r are all strongly suppressed. The local endpoint operators therefor remain aligned with the intended second Majorana component, rather than rotating into a micture of the two components. For $U = 0.0$ and $U = 0.1$, these coefficients are close to ± 1 across all blocks. Small departures become visible at $U = 0.5$, while $U = 1.0$ and $U = 2.0$ show larger block-dependent variations in magnitude.

The coefficient table also reveals that different blocks within the same sector can select different relative exchange orientations. Since the first-component coefficients vanish at the saved precision, the fitted operators in block r can be written as

$$\Gamma^{(r)}_{B,\mathrm{fit}} = b_{B,r} \gamma^{(r)}_{B2}, \qquad \Gamma^{(r)}_{C,\mathrm{fit}} = b_{C,r} \gamma^{(r)}_{C2}, \tag{4.4}$$

where $\gamma^{(r)}_{B2}$ and $\gamma^{(r)}_{C2}$ are the corresponding energy-level Majorana blocks. The fitted target unitary within this block is then

$$\begin{aligned} U_r &= \exp\left[-\frac{\pi}{4} b_{B,r} b_{C,r} \gamma^{(r)}_{B2} \gamma^{(r)}_{C2}\right] \\ &= \exp\left[-s_r \alpha_r \frac{\pi}{4} \gamma^{(r)}_{B2} \gamma^{(r)}_{C2}\right], \qquad s_r = \mathrm{sgn}(b_{B,r} b_{C,r}), \quad \alpha_r = \left|b_{B,r} b_{C,r}\right|. \end{aligned} \tag{4.5}$$

The sign s_r determines the exchange orientation relative to the fixed energy-level convention, while the magnitude α_r determines the rotation angle generated within the block. When $\alpha_r \approx 1$ and the projected operators satisfy the ideal Majorana algebra, the two orientations reduce up to a global phase and in the corresponding occupation basis, to

$$s_r = +1: \quad U_r \sim \begin{pmatrix} 1 & 0 \\ 0 & i \end{pmatrix} = \sqrt{Z}, \qquad s_r = -1: \quad U_r \sim \begin{pmatrix} 1 & 0 \\ 0 & -i \end{pmatrix} = \sqrt{Z}^{\dagger}. \tag{4.6}$$

Consequently, a sector containing both signs does not implement one uniform exchange orientation relative to the energy-level convention. Instead, its fitted target is a direct sum of block-dependent exchange orientations. The individual signs depend on the chosen Majorana sign convention, but their blockwise pattern records how the projected local channels are oriented relative to the fixed energy-level construction.

Figure 4.16 summarizes the sector-resolved self-target errors for the projected local-operator model, the block-matched local-operator model, and the energy-level Majorana baseline. For each interaction strength, the bar shows the unweighted mean across the identified sectors, the white diamond marks the ground-sector error, and the black bar shows the sector-to-sector standard deviation. Within these respective target comparisons, the $U = 0.1$ and $U = 0.0$ configurations have the lowest mean deviations. At $U = 0.5$, the local-operator constructions show a noticeable increase in their mean errors, while the energy-level Majorana baseline remains accurate. The worst and most sector-dependent performance occurs at $U = 1.0$, while $U = 2.0$ is accurate in the energy-level baseline but less accurate in both local-operator constructions.

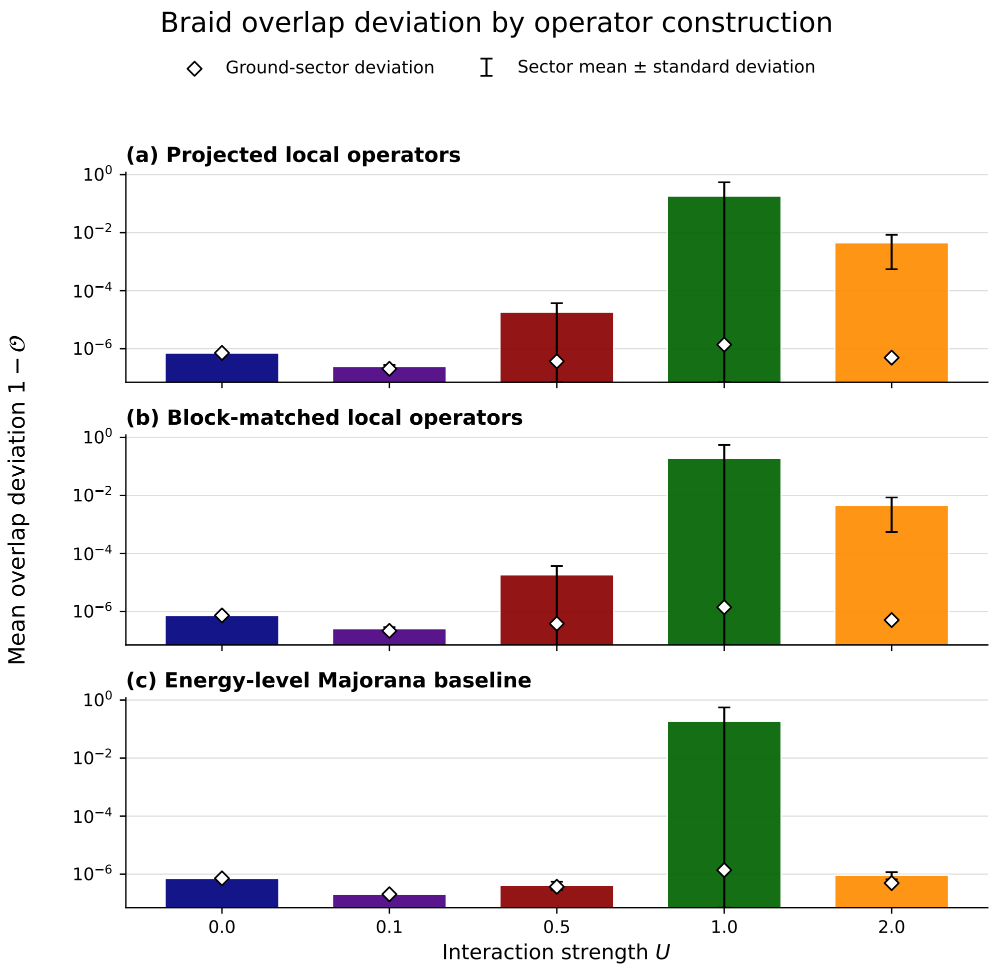

Figure 4.16: Mean self-target braid error across the tested sectors for a) the projected local-operator model, b) the block-matched local-operator model, and c) the energy-level Majorana baseline. The ground-sector error is marked by a white diamond, and the standard deviation across sectors is shown by a black bar.

4.6 Summary of Main Findings

The optimization study shows that the PMM chains recover the expected limits while also producing nontrivial interacting configurations. For the non-interacting two-dot system, the optimizer reproduces the known sweet spot, with nearly degenerate parity pairs and edge-localized Majorana polarization. For longer chains, the optimized parameters become increasingly inhomogeneous. In particular, the interior onsite energies are often shifted more strongly than the edge onsite energies. This interior detuning is correlated with favorable edge-localization diagnostics, but a controlled parameter scan would be needed to determine whether it directly suppresses bulk Majorana weight.

The post-optimization diagnostics identify the three-dot system at $U = 0.1$ as the strongest interacting braiding candidate. This configuration combines a small ground-state splitting, a large excitation gap, strong opposite-sign edge Majorana polarization, and strongly suppressed bulk polarization. The four-dot system gives a smaller charge difference in the representative weakly interacting case, but it also shows larger ground-state splitting and stronger bulk polarization. Among the tested configurations, the three-dot system therefore provides the best overall compromise between spectral degeneracy and spatial localization. It also decreases the computational cost of the sector-resolved braiding calculations compared with the four-dot system.

The low-energy braiding calculations validate this choice. After projection to the lowest manifold, the optimized three-dot building block reproduces the ideal four-Majorana exchange with high accuracy. The projected microscopic junction operators are dominated by the intended Majorana bilinears, and the ideal four-Majorana reference model, the projected microscopic junction model, and the projected local-operator model give nearly identical braid diagnostics in the lowest projected space. This shows that the optimized interacting PMM system can realize the expected Majorana exchange within the controlled low-energy projected model.

Within their respective target comparisons, the sector-resolved scans show accurate braiding across all identified energy sectors for the non-interacting and weakly interacting configurations. The energy-level Majorana baseline remains stable for $U = 0.0$, $U = 0.1$, $U = 0.5$, and $U = 2.0$, while the selected $U = 1.0$ configuration shows large variations between sectors. The direct and block-matched local-operator models are more sensitive in their self-target errors: these errors increase already at $U = 0.5$, and both $U = 1.0$ and $U = 2.0$ contain less accurate sectors.

The block-matching analysis shows that the projected local operators can be represented essentially exactly within the energy-level Majorana basis. Within each block, the local endpoint operator remains aligned with the intended second Majorana component, while its magnitude and sign can vary between blocks. The relative signs of the two exchanged operators determine whether a block follows the same or opposite exchange orientation relative to the fixed energy-level convention. A sector containing both orientations therefore does not implement one uniform logical exchange across all its internal blocks. Despite resolving this structure, block matching does not substantially change the self-target braid deviations relative to the direct projected local-operator model.

The especially poor and irregular $U = 1.0$ result should not be interpreted as evidence that $U = t = 1$ is generally worse for braiding than $U = 2t$. Since every interaction strength uses a separately optimized parameter set, the selected $U = 1.0$ configuration may represent a poorer local minimum of the optimization landscape.

For configurations that support a stable exchange channel, the self-target braid accuracy is relatively insensitive to the chosen energy sector. However, the strongly sector-dependent $U = 1.0$ result shows that this behavior is not guaranteed and depends on the selected optimized configuration. The differing sector dependence of the energy-level and local constructions motivates a common-target comparison before drawing conclusions about how accurately a projected local channel accesses the energy-level exchange.

Chapter 5

Discussion

This section interprets the numerical results in relation to the main goal of the thesis: to test whether numerical optimization can identify favorable parameter regimes in interacting quantum-dot-superconductor chains that support Majorana-like low-energy physics and useful braiding behavior. It also discusses the implications of extending the braiding analysis beyond the ground-state manifold.

5.1 Parameter Optimization Scheme

For a simple two-dot system, the optimizer managed to approach the analytically known sweet spot, which provides a reassuring sanity check. In the non-interacting case, the gradient-based search found a parameter regime that matches the known sweet spot up to a numerical tolerance of approximately 10^{-3}. This is a strong indication that the optimization procedure is working correctly, at least in the non-interacting limit.

For weakly interacting two-dot systems, the optimizer found parameter sets that are also close to the expected interacting sweet-spot conditions, where the ground-state degeneracy should be preserved. In particular, the optimized onsite energies remain close to $\epsilon_1, \epsilon_2 \approx -U/2$, with $\Delta \sim t$. This is consistent with the known interacting sweet-spot condition for two-dot systems, and suggests that the optimization is correctly identifying parameter regimes that satisfy the chosen Majorana-based criteria in the presence of weak interactions. For stronger interactions, the optimizer had more difficulty finding low-loss configurations. It also tended to push $\Delta > t$, while the relation between ϵ_i and U remained close to the expected sweet-spot condition. This trend is consistent with the constrained parameter space becoming less able to maintain even–odd degeneracy across the relevant energy levels when the interaction strength grows relative to the fixed hopping scale. However, the present searches cannot separate such a physical limitation from the increased difficulty of optimizing a more complicated loss landscape.

Beyond the two-dot sweet spot, the analytical solution rapidly becomes more complicated. Without assuming symmetry or homogeneity, the number of parameters grows with the system size, and solving the sweet-spot conditions analytically becomes increasingly difficult. This makes numerical optimization a well-motivated approach for larger systems. For the three- and four-dot systems, the optimizer found parameter regimes that are not close to a simple homogeneous sweet-spot picture. Instead, the optimized parameters show a clear trend toward strong inhomogeneity, with large-magnitude onsite potentials in the middle of the chain and, for the four-dot system, enhanced pairing amplitudes near the center. This suggests that nontrivial inhomogeneous configurations can satisfy the chosen Majorana-based criteria, but it does not by itself establish why those profiles are favorable.

The parameter searches across all system sizes show that strong interactions make it substantially harder to find low-loss configurations. This is consistent with the expectation that strong interactions can destabilize Majorana-like physics, but optimization limitations cannot be excluded. Distinguishing these explanations would require more extensive searches or controlled comparisons of the loss landscape at different interaction strengths. However, at lower interaction strengths, especially in the three-dot system, the optimizer was able to find favorable low-loss configurations. This indicates that weakly interacting parameter regimes can satisfy the combined spectral and local criteria used in the optimization. For the four-dot system, the optimizer struggled more to find low-loss configurations even at weak interactions. This could reflect limitations of the optimization strategy, but it could also indicate that the parameter space becomes more complex and potentially more constrained as the system size increases. Since the optimization budget was kept fixed across all system sizes, the larger systems may not have been explored as thoroughly as the smaller ones. It would therefore be interesting to test whether increasing the number of iterations or using a more global search strategy could help the optimizer find better configurations for the four-dot system, especially in the weak-interaction regime.

One of the more interesting findings for chains longer than two dots was the emergence of large-magnitude onsite potentials in the middle of the chain. This was not expected from the simple sweet-spot picture. The resulting interior detuning is correlated with favorable edge-localization diagnostics and is consistent with suppressing bulk Majorana weight. A controlled scan of the interior potentials would be needed to establish whether this is the mechanism responsible. Subject to that qualification, the result motivates studying controlled inhomogeneity as a possible design parameter rather than treating it only as an imperfection.

5.2 Braiding Analysis

The braiding results are mainly split into two parts. The low-energy benchmark compares the ideal four-Majorana reference model with projected microscopic junction and local-operator models. The sector-resolved analysis then compares an energy-level Majorana baseline with direct and block-matched projected local-operator constructions.

The main difference between the two constructions is what they test. The energy-level construction tests whether Majorana operators built from paired eigenstates define an internally consistent effective exchange channel. The local-operator constructions instead test whether braids built from projected endpoint operators accurately execute targets defined from those local operators. In the sector-resolved results, each construction is compared with its own target. The reported self-target errors can therefore reveal sector dependence within a construction, but they do not by themselves determine whether the local constructions reproduce the energy-level exchange channel. Establishing that stronger comparison requires evaluating all constructions against a common target.

The first part of the analysis tests whether braiding can be performed in the ground-state manifold using the Majorana modes generated by the optimized parameters. Based on the post-optimization results, the optimized $U = 0.1$ three-dot system was selected as the most promising candidate for the interacting braiding analysis. The results show that the ground-state manifold is well isolated, and that the projected junction operators closely match the ideal Majorana bilinears. This is a strong indication that the optimized parameters support Majorana-like physics in the ground-state manifold, and that the braiding protocol can be successfully implemented within this projected model.

After verifying the braid in the ground-state manifold, we extended the analysis to higher energy sectors and compared the interaction strengths $\frac{U}{t} = 0.0, 0.1, 0.5, 1.0, 2.0$. This was done in three parts. First, the energy-level Majorana baseline tested whether paired eigenstates define a stable exchange channel in each sector. Second, the direct projected local-operator construction tested whether a microscopic endpoint operator can perform its corresponding self-target braid in the same sectors. Finally, the block-matched local-operator construction resolved the internal labelled blocks of each sector and fitted the local operator independently within them.

These results are shown in Figs. 4.11, 4.12, 4.15 and 4.16. The reported self-target braids work well across all sectors for the non-interacting and weakly interacting configurations. This is consistent with the optimization scheme favoring even–odd degeneracy across the spectrum while also favoring gaps between neighboring parity pairs.

The energy-level Majorana baseline is stable for most tested interaction strengths, but the $U = 1.0$ configuration is strongly sector dependent. Some of its sectors braid accurately, while others fail by several orders of magnitude. Since the $U = 2.0$ energy-level baseline performs much better, interaction strength alone does not explain the ordering of the selected configurations. The irregular $U = 1.0$ result may reflect its particular optimized parameters, non-monotonic interaction effects, or both. It remains possible that a better $U = 1.0$ minimum exists elsewhere in the optimization landscape.

The direct and block-matched local-operator constructions show a stronger dependence of their self-target errors on both interaction strength and sector. The $U = 0.5$ case, which remains in the low-deviation regime for the energy-level Majorana baseline, moves into the intermediate-deviation regime for both local constructions. The ground-state sector is nevertheless among the lower-error sectors in most cases. Within the self-target comparison used here, the block-matched local-operator braid gives almost the same deviations as the direct local-operator braid. Resolving the internal block structure therefore does not by itself remove the observed self-target sensitivity, although this comparison does not establish how either local construction agrees with the energy-level target.

The block-dependent exchange orientations also have consequences for quantum computation. In the successful low-interaction cases, where the fitted coefficient products have magnitude close to one, opposite exchange orientations correspond to $\sqrt{Z}$ and $\sqrt{Z}^\dagger$. These gates differ by a relative phase rather than an irrelevant global phase:

$$\sqrt{Z} : \phi \mapsto \phi + \frac{\pi}{2}, \qquad \sqrt{Z}^\dagger : \phi \mapsto \phi - \frac{\pi}{2}. \tag{5.1}$$

They therefore rotate a logical qubit in opposite directions around the Bloch-sphere z-axis and can produce distinguishable outcomes in later interference measurements. If computation is restricted to one known block, its exchange orientation can be tracked and incorporated into the logical gate convention. However, a coherent state occupying blocks with different orientations experiences a block-dependent logical operation:

$$\begin{aligned} U_{\text{braid}} &= U_1 \oplus U_2 \oplus U_3, \\ |\Psi\rangle &= a\,|\psi_1\rangle + b\,|\psi_2\rangle + c\,|\psi_3\rangle\,, \\ U_{\text{braid}}\,|\Psi\rangle &= aU_1\,|\psi_1\rangle + bU_2\,|\psi_2\rangle + cU_3\,|\psi_3\rangle\,. \end{aligned} \tag{5.2}$$

If the block label is unobserved, this can appear as dephasing of the logical qubit. Reliable computation across a larger energy sector would therefore require control of the occupied block, tracking of its exchange orientation, or a protocol producing a uniform orientation across all relevant blocks.

The main implication is that, particularly in the low-interaction configurations, the projected local operators can be represented essentially exactly within the energy-level Majorana basis. However, their block-dependent exchange orientations mean that this operator-level agreement does not guarantee one uniform logical operation across an entire higher-dimensional sector. Predictable braiding across such a sector therefore requires knowledge or control of the occupied block.

5.3 Limitations and Outlook

The main limitation of the present study is that the optimization procedure does not locate exact sweet spots; it finds numerically favorable minima of the chosen loss function. In a complicated loss landscape, a low loss may represent a local minimum even when a better solution exists elsewhere. Moreover, the spectral, charge, and Majorana-polarization diagnostics are also components of the optimization objective. Their favorable values confirm that the optimizer satisfies its intended criteria, but they are not fully independent validation of the resulting configurations. The low-energy microscopic-junction braid provides a more independent test, since it is not directly optimized. There is also no direct correspondence between the numerical value of the loss and the expected error of a later observable. The results should therefore be interpreted as evidence for approximate Majorana-like behavior within the tested models, not as proof of exact topological protection.

A second limitation is that the present braiding analysis is still highly controlled. The clean agreement found in the lowest manifold relies on projection to a small low-energy space and on an idealized, disorder-free setting. The study does not yet include disorder, quasiparticle poisoning, nonadiabatic driving errors, or explicit coupling to thermal environments. Those effects are precisely the kinds of ingredients that would determine whether the optimized configurations remain useful in a more realistic device-level setting.

Within the controlled sector-resolved projected models, a potentially useful finding is that the braiding protocol can also be implemented in higher-energy sectors. For configurations with a stable exchange channel, the self-target results are relatively insensitive to the selected sector. The strongly sector-dependent $U = 1.0$ result shows that this behavior is not guaranteed and may depend strongly on the optimized parameter set. Whether such higher-energy sectors can be prepared and manipulated in a realistic device remains an open question. The repeated interior detuning in the optimized parameters also motivates a controlled study of whether inhomogeneous onsite potentials can suppress bulk Majorana weight and improve edge localization.

Chapter 6

Conclusion

This thesis investigated whether interacting quantum-dot chains can be numerically optimized to realize Majorana-like low-energy states and whether the resulting configurations support a useful braiding protocol. The optimization reproduces the known non-interacting two-dot sweet spot and, for longer chains, identifies more inhomogeneous parameter profiles with strong central structure in the onsite energies and pairing amplitudes. In particular, the recurring interior detuning is correlated with favorable edge-localization diagnostics and suppressed bulk Majorana polarization, although a controlled parameter scan would be required to establish whether it is directly responsible for this behavior.

Among the tested interacting configurations, the weakly interacting three-dot system provides the strongest overall braiding candidate. It combines a small ground-state splitting, a large excitation gap, edge-localized Majorana polarization, and strongly suppressed bulk polarization. Stronger interactions and larger systems were generally more difficult to optimize within the searches performed here. However, the optimization combines local gradient-based updates, repeated restarts, and basin hopping, and the finite numerical search cannot guarantee that the global minimum was found. The optimized configurations should therefore be interpreted as promising parameter regimes rather than proof of perfectly robust Majorana states.

Within the lowest projected manifold, the optimized three-dot system reproduces the expected Majorana exchange with high accuracy. The projected microscopic junction operators are dominated by the intended Majorana bilinears, and the ideal four-Majorana reference, projected microscopic junction, and projected local-operator models give nearly identical braid diagnostics. This demonstrates that the optimized interacting PMM system can support the expected exchange behavior within the controlled low-energy projected model.

The sector-resolved analysis shows that self-target braiding remains accurate across all identified sectors for the non-interacting and weakly interacting configurations. The selected stronger-interaction local-operator constructions contain less accurate and more sector-dependent results. However, the stable $U = 2.0$ energy-level baseline and irregular $U = 1.0$ result show that interaction strength alone does not explain the observed ordering. The quality of the selected optimized configuration and the distinction between the energy-level and local exchange constructions are also important.

The block-matching analysis shows that the projected local operators can be represented essentially exactly within the energy-level Majorana basis. However, different internal blocks can follow opposite exchange orientations relative to the fixed energy-level convention. A higher-dimensional sector therefore does not necessarily implement one uniform logical gate, and predictable braiding across such a sector requires knowledge or control of the occupied block.

Overall, optimization can identify interacting microscopic PMM configurations that reproduce effective Majorana exchange within controlled projected models. Extending this behavior beyond one known low-energy block requires additional control over the occupied block and exchange orientation, as well as a common-target comparison to determine how accurately projected local channels reproduce the energy-level exchange. Natural next steps are to improve the optimization coverage, test the role of controlled interior detuning, and include disorder, noise, finite-temperature effects, and nonadiabatic driving in the braiding analysis.

Appendix A

Block Labeling Method

The higher-dimensional energy sectors can contain several independent subspaces associated with different combinations of subsystem energy levels. This is especially clear in the non-interacting and weakly interacting configurations, where a 24-dimensional sector commonly decomposes into three 8-dimensional blocks and a 72-dimensional sector into nine such blocks. The 24-dimensional example is illustrated in Figs. 4.13 and 4.14. At stronger interaction strengths, an energy sector may intersect only part of one of these blocks, so neither the sector dimension nor every labelled block is necessarily a multiple of eight.

The blocks are identified from the energy levels of the three individual subsystems. For a single three-dot subsystem, the four even-parity eigenstates are paired with the four odd-parity eigenstates by nearest energy. For pair ℓ, define the projector

$$Q_\ell = |e_\ell\rangle \langle e_\ell| + |o_\ell\rangle \langle o_\ell| , \qquad \ell = 0, 1, 2, 3. \tag{A.1}$$

The single-subsystem label operator is then

$$L_{\text{single}} = \sum_{\ell=0}^{3} \ell Q_\ell. \tag{A.2}$$

Its eigenvalue records which even–odd energy pair a state belongs to without distinguishing the two parity states within that pair.

To label the three-subsystem product states, L_{single} is embedded separately on subsystems A, B, and C:

$$\begin{aligned} L_A &= L_{\text{single}} \otimes I \otimes I, \\ L_B &= I \otimes L_{\text{single}} \otimes I, \\ L_C &= I \otimes I \otimes L_{\text{single}}. \end{aligned} \tag{A.3}$$

These operators are combined using a positional label,

$$L_{\text{full}} = L_A + BL_B + B^2 L_C, \qquad B = 5. \tag{A.4}$$

Because B is larger than the greatest single-subsystem label, each eigenvalue

$$\lambda(\ell_A, \ell_B, \ell_C) = \ell_A + B\ell_B + B^2 \ell_C \tag{A.5}$$

uniquely identifies the tuple (ℓ_A, ℓ_B, ℓ_C). In the full Hilbert space, each tuple spans the eight product states obtained by independently choosing even or odd parity in the three subsystems.

Let V_g contain an orthonormal basis for energy sector g. The full label operator is projected into this sector as

$$L_g = V_g^\dagger L_{\text{full}} V_g. \tag{A.6}$$

Diagonalizing L_g,

$$L_g w_i = \lambda_i w_i, \tag{A.7}$$

groups the sector basis vectors by their encoded subsystem labels. Vectors with the same eigenvalue are collected in a block basis W_r. For an operator O_g already projected into sector g, its restriction to block r is

$$O_g^{(r)} = W_r^\dagger O_g W_r. \tag{A.8}$$

Within each labelled block, the projected local endpoint operator is fitted independently to the two corresponding energy-level Majorana components, as defined in Eq. (3.30). The fitted blocks are then reassembled into an operator on the complete sector. The label construction only identifies the independent blocks; it does not impose the Majorana algebra, which is checked separately after projection. In well defined cases, the block-matched operator is simply a signed version of the energy-level Majorana operator, but for stronger interacting systems, the block-matched operator can deviate. Numerical results for the block-matched coefficients are reported in Chapter C.

Appendix B

Energy Sectors

The figures below are displaying the full system energy spectrum and the ground state energy splitting after Hamiltonian initiation for the three identical three dot systems at each of the tested interaction strengths. The system configuration and optimization results are reported in Table 4.1. The leftmost Y-axis of the figures are a measure of the energy of the system. The rightmost Y-axis of the figures show the count of states within each energy sector. The figures are split into even and odd subspaces, with a dashed line in between to indicate the degeneracy of the systems.

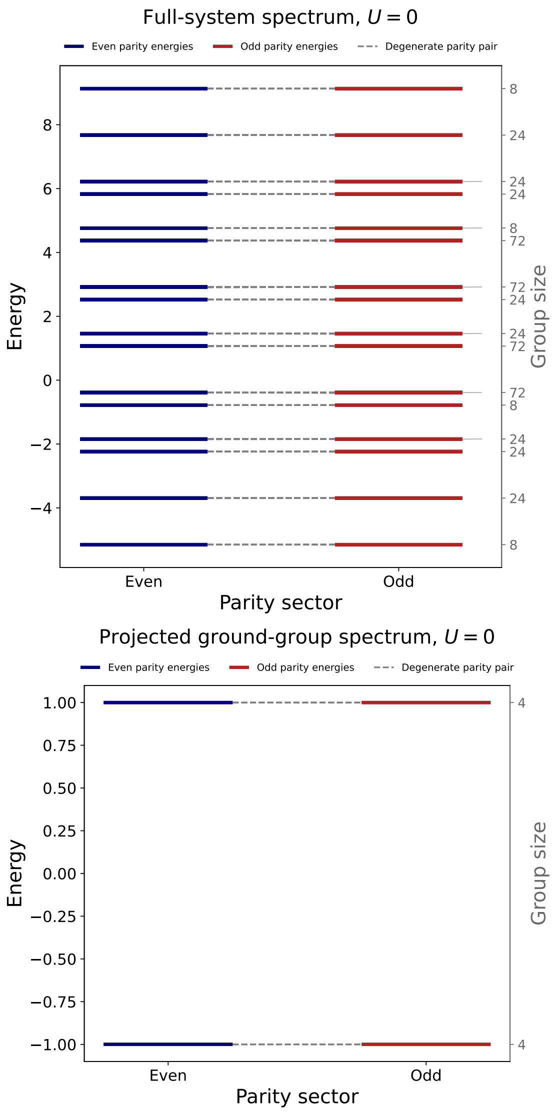

Figure B.1: Energy spectra and ground state splitting after Hamiltonian initiation. System configuration and optimization results are reported in Table 4.1. These figures are for mode $U = 0.0$, for the three identical three dot systems.

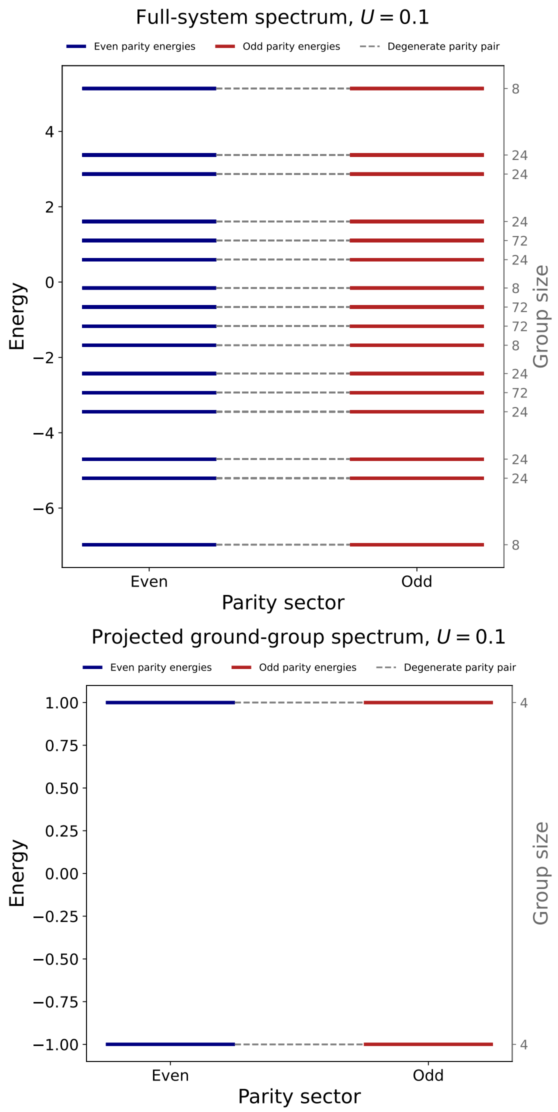

Figure B.2: Energy spectra and ground state splitting after Hamiltonian initiation. System configuration and optimization results are reported in Table 4.1. These figures are for mode $U = 0.1$, for the three identical three dot systems.

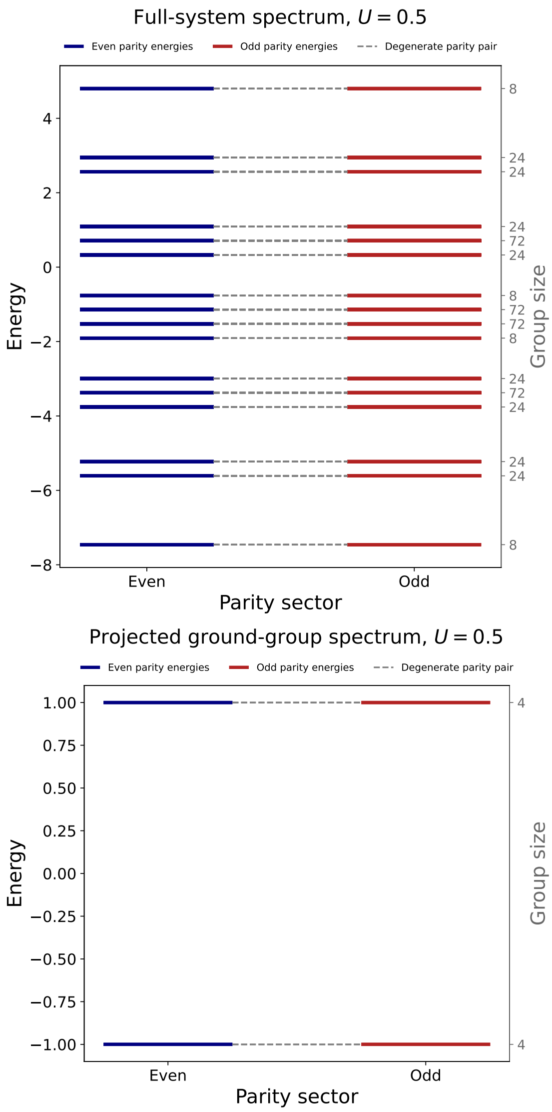

Figure B.3: Energy spectra and ground state splitting after Hamiltonian initiation. System configuration and optimization results are reported in Table 4.1. These figures are for mode $U = 0.5$, for the three identical three dot systems.

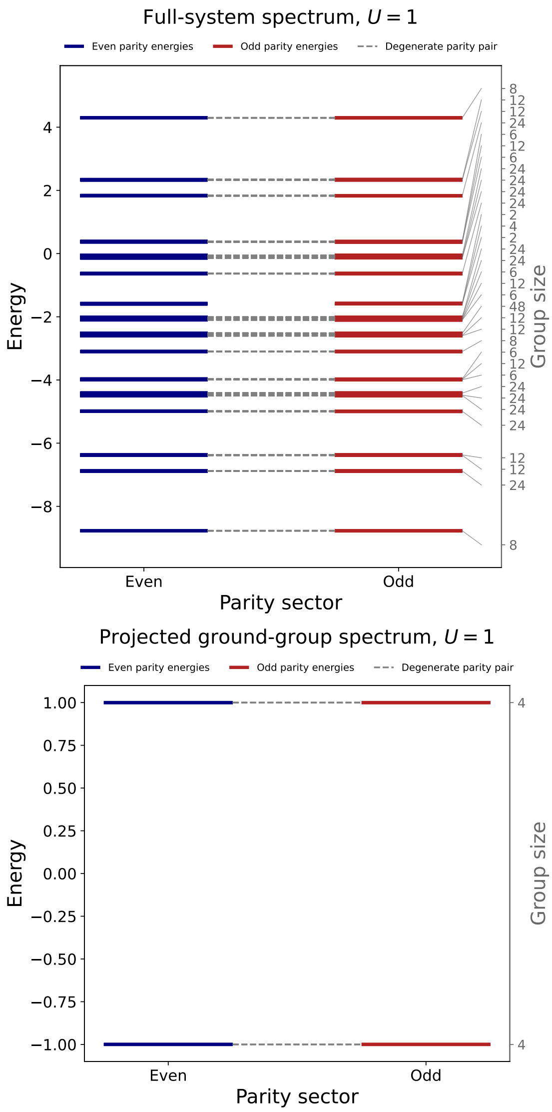

Figure B.4: Energy spectra and ground state splitting after Hamiltonian initiation. System configuration and optimization results are reported in Table 4.1. These figures are for mode $U = 1.0$, for the three identical three dot systems.

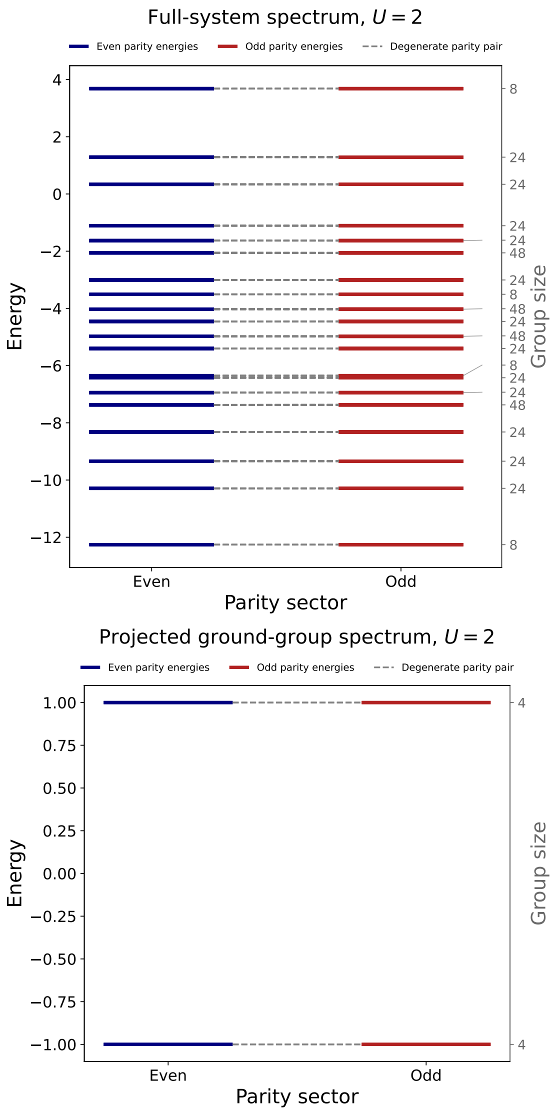

Figure B.5: Energy spectra and ground state splitting after Hamiltonian initiation. System configuration and optimization results are reported in Table 4.1. These figures are for mode $U = 2.0$, for the three identical three dot systems.

Appendix C

Block-Matched Coefficients

For each selected energy sector, the projected local endpoint operators were fitted independently within the labelled blocks to the two corresponding energy-level Majorana components. The fitted first-component coefficients vanish at the saved precision, while the second-component coefficients are close to either $+1$ or -1 in most blocks. The blockwise coefficients and their relative signs are reported in Table C.1.

Table C.1: Blockwise coefficients of the projected local endpoint operators in the energy-level Majorana basis. The entries list the fitted coefficient multiplying the second energy-level Majorana in each labelled block, in the block order used by the fit. The fitted coefficients of the first Majorana component vanish at the saved precision. The final column gives the sign of the product $b_{B,r}b_{C,r}$, which determines whether the two local operators select the same (+) or opposite (−) relative orientation in block r.

U/t	Sector	Dim.	$b_{B,r}$	$b_{C,r}$	$\mathrm{sgn}(b_{B,r}b_{C,r})$
0.0	0	8	[+1.00]	[+1.00]	[+]
0.0	1	24	[+1.00, −1.00, +1.00]	[+1.00, +1.00, −1.00]	[+, −, −]
0.0	2	24	[−1.00, +1.00, −1.00]	[+1.00, −1.00, −1.00]	[−, −, +]
0.0	3	24	[+1.00, +1.00, +1.00]	[+1.00, +1.00, +1.00]	[+, +, +]
0.0	4	8	[−1.00]	[−1.00]	[+]
0.0	5	72	[+1.00, −1.00, +1.00, −1.00, +1.00, +1.00, +1.00, −1.00, +1.00]	[+1.00, +1.00, +1.00, +1.00, −1.00, −1.00, +1.00, +1.00, −1.00]	[+, −, +, −, −, −, +, −, −]
0.0	6	72	[−1.00, −1.00, +1.00, −1.00, +1.00, −1.00, −1.00, +1.00, −1.00]	[+1.00, +1.00, −1.00, −1.00, −1.00, −1.00, +1.00, −1.00, −1.00]	[−, −, −, +, −, +, −, −, +]
0.0	7	24	[+1.00, +1.00, +1.00]	[+1.00, +1.00, +1.00]	[+, +, +]
0.0	8	24	[−1.00, −1.00, −1.00]	[−1.00, −1.00, −1.00]	[+, +, +]
0.0	9	72	[+1.00, −1.00, +1.00, +1.00, −1.00, +1.00, −1.00, +1.00, +1.00]	[+1.00, +1.00, −1.00, +1.00, +1.00, +1.00, +1.00, −1.00, −1.00]	[+, −, −, +, −, +, −, −, −]
0.0	10	72	[−1.00, +1.00, −1.00, −1.00, −1.00, +1.00, −1.00, +1.00, −1.00]	[+1.00, −1.00, −1.00, +1.00, +1.00, −1.00, −1.00, −1.00, −1.00]	[−, −, +, −, −, −, +, −, +]
0.0	11	8	[+1.00]	[+1.00]	[+]
0.0	12	24	[−1.00, −1.00, −1.00]	[−1.00, −1.00, −1.00]	[+, +, +]
0.0	13	24	[+1.00, −1.00, +1.00]	[+1.00, +1.00, −1.00]	[+, −, −]
0.0	14	24	[−1.00, +1.00, −1.00]	[+1.00, −1.00, −1.00]	[−, −, +]
0.0	15	8	[−1.00]	[−1.00]	[+]
0.1	0	8	[−1.00]	[−1.00]	[+]
0.1	1	24	[−1.00, +1.00, −1.00]	[−1.00, −1.00, +1.00]	[+, −, −]
0.1	2	24	[−1.00, −1.00, −1.00]	[−1.00, −1.00, −1.00]	[+, +, +]
0.1	3	24	[+1.00, −1.00, +1.00]	[−1.00, +1.00, +1.00]	[−, −, +]
0.1	4	72	[−1.00, +1.00, −1.00, +1.00, −1.00, −1.00, −1.00, +1.00, −1.00]	[−1.00, −1.00, −1.00, −1.00, +1.00, +1.00, −1.00, −1.00, +1.00]	[+, −, +, −, −, −, +, −, −]
0.1	5	24	[−1.00, −1.00, −1.00]	[−1.00, −1.00, −1.00]	[+, +, +]
0.1	6	8	[+1.00]	[+1.00]	[+]
0.1	7	72	[+1.00, +1.00, −1.00, +1.00, −1.00, +1.00, +1.00, −1.00, +1.00]	[−1.00, −1.00, +1.00, +1.00, +1.00, +1.00, −1.00, +1.00, +1.00]	[−, −, −, +, −, +, −, −, +]
0.1	8	72	[−1.00, +1.00, −1.00, −1.00, +1.00, −1.00, +1.00, −1.00, −1.00]	[−1.00, −1.00, +1.00, −1.00, −1.00, −1.00, −1.00, +1.00, +1.00]	[+, −, −, +, −, +, −, −, −]
0.1	9	8	[−1.00]	[−1.00]	[+]
0.1	10	24	[+1.00, +1.00, +1.00]	[+1.00, +1.00, +1.00]	[+, +, +]

Continued on next page

Table C.1 continued

U/t	Sector	Dim.	$b_{B,r}$	$b_{C,r}$	$\mathrm{sgn}(b_{B,r}b_{C,r})$
0.1	11	72	[+1.00, −1.00, +1.00, +1.00, +1.00, −1.00, +1.00, −1.00, +1.00]	[−1.00, +1.00, +1.00, −1.00, −1.00, +1.00, +1.00, +1.00, +1.00]	[−, −, +, −, −, −, +, −, +]
0.1	12	24	[−1.00, +1.00, −1.00]	[−1.00, −1.00, +1.00]	[+, −, −]
0.1	13	24	[+1.00, +1.00, +1.00]	[+1.00, +1.00, +1.00]	[+, +, +]
0.1	14	24	[+1.00, −1.00, +1.00]	[−1.00, +1.00, +1.00]	[−, −, +]
0.1	15	8	[+1.00]	[+1.00]	[+]
0.5	0	8	[+1.00]	[+1.00]	[+]
0.5	1	24	[+1.01, +0.99, +1.01]	[+1.01, +1.01, +0.99]	[+, +, +]
0.5	2	24	[+1.00, −1.00, +1.00]	[+1.00, +1.00, −1.00]	[+, −, −]
0.5	3	24	[+0.99, +1.01, +0.99]	[+1.01, +0.99, +0.99]	[+, +, +]
0.5	4	72	[+1.00, +0.98, −1.01, −1.01, +1.00, −1.01, +1.00, +0.98, +1.00]	[+1.00, +1.00, +1.00, +1.00, +0.98, +0.98, −1.01, −1.01, −1.01]	[+, +, −, −, +, −, −, −, −]
0.5	5	24	[−1.00, +1.00, −1.00]	[+1.00, −1.00, −1.00]	[−, −, +]
0.5	6	8	[+1.00]	[+1.00]	[+]
0.5	7	72	[+0.99, −1.01, +1.01, +0.99, −1.01, −1.01, +0.99, +1.01, +0.99]	[+1.01, +1.01, +0.99, +0.99, +0.99, +0.99, −1.01, −1.01, −1.01]	[+, −, +, +, −, −, −, −, −]
0.5	8	72	[−1.00, −1.00, −1.00, +1.00, +0.98, −1.00, −1.00, +1.00, −1.00]	[+1.00, +1.00, +0.98, −1.00, −1.00, −1.00, −1.00, −1.00, −1.00]	[−, −, −, −, −, +, +, −, +]
0.5	9	8	[−1.00]	[−1.00]	[+]
0.5	10	24	[+0.99, −1.02, +0.99]	[+0.99, +0.99, −1.02]	[+, −, −]
0.5	11	72	[−1.01, −1.01, −1.01, +0.98, −1.01, +1.00, +0.98, −1.01, −1.01]	[+1.00, +0.98, +0.98, −1.01, −1.01, −1.01, −1.01, −1.01, −1.01]	[−, −, −, −, +, −, −, +, +]
0.5	12	24	[−1.00, −1.00, −1.00]	[−1.00, −1.00, −1.00]	[+, +, +]
0.5	13	24	[−1.01, +0.98, −1.01]	[+0.98, −1.01, −1.01]	[−, −, +]
0.5	14	24	[−1.00, −1.00, −1.00]	[−1.00, −1.00, −1.00]	[+, +, +]
0.5	15	8	[−1.00]	[−1.00]	[+]
1.0	0	8	[+1.00]	[+1.00]	[+]
1.0	1	24	[+1.02, +0.95, +1.02]	[+1.02, +1.02, +0.95]	[+, +, +]
1.0	2	12	[+1.22, +1.23, +1.22]	[+1.22, +1.22, +1.23]	[+, +, +]
1.0	3	12	[+1.22, +1.23, +1.22]	[+1.22, +1.22, +1.23]	[+, +, +]
1.0	4	24	[+0.98, +1.05, +0.98]	[+1.05, +0.98, +0.98]	[+, +, +]
1.0	5	24	[+1.18, +1.27, +1.27, +1.27, +1.27, +1.18]	[+1.27, +1.27, +1.18, +1.18, +1.27, +1.27]	[+, +, +, +, +, +]
1.0	6	24	[+1.18, +1.27, +1.27, +1.27, +1.27, +1.18]	[+1.27, +1.27, +1.18, +1.18, +1.27, +1.27]	[+, +, +, +, +, +]
1.0	7	24	[+0.99, −1.01, +0.99]	[+0.99, +0.99, −1.01]	[+, −, −]
1.0	8	6	[+1.74, +1.73, +1.74]	[+1.73, +1.75, +1.74]	[+, +, +]
1.0	9	12	[+1.74, +1.73, +1.74]	[+1.73, +1.75, +1.74]	[+, +, +]
1.0	10	6	[+1.71, +1.73, +1.73]	[+1.73, +1.74, +1.75]	[+, +, +]
1.0	11	8	[+1.00]	[+1.00]	[+]

Continued on next page

Table C.1 continued

U/t	Sector	Dim.	$b_{B,r}$	$b_{C,r}$	$\mathrm{sgn}(b_{B,r}b_{C,r})$
1.0	12	12	[+1.22, +1.32, +1.22]	[+1.22, +1.22, +1.32]	[+, +, +]
1.0	13	12	[+1.22, +1.31, +1.22]	[+1.22, +1.22, +1.32]	[+, +, +]
1.0	14	48	[+0.95, −1.04, +1.01, −1.04, +1.01, +0.95]	[+1.01, +1.01, +0.95, +0.95, −1.04, −1.04]	[+, −, +, −, −, −]
1.0	15	6	[+1.86, +1.73, +1.86]	[+1.73, +1.84, +1.84]	[+, +, +]
1.0	16	12	[+1.87, +1.73, +1.86]	[+1.73, +1.86, +1.86]	[+, +, +]
1.0	17	6	[+1.87, +1.73, +1.86]	[+1.73, +1.86, +1.89]	[+, +, +]
1.0	18	24	[+1.22, −1.24, +1.21, −1.24, +1.21, +1.21]	[+1.21, +1.21, +1.22, +1.21, −1.24, −1.24]	[+, −, +, −, −, −]
1.0	19	24	[+1.21, −1.24, +1.21, −1.24, +1.21, +1.22]	[+1.21, +1.21, +1.21, +1.21, −1.24, −1.24]	[+, −, +, −, −, −]
1.0	20	2	[+1.02]	[+4.81]	[+]
1.0	21	4	[+1.25]	[+1.66]	[+]
1.0	22	2	[+1.78]	[+1.21]	[+]
1.0	23	24	[+0.97, −1.06, +0.97]	[+0.97, +0.97, −1.06]	[+, −, −]
1.0	24	24	[+1.26, −1.28, +1.17, −1.28, +1.17, +1.25]	[+1.17, +1.17, +1.26, +1.26, −1.28, −1.28]	[+, −, +, −, −, −]
1.0	25	24	[+1.26, −1.28, +1.17, −1.28, +1.17, +1.25]	[+1.17, +1.17, +1.25, +1.25, −1.28, −1.28]	[+, −, +, −, −, −]
1.0	26	24	[−1.01, +0.99, −1.01]	[+0.99, −1.01, −1.01]	[−, −, +]
1.0	27	6	[+1.70, −1.73, +1.70]	[+1.70, +1.70, −1.73]	[+, −, −]
1.0	28	12	[+1.70, −1.73, +1.70]	[+1.70, +1.70, −1.73]	[+, −, −]
1.0	29	6	[+1.71, −1.73, +1.71]	[+1.72, +1.70, −1.73]	[+, −, −]
1.0	30	24	[−1.03, +0.94, −1.03]	[+0.94, −1.03, −1.03]	[−, −, +]
1.0	31	12	[−1.22, +1.20, −1.22]	[+1.20, −1.22, −1.22]	[−, −, +]
1.0	32	12	[−1.22, +1.20, −1.22]	[+1.19, −1.22, −1.22]	[−, −, +]
1.0	33	8	[−1.00]	[−1.00]	[+]
2.0	0	8	[−1.00]	[−1.00]	[+]
2.0	1	24	[−1.02, +0.96, −1.02]	[−1.02, −1.02, +0.96]	[+, −, −]
2.0	2	24	[−1.11, +0.74, −1.11]	[−1.11, −1.11, +0.74]	[+, −, −]
2.0	3	24	[+0.98, −1.04, +0.98]	[−1.04, +0.98, +0.98]	[−, −, +]
2.0	4	48	[+1.07, +0.76, −1.14, +0.76, −1.14, +1.07]	[−1.14, −1.14, +1.07, +1.07, +0.76, +0.76]	[−, −, −, +, −, +]
2.0	5	24	[−1.09, −0.80, −1.09]	[−1.09, −1.09, −0.80]	[+, +, +]
2.0	6	24	[+0.84, −1.26, +0.84]	[−1.26, +0.84, +0.84]	[−, −, +]
2.0	7	8	[+1.00]	[+1.00]	[+]
2.0	8	24	[+1.09, +0.78, +1.09]	[+1.09, +1.09, +0.78]	[+, +, +]
2.0	9	48	[+1.05, −0.82, −1.11, −0.82, −1.11, +1.05]	[−1.11, −1.11, +1.05, +1.05, −0.82, −0.82]	[−, +, −, −, +, −]
2.0	10	24	[+0.87, +1.22, +0.87]	[+1.22, +0.87, +0.87]	[+, +, +]
2.0	11	48	[+0.82, −0.90, −1.23, −0.90, −1.23, +0.82]	[−1.23, −1.23, +0.82, +0.82, −0.90, −0.90]	[−, +, −, −, +, −]
2.0	12	8	[+1.00]	[+1.00]	[+]
2.0	13	24	[+1.07, −0.84, +1.07]	[+1.07, +1.07, −0.84]	[+, −, −]
2.0	14	48	[+0.84, −0.93, +1.19, −0.93, +1.19, +0.84]	[+1.19, +1.19, +0.84, +0.84, −0.93, −0.93]	[+, −, +, −, −, −]
2.0	15	24	[−0.88, −1.20, −0.88]	[−1.20, −0.88, −0.88]	[+, +, +]
2.0	16	24	[+0.97, −1.07, +0.97]	[+0.97, +0.97, −1.07]	[+, −, −]

Continued on next page

Table C.1 continued

U/t	Sector	Dim.	$b_{B,r}$	$b_{C,r}$	$\mathrm{sgn}(b_{B,r}b_{C,r})$
2.0	17	24	$[-0.91,\ +1.16,\ -0.91]$	$[+1.16,\ -0.91,\ -0.91]$	$[-,\ -,\ +]$
2.0	18	24	$[-1.03,\ +0.93,\ -1.03]$	$[+0.93,\ -1.03,\ -1.03]$	$[-,\ -,\ +]$
2.0	19	8	$[-1.00]$	$[-1.00]$	$[+]$

Bibliography

[1] Ramón Aguado et al., eds. *New Trends and Platforms for Quantum Technologies*. Vol. 1025. Lecture Notes in Physics. Cham: Springer Nature Switzerland, 2024. ISBN: 978-3-031-55656-2 978-3-031-55657-9. DOI: 10.1007/978-3-031-55657-9. URL: https://link.springer.com/10.1007/978-3-031-55657-9 (visited on 06/13/2026).

[2] Anton Akhmerov, Jay Sau, and Bernard van Heck. *Bulk-Edge Correspondence in the Kitaev Chain*. Course material licensed under CC BY-SA 4.0. Topology in Condensed Matter. URL: https://topocondmat.org/w1-topointro/d-1/ (visited on 06/14/2026).

[3] Jason Alicea. "New Directions in the Pursuit of Majorana Fermions in Solid State Systems". Version 1. In: (2012). DOI: 10.48550/ARXIV.1202.1293. URL: https://arxiv.org/abs/1202.1293 (visited on 06/14/2026).

[4] Jason Alicea. "New Directions in the Pursuit of Majorana Fermions in Solid State Systems". In: *Reports on Progress in Physics* 75.7 (July 1, 2012), p. 076501. ISSN: 0034-4885, 1361-6633. DOI: 10.1088/0034-4885/75/7/076501. arXiv: 1202.1293 [cond-mat.supr-con]. URL: http://arxiv.org/abs/1202.1293 (visited on 06/14/2026).

[5] Nur R. Ayukaryana, Mohammad H. Fauzi, and Eddwi H. Hasdeo. "The Quest and Hope of Majorana Zero Modes in Topological Superconductor for Fault-Tolerant Quantum Computing: An Introductory Overview". In: PROCEEDINGS OF THE 4TH INTERNATIONAL SEMINAR ON METALLURGY AND MATERIALS (ISMM2020): Accelerating Research and Innovation on Metallurgy and Materials for Inclusive and Sustainable Industry. Tangerang Selatan, Indonesia, 2021, p. 020007. DOI: 10.1063/5.0059974. URL: https://pubs.aip.org/aip/acp/article/783518 (visited on 06/13/2026).

[6] Nur R. Ayukaryana, Mohammad H. Fauzi, and Eddwi H. Hasdeo. "The Quest and Hope of Majorana Zero Modes in Topological Superconductor for Fault-Tolerant Quantum Computing: An Introductory Overview". In: PROCEEDINGS OF THE 4TH INTERNATIONAL SEMINAR ON METALLURGY AND MATERIALS (ISMM2020): Accelerating Research and Innovation on Metallurgy and Materials for Inclusive and Sustainable Industry. Tangerang Selatan, Indonesia, 2021, p. 020007. DOI: 10.1063/5.0059974. URL: https://pubs.aip.org/aip/acp/article/783518 (visited on 06/13/2026).

[7] Stephen D. Bartlett, Terry Rudolph, and Robert W. Spekkens. "Reference Frames, Superselection Rules, and Quantum Information". In: *Reviews of Modern Physics* 79.2 (Apr. 5, 2007), pp. 555–609. ISSN: 0034-6861, 1539-0756. DOI: 10.1103/RevModPhys.79.555. URL: https://link.aps.org/doi/10.1103/RevModPhys.79.555 (visited on 06/14/2026).

[8] Carlo Beenakker. "Search for Non-Abelian Majorana Braiding Statistics in Superconductors". In: *SciPost Physics Lecture Notes* (Aug. 4, 2020), p. 15. ISSN: 2590-1990. DOI: 10.21468/SciPostPhysLectNotes.15. URL: https://scipost.org/10.21468/SciPostPhysLectNotes.15 (visited on 06/16/2026).

[9] Piet W. Brouwer. "Enter the Majorana Fermion". In: *Science* 336.6084 (May 25, 2012), pp. 989–990. ISSN: 0036-8075, 1095-9203. DOI: 10.1126/science.1223302. URL: https://www.science.org/doi/10.1126/science.1223302 (visited on 06/13/2026).

[10] G. Castagnoli and M. Rasetti. "The Notions of Symmetry and Computational Feedback in the Paradigm of Steady, Simultaneous Quantum Computation". In: *International Journal of Theoretical Physics* 32.12 (Dec. 1993), pp. 2335–2347. ISSN: 0020-7748, 1572-9575. DOI: 10.1007/BF00673003. URL: http://link.springer.com/10.1007/BF00673003 (visited on 06/13/2026).

[11] Charles Dugas et al. "Incorporating Second-Order Functional Knowledge for Better Option Pricing". In: *Advances in Neural Information Processing Systems*. Ed. by T. Leen, T. Dietterich, and V. Tresp. Vol. 13. MIT Press, 2000. URL: https://proceedings.neurips.cc/paper_files/paper/2000/file/44968aece94f667e4095002d140b5896-Paper.pdf.

[12] Tom Dvir et al. "Realization of a Minimal Kitaev Chain in Coupled Quantum Dots". In: *Nature* 614.7948 (Feb. 16, 2023), pp. 445–450. ISSN: 0028-0836, 1476-4687. DOI: 10.1038/s41586-022-05585-1. URL: https://www.nature.com/articles/s41586-022-05585-1 (visited on 06/14/2026).

[13] Tom Dvir et al. "Realization of a Minimal Kitaev Chain in Coupled Quantum Dots". In: *Nature* 614.7948 (Feb. 2023), pp. 445–450. ISSN: 1476-4687. DOI: 10.1038/s41586-022-05585-1. URL: https://www.nature.com/articles/s41586-022-05585-1 (visited on 05/14/2026).

[14] Nikhil Harle, Oles Shtanko, and Ramis Movassagh. "Observing and Braiding Topological Majorana Modes on Programmable Quantum Simulators". In: *Nature Communications* 14.1 (Apr. 21, 2023), p. 2286. ISSN: 2041-1723. DOI: 10.1038/s41467-023-37725-0. URL: https://www.nature.com/articles/s41467-023-37725-0 (visited on 06/13/2026).

[15] Fabian Hassler. *Topological Quantum Computing*. Oct. 21, 2024. DOI: 10.48550/arXiv.2410.13547. arXiv: 2410.13547 [quant-ph]. URL: http://arxiv.org/abs/2410.13547 (visited on 06/16/2026). Pre-published.

[16] Raúl Hidalgo-Sacoto, Thomas Busch, and D. Blume. "Two Identical One-Dimensional Anyons with Zero-Range Interactions: Exchange Statistics, Scattering Theory, and Anyon-Anyon Mapping". In: *Physical Review A* 112.6 (Dec. 11, 2025), p. 063310. ISSN: 2469-9926, 2469-9934. DOI: 10.1103/h2vs-ll9d. URL: https://link.aps.org/doi/10.1103/h2vs-ll9d (visited on 06/14/2026).

[17] P. Jordan and E. Wigner. "Uber das Paulische Aquivalenzverbot". In: *Zeitschrift for Physik* 47.9–10 (Sept. 1928), pp. 631–651. ISSN: 1434-6001, 1434-601X. DOI: 10.1007/BF01331938. URL: http://link.springer.com/10.1007/BF01331938 (visited on 06/14/2026).

[18] Tosio Kato. "On the Adiabatic Theorem of Quantum Mechanics". In: *Journal of the Physical Society of Japan* 5.6 (Nov. 15, 1950), pp. 435–439. ISSN: 0031-9015, 1347-4073. DOI: 10.1143/JPSJ.5.435. URL: http://journals.jps.jp/doi/10.1143/JPSJ.5.435 (visited on 06/14/2026).

[19] G. Kells, N. Moran, and D. Meidan. "Localization Enhanced and Degraded Topological Order in Interacting p -Wave Wires". In: *Physical Review B* 97.8 (Feb. 20, 2018), p. 085425. ISSN: 2469-9950, 2469-9969. DOI: 10.1103/PhysRevB.97.085425. URL: https://link.aps.org/doi/10.1103/PhysRevB.97.085425 (visited on 06/16/2026).

[20] Alexei Kitaev. "Unpaired Majorana Fermions in Quantum Wires". In: *Physics-Uspekhi* 44 (10S Oct. 1, 2001), pp. 131–136. ISSN: 1468-4780. DOI: 10.1070/1063-7869/44/10S/S29. arXiv: cond-mat/0010440. URL: http://arxiv.org/abs/cond-mat/0010440 (visited on 06/13/2026).

[21] Alexei Kitaev. "Unpaired Majorana Fermions in Quantum Wires". In: *Physics-Uspekhi* 44 (10S Oct. 1, 2001), pp. 131–136. ISSN: 1468-4780. DOI: 10.1070/1063-7869/44/10S/S29. arXiv: cond-mat/0010440. URL: http://arxiv.org/abs/cond-mat/0010440 (visited on 06/14/2026).

[22] Ville Lahtinen and Jiannis Pachos. "A Short Introduction to Topological Quantum Computation". In: *SciPost Physics* 3.3 (Sept. 9, 2017), p. 021. ISSN: 2542-4653. DOI: 10.21468/SciPostPhys.3.3.021. URL: https://scipost.org/10.21468/SciPostPhys.3.3.021 (visited on 06/14/2026).

[23] Martin Leijnse and Karsten Flensberg. "Introduction to Topological Superconductivity and Majorana Fermions". In: *Semiconductor Science and Technology* 27.12 (Dec. 5, 2012), p. 124003. ISSN: 0268-1242, 1361-6641. DOI: 10.1088/0268-1242/27/12/124003. URL: https://iopscience.iop.org/article/10.1088/0268-1242/27/12/124003 (visited on 06/16/2026).

[24] Martin Leijnse and Karsten Flensberg. "Parity Qubits and Poor Man's Majorana Bound States in Double Quantum Dots". In: *Physical Review B* 86.13 (Oct. 23, 2012), p. 134528. ISSN: 1098-0121, 1550-235X. DOI: 10.1103/PhysRevB.86.134528. arXiv: 1207.4299 [cond-mat.mes-hall]. URL: http://arxiv.org/abs/1207.4299 (visited on 06/13/2026).

[25] Martin Leijnse and Karsten Flensberg. "Parity Qubits and Poor Man's Majorana Bound States in Double Quantum Dots". In: *Physical Review B* 86.13 (Oct. 23, 2012), p. 134528. ISSN: 1098-0121, 1550-235X. DOI: 10.1103/PhysRevB.86.134528. URL: https://link.aps.org/doi/10.1103/PhysRevB.86.134528 (visited on 06/14/2026).

[26] Martin Leijnse and Karsten Flensberg. "Parity Qubits and Poor Man's Majorana Bound States in Double Quantum Dots". In: *Physical Review B* 86.13 (Oct. 23, 2012), p. 134528. ISSN: 1098-0121, 1550-235X. DOI: 10.1103/PhysRevB.86.134528. URL: https://link.aps.org/doi/10.1103/PhysRevB.86.134528 (visited on 05/14/2026).

[27] Biao Lian et al. "Topological Quantum Computation Based on Chiral Majorana Fermions". In: *Proceedings of the National Academy of Sciences* 115.43 (Oct. 23, 2018), pp. 10938–10942. ISSN: 0027-8424, 1091-6490. DOI: 10.1073/pnas.1810003115. URL: https://pnas.org/doi/full/10.1073/pnas.1810003115 (visited on 06/13/2026).

[28] Ettore Majorana. "Teoria simmetrica dell'elettrone e del positrone". In: *Il Nuovo Cimento* 14.4 (Apr. 1937), pp. 171–184. ISSN: 0029-6341, 1827-6121. DOI: 10.1007/BF02961314. URL: http://link.springer.com/10.1007/BF02961314 (visited on 06/13/2026).

[29] V. Mourik et al. “Signatures of Majorana Fermions in Hybrid Superconductor-Semiconductor Nanowire Devices”. In: *Science* 336.6084 (May 25, 2012), pp. 1003–1007. ISSN: 0036-8075, 1095-9203. DOI: 10.1126/science.1222360. URL: https://www.science.org/doi/10.1126/science.1222360 (visited on 06/13/2026).

[30] Dana Najjar. *Milestone Evidence for Anyons, a Third Kingdom of Particles*. Illustration by 5W Infographics. Quanta Magazine. May 12, 2020. URL: https://www.quantamagazine.org/milestone-evidence-for-anyons-a-third-kingdom-of-particles-20200512/ (visited on 06/14/2026).

[31] Boris Narozhny. “Majorana Fermions in the Nonuniform Ising-Kitaev Chain: Exact Solution”. In: *Scientific Reports* 7.1 (May 3, 2017), p. 1447. ISSN: 2045-2322. DOI: 10.1038/s41598-017-01413-z. URL: https://www.nature.com/articles/s41598-017-01413-z (visited on 06/13/2026).

[32] Chetan Nayak et al. “Non-Abelian Anyons and Topological Quantum Computation”. In: *Reviews of Modern Physics* 80.3 (Sept. 12, 2008), pp. 1083–1159. ISSN: 0034-6861, 1539-0756. DOI: 10.1103/RevModPhys.80.1083. URL: https://link.aps.org/doi/10.1103/RevModPhys.80.1083 (visited on 06/14/2026).

[33] Michael A Nielsen et al. “The Fermionic canonical commutation relations and the Jordan-Wigner transform”. In: *School of Physical Sciences The University of Queensland* 59 (2005), p. 75.

[34] Maximilian Nitsch et al. “Adiabatic Nonabelian Braiding of Imperfect Majoranas”. In: (2025). DOI: 10.48550/arXiv.2507.11039. arXiv: 2507.11039 [cond-mat.mes-hall].

[35] William Samuelson et al. “Quantifying Robustness and Locality of Majorana Bound States in Interacting Systems”. In: *PRX Quantum* 7.2 (May 28, 2026), p. 020341. ISSN: 2691-3399. DOI: 10.1103/pw51-tnsz. URL: https://link.aps.org/doi/10.1103/pw51-tnsz (visited on 06/16/2026).

[36] J E Sanches et al. “Revisiting the Poor Man’s Majoranas: The Spin–Exchange Induced Spillover Effect”. In: *Journal of Physics: Condensed Matter* 38.2 (Jan. 16, 2026), p. 023001. ISSN: 0953-8984, 1361-648X. DOI: 10.1088/1361-648X/ae2f13. URL: https://iopscience.iop.org/article/10.1088/1361-648X/ae2f13 (visited on 06/13/2026).

[37] J E Sanches et al. “Revisiting the Poor Man’s Majoranas: The Spin–Exchange Induced Spillover Effect”. In: *Journal of Physics: Condensed Matter* 38.2 (Jan. 16, 2026), p. 023001. ISSN: 0953-8984, 1361-648X. DOI: 10.1088/1361-648X/ae2f13. URL: https://iopscience.iop.org/article/10.1088/1361-648X/ae2f13 (visited on 06/13/2026).

[38] Sankar Das Sarma, Michael Freedman, and Chetan Nayak. “Majorana Zero Modes and Topological Quantum Computation”. In: *npj Quantum Information* 1.1 (Oct. 27, 2015), p. 15001. ISSN: 2056-6387. DOI: 10.1038/npjqi.2015.1. URL: https://www.nature.com/articles/npjqi20151 (visited on 06/14/2026).

[39] Sankar Das Sarma, Michael Freedman, and Chetan Nayak. “Majorana Zero Modes and Topological Quantum Computation”. In: *npj Quantum Information* 1.1 (Oct. 27, 2015), p. 15001. ISSN: 2056-6387. DOI: 10.1038/npjqi.2015.1. URL: https://www.nature.com/articles/npjqi20151 (visited on 06/14/2026).

[40] Jay D. Sau and S. Das Sarma. "Realizing a Robust Practical Majorana Chain in a Quantum-Dot-Superconductor Linear Array". In: *Nature Communications* 3.1 (July 17, 2012), p. 964. ISSN: 2041-1723. DOI: 10.1038/ncomms1966. URL: https://www.nature.com/articles/ncomms1966 (visited on 06/14/2026).

[41] Kirill Shtengel. "A Home for Anyon?" In: *Nature Physics* 3.11 (Nov. 2007), pp. 763–763. ISSN: 1745-2473, 1745-2481. DOI: 10.1038/nphys767. URL: https://www.nature.com/articles/nphys767 (visited on 06/14/2026).

[42] Viktor Svensson and Martin Leijnse. "Quantum Dot Based Kitaev Chains: Majorana Quality Measures and Scaling with Increasing Chain Length". In: *Physical Review B* 110.15 (Oct. 25, 2024), p. 155436. ISSN: 2469-9950, 2469-9969. DOI: 10.1103/PhysRevB.110.155436. URL: https://link.aps.org/doi/10.1103/PhysRevB.110.155436 (visited on 06/13/2026).

[43] Viktor Svensson and Martin Leijnse. "Quantum Dot Based Kitaev Chains: Majorana Quality Measures and Scaling with Increasing Chain Length". In: *Physical Review B* 110.15 (Oct. 25, 2024), p. 155436. ISSN: 2469-9950, 2469-9969. DOI: 10.1103/PhysRevB.110.155436. URL: https://link.aps.org/doi/10.1103/PhysRevB.110.155436 (visited on 06/14/2026).

[44] Zhi-Lei Zhang, Guo-Jian Qiao, and C. P. Sun. "Poor Man's Majorana Modes in Two Quantum Dots Dressed by Superconducting Quasiexcitations". In: *Physical Review B* 113.19 (May 15, 2026), p. 195416. ISSN: 2469-9950, 2469-9969. DOI: 10.1103/qp5g-942h. URL: https://link.aps.org/doi/10.1103/qp5g-942h (visited on 06/13/2026).

[45] Guoyi Zhu, Ruirui Wang, and Guangming Zhang. "Majorana Fermions and Topological Quantum Computation". In: *Physics* 46.3 (2017), p. 154. DOI: 10.7693/wl20170303. URL: http://www.wuli.ac.cn/CN/10.7693/wl20170303 (visited on 06/13/2026).